
```
Package xAct`xPerm` version 1.2.3, {2015, 8, 23}
```

```
Copyright (C) 2003–2020, Jose M. Martin–Garcia, under the General Public License.
```

```
Connecting to external linux executable...
```

```
Connection established.
```

```
Package xAct`xTensor` version 1.2.0, {2021, 10, 17}
```

```
Copyright (C) 2002–2021, Jose M. Martin–Garcia, under the General Public License.
```

```
Package xAct`xPlain` version 0.0.0–developer, {2025, 5, 8}
```

```
Copyright © 2023, Will Barker and Sebastian Zell, under the General Public License.
```

```
These packages come with ABSOLUTELY NO WARRANTY; for details type
```

```
Disclaimer[]]. This is free software, and you are welcome to redistribute  
it under certain conditions. See the General Public License for details.
```

Calibration

```
Get["xAct`PSALTer`"];
```

```
Package xAct`SymManipulator` version 0.9.5, {2021, 9, 14}
```

```
Copyright (C) 2011–2021, Thomas Bäckdahl, under the General Public License.
```

```
Package xAct`xPert` version 1.0.6, {2018, 2, 28}
```

Copyright (C) 2005–2020, David Brizuela, Jose M. Martin–Garcia
and Guillermo A. Mena Marugan, under the General Public License.

** Variable \$CovDFormat changed from Prefix to Postfix

** Option AllowUpperDerivatives of ContractMetric changed from False to True

** Option MetricOn of MakeRule changed from None to All

** Option ContractMetrics of MakeRule changed from False to True

Package xAct`Invar` version 2.0.5, {2013, 7, 1}

Copyright (C) 2006–2020, J. M. Martin–Garcia,
D. Yllanes and R. Portugal, under the General Public License.

** DefConstantSymbol: Defining constant symbol sigma.

** DefConstantSymbol: Defining constant symbol dim.

** Option CurvatureRelations of DefCovD changed from True to False

** Variable \$CommuteCovDsOnScalars changed from True to False

Package xAct`xCoba` version 0.8.6, {2021, 2, 28}

Copyright (C) 2005–2021, David Yllanes and
Jose M. Martin–Garcia, under the General Public License.

Package xAct`xTras` version 1.4.2, {2014, 10, 30}

Copyright (C) 2012–2014, Teake Nutma, under the General Public License.

** Variable \$CovDFormat changed from Postfix to Prefix

** Option CurvatureRelations of DefCovD changed from False to True

Package xAct`PSALter` version 2.0.1, {2025, 6, 5}

Copyright © 2022, Will Barker, Giorgos Karananas, Carlo Marzo,
Claire Rigouzzo and Haochen Tu, under the General Public License.



These packages come with ABSOLUTELY NO WARRANTY; for details type
Disclaimer`[]`. This is free software, and you are welcome to redistribute
it under certain conditions. See the General Public License for details.

Fundamental field	Symmetries
$\mathcal{K}_{\alpha\beta\chi}$	Symmetry[3, $\mathcal{K}^{\bullet 1\bullet 2\bullet 3}_{\bullet}$, { $\bullet 1 \rightarrow -a$, $\bullet 2 \rightarrow -b$, $\bullet 3 \rightarrow -c$ }, StrongGenSet[{2, 3}, GenSet[-
SO(3) irrep	Symmetries
$\mathcal{K}^{\#1}_{0^+}$	Symmetry[0, $\mathcal{K}^{\#1}_{0^+}$, {}, StrongGenSet[{}], GenSet[{}]]
$\mathcal{K}^{\#1}_{0^+ \alpha\beta\chi}$	Symmetry[3, $\mathcal{K}^{\#1\bullet 1\bullet 2\bullet 3}_{0^+}$, { $\bullet 1 \rightarrow -a$, $\bullet 2 \rightarrow -b$, $\bullet 3 \rightarrow -c$ }, StrongGenSet[{1, 2, 3}, GenS
$\mathcal{K}^{\#1}_{1^+ \alpha\beta}$	Symmetry[2, $\mathcal{K}^{\#1\bullet 1\bullet 2}_{1^+}$, { $\bullet 1 \rightarrow -a$, $\bullet 2 \rightarrow -b$ }, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]
$\mathcal{K}^{\#2}_{1^+ \alpha\beta}$	Symmetry[2, $\mathcal{K}^{\#2\bullet 1\bullet 2}_{1^+}$, { $\bullet 1 \rightarrow -a$, $\bullet 2 \rightarrow -b$ }, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]
$\mathcal{K}^{\#1}_{1^+ \alpha}$	Symmetry[1, $\mathcal{K}^{\#1\bullet 1}_{1^+}$, { $\bullet 1 \rightarrow -a$ }, StrongGenSet[{}], GenSet[{}]]
$\mathcal{K}^{\#2}_{1^+ \alpha}$	Symmetry[1, $\mathcal{K}^{\#2\bullet 1}_{1^+}$, { $\bullet 1 \rightarrow -a$ }, StrongGenSet[{}], GenSet[{}]]
$\mathcal{K}^{\#1}_{2^+ \alpha\beta}$	Symmetry[2, $\mathcal{K}^{\#1\bullet 1\bullet 2}_{2^+}$, { $\bullet 1 \rightarrow -a$, $\bullet 2 \rightarrow -b$ }, StrongGenSet[{1, 2}, GenSet[(1,2)]]]
$\mathcal{K}^{\#1}_{2^+ \alpha\beta\chi}$	Symmetry[3, $\mathcal{K}^{\#1\bullet 1\bullet 2\bullet 3}_{2^+}$, { $\bullet 1 \rightarrow -a$, $\bullet 2 \rightarrow -b$, $\bullet 3 \rightarrow -c$ }, StrongGenSet[{1, 2}, GenSet

****** xAct`PSALTER`Private`Diagnostic: the value of the variable
xAct`PSALTER`Private`CrossingRules\$102369 will now be printed.

```

{ HoldPattern[ $\partial^u \partial^\omega \mathcal{K}^{\alpha\beta\chi} \partial^\psi \partial^\pi \mathcal{K}^{\xi\phi\zeta}$ ] :=>
  Module[{ },  $k^\pi k^\omega k^u k^\psi \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^\dagger^{\alpha\beta\chi}$ ],
  HoldPattern[ $\partial^\omega \mathcal{K}^{\alpha\beta\chi} \partial^\psi \partial^\pi \mathcal{K}^{\xi\phi\zeta}$ ] :=>
  Module[{ },  $-i k^\pi k^\omega k^\psi \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^\dagger^{\alpha\beta\chi}$ ],
  HoldPattern[ $\partial^\pi \mathcal{K}^{\xi\phi\zeta} \partial^u \partial^\omega \mathcal{K}^{\alpha\beta\chi}$ ] :=>
  Module[{ },  $i k^\pi k^\omega k^u \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^\dagger^{\alpha\beta\chi}$ ],
  HoldPattern[ $\partial^\pi \mathcal{K}^{\xi\phi\zeta} \partial^\omega \mathcal{K}^{\alpha\beta\chi}$ ] :=> Module[{ },  $k^\pi k^\omega \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^\dagger^{\alpha\beta\chi}$ ],
  HoldPattern[ $\mathcal{K}^{\xi\phi\zeta} \partial^\omega \partial^\pi \mathcal{K}^{\alpha\beta\chi}$ ] :=> Module[{ },  $-k^\pi k^\omega \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^\dagger^{\alpha\beta\chi}$ ],
  HoldPattern[ $\mathcal{K}^{\alpha\beta\chi} \partial^\omega \partial^\pi \mathcal{K}^{\xi\phi\zeta}$ ] :=> Module[{ },  $-k^\pi k^\omega \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^\dagger^{\alpha\beta\chi}$ ],
  HoldPattern[ $\mathcal{K}^{\alpha\beta\chi} \partial^\pi \mathcal{K}^{\xi\phi\zeta}$ ] :=> Module[{ },  $-i k^\pi \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^\dagger^{\alpha\beta\chi}$ ],
  HoldPattern[ $\mathcal{K}^{\alpha\beta\chi} \mathcal{K}^{\xi\phi\zeta}$ ] :=> Module[{ },  $\mathcal{K}^{\xi\phi\zeta} \mathcal{K}^\dagger^{\alpha\beta\chi}$ ] }

```

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`ToMomentumExpr\$102369 will now be printed.

$$^{(2)}_{K_1} \mathcal{K}_{\beta\alpha\chi} \mathcal{K}^\dagger^{\alpha\beta\chi}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`ToMomentumExpr\$102369 will now be printed.

$$^{(2)}_{K_1} \mathcal{K}_{\beta\alpha\chi} \mathcal{K}^\dagger^{\alpha\beta\chi}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`ToMomentumExpr\$102369 will now be printed.

$$^{(2)}_{K_1} \mathcal{K}^{\alpha\beta\chi} \mathcal{K}^\dagger_{\beta\alpha\chi}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`ToMomentumExpr\$102369 will now be printed.

$$\begin{aligned}
& -K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{\theta^-}^{\dagger 1} \dagger_{\alpha \beta \chi} + \frac{K_1^{(2)} \mathcal{K}_{\theta^+}^{\dagger 1} \mathcal{K}_{\theta^+}^{\dagger 1} \dagger}{3} + \\
& -\frac{1}{2} K_1^{(2)} \mathcal{K}_{1^-}^{\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger 1} \dagger_{\alpha} - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 1} \dagger_{\alpha \beta} - \\
& -\frac{4}{3} K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \dagger_{\alpha \beta \chi} \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi + \frac{8}{9} K_1^{(2)} \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger_{\alpha \beta \chi} - \\
& -\frac{4}{3} K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger_{\alpha \beta \chi} + K_1^{(2)} \mathcal{K}_{2^+}^{\dagger 1} \alpha \beta \mathcal{K}_{2^+}^{\dagger 1} \dagger_{\alpha \beta} + \\
& K_1^{(2)} \mathcal{K}_{1^-}^{\dagger 2} \alpha \mathcal{K}_{1^-}^{\dagger 2} \dagger_{\alpha} - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \dagger_{\alpha \beta} \mathcal{K}_{1^+}^{\dagger 2} \alpha \beta - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 2} \dagger_{\alpha \beta}
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`ToMomentumExpr\$102369 will now be printed.

$$\begin{aligned}
& -K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{\theta^-}^{\dagger 1} \dagger_{\alpha \beta \chi} + \frac{K_1^{(2)} \mathcal{K}_{\theta^+}^{\dagger 1} \mathcal{K}_{\theta^+}^{\dagger 1} \dagger}{3} + \\
& -\frac{1}{2} K_1^{(2)} \mathcal{K}_{1^-}^{\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger 1} \dagger_{\alpha} - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 1} \dagger_{\alpha \beta} - \\
& -\frac{4}{3} K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \dagger_{\alpha \beta \chi} \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi + \frac{8}{9} K_1^{(2)} \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger_{\alpha \beta \chi} - \\
& -\frac{4}{3} K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger_{\alpha \beta \chi} + K_1^{(2)} \mathcal{K}_{2^+}^{\dagger 1} \alpha \beta \mathcal{K}_{2^+}^{\dagger 1} \dagger_{\alpha \beta} + \\
& K_1^{(2)} \mathcal{K}_{1^-}^{\dagger 2} \alpha \mathcal{K}_{1^-}^{\dagger 2} \dagger_{\alpha} - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \dagger_{\alpha \beta} \mathcal{K}_{1^+}^{\dagger 2} \alpha \beta - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 2} \dagger_{\alpha \beta}
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`ToMomentumExpr\$102369 will now be printed.

$$\begin{aligned}
& -K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{\theta^-}^{\dagger 1} \dagger_{\alpha \beta \chi} + \frac{K_1^{(2)} \mathcal{K}_{\theta^+}^{\dagger 1} \mathcal{K}_{\theta^+}^{\dagger 1} \dagger}{3} + \\
& -\frac{1}{2} K_1^{(2)} \mathcal{K}_{1^-}^{\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger 1} \dagger_{\alpha} - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 1} \dagger_{\alpha \beta} - \\
& -\frac{4}{3} K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \dagger_{\alpha \beta \chi} \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi + \frac{8}{9} K_1^{(2)} \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger_{\alpha \beta \chi} - \\
& -\frac{4}{3} K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger_{\alpha \beta \chi} + K_1^{(2)} \mathcal{K}_{2^+}^{\dagger 1} \alpha \beta \mathcal{K}_{2^+}^{\dagger 1} \dagger_{\alpha \beta} + \\
& K_1^{(2)} \mathcal{K}_{1^-}^{\dagger 2} \alpha \mathcal{K}_{1^-}^{\dagger 2} \dagger_{\alpha} - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \dagger_{\alpha \beta} \mathcal{K}_{1^+}^{\dagger 2} \alpha \beta - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 2} \dagger_{\alpha \beta}
\end{aligned}$$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable xAct`PSALTER`Private`\$ParityTranslate will now be printed.

$\langle | 1 \rightarrow \text{xAct`PSALTER`Private`Even}, -1 \rightarrow \text{xAct`PSALTER`Private`Odd} | \rangle$

****** xAct`PSALTER`Private`Diagnostic:

the value of the variable $\{ \}$ will now be printed.

$\{ \}$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable xAct`PSALTER`Private`OmittedTensors\$102369 will now be printed.

$\{ \}$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable xAct`PSALTER`Private`OmittedTensors\$102369 will now be printed.

$\{ \}$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable xAct`PSALTER`Private`OmittedTensors\$102369 will now be printed.

$\{ \}$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable xAct`PSALTER`Private`ToMomentumExpr\$102369 will now be printed.

$$\begin{aligned}
& -K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{\theta^-}^{\dagger 1} \dagger \alpha \beta \chi + \frac{K_1^{(2)} \mathcal{K}_{\theta^+}^{\dagger 1} \mathcal{K}_{\theta^+}^{\dagger 1} \dagger}{3} + \\
& -\frac{1}{2} K_1^{(2)} \mathcal{K}_{1^-}^{\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger 1} \dagger \alpha - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 1} \dagger \alpha \beta - \\
& -\frac{4}{3} K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \dagger \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi + \frac{8}{9} K_1^{(2)} \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger \alpha \beta \chi - \\
& -\frac{4}{3} K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger \alpha \beta \chi + K_1^{(2)} \mathcal{K}_{2^+}^{\dagger 1} \alpha \beta \mathcal{K}_{2^+}^{\dagger 1} \dagger \alpha \beta + \\
& K_1^{(2)} \mathcal{K}_{1^-}^{\dagger 2} \alpha \mathcal{K}_{1^-}^{\dagger 2} \dagger \alpha - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \dagger \alpha \beta \mathcal{K}_{1^+}^{\dagger 2} \alpha \beta - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 2} \dagger \alpha \beta
\end{aligned}$$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable
xAct`PSALTER`Private`DecomposeFieldsdLagrangian will now be printed.

$$\begin{aligned}
& -K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{\theta^-}^{\dagger 1} \dagger \alpha \beta \chi + \frac{K_1^{(2)} \mathcal{K}_{\theta^+}^{\dagger 1} \mathcal{K}_{\theta^+}^{\dagger 1} \dagger}{3} + \\
& -\frac{1}{2} K_1^{(2)} \mathcal{K}_{1^-}^{\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger 1} \dagger \alpha - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 1} \dagger \alpha \beta - \\
& -\frac{4}{3} K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \dagger \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi + \frac{8}{9} K_1^{(2)} \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger \alpha \beta \chi - \\
& -\frac{4}{3} K_1^{(2)} \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger \alpha \beta \chi + K_1^{(2)} \mathcal{K}_{2^+}^{\dagger 1} \alpha \beta \mathcal{K}_{2^+}^{\dagger 1} \dagger \alpha \beta + \\
& K_1^{(2)} \mathcal{K}_{1^-}^{\dagger 2} \alpha \mathcal{K}_{1^-}^{\dagger 2} \dagger \alpha - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \dagger \alpha \beta \mathcal{K}_{1^+}^{\dagger 2} \alpha \beta - K_1^{(2)} \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 2} \dagger \alpha \beta
\end{aligned}$$

****** xAct`PSALTER`Private`Diagnostic: the value of the
variable xAct`PSALTER`Private`CouplingAssumptions will now be printed.

$$\left\{ K_1^{(2)} \in \mathbb{R} \right\}$$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable
xAct`PSALTER`Private`Class\$103292[xAct`PSALTER`Private`InvariantToConstantRules]
will now be printed.

$$\left\{ \text{HoldPattern}[\mathcal{K}_{\theta^+}^{\dagger 1} \mathcal{K}_{\theta^+}^{\dagger 1} \dagger] \rightarrow \text{Module}[\{\},$$

[illegible]

[illegible]

$$\begin{aligned} & \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\ \text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\beta\underline{\chi}}}{}_{\underline{\alpha}}\mathcal{K}^{\ddag_1}_-{}^{\underline{\beta}}_{\underline{\alpha}}\mathcal{K}^{\ddag_1}_+{}^\dagger\right]&\rightarrow\text{Module}[\{\}, \\ & \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\ \text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\alpha}}{}^{\underline{\beta\underline{\chi}}}{}_{\underline{\chi}}\mathcal{K}^{\ddag_1}_-{}^{\underline{\alpha}}_{\underline{\chi}}\mathcal{K}^{\ddag_1}_+{}^\dagger\right]&\rightarrow\text{Module}[\{\}, \\ & \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\ \text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\alpha}}{}^{\underline{\beta}}_{\underline{\chi}}\mathcal{K}^{\ddag_1}_-{}^{\underline{\chi\underline{\alpha}}}{}_{\underline{\beta}}\mathcal{K}^{\ddag_1}_+{}^\dagger\right]&\rightarrow\text{Module}[\{\}, \\ & \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\ \text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\alpha\underline{\beta}}}{}^{\underline{\chi}}_{\underline{\chi}}\mathcal{K}^{\ddag_1}_-{}^{\underline{\alpha\underline{\beta}}}{}_{\underline{\chi}}\mathcal{K}^{\ddag_1}_+{}^\dagger\right]&\rightarrow\text{Module}[\{\}, \\ & \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\ \text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\alpha\underline{\beta\underline{\chi}}}}\mathcal{K}^{\ddag_1}_-{}^{\underline{\chi\underline{\alpha\underline{\beta}}}}\mathcal{K}^{\ddag_1}_+{}^\dagger\right]&\rightarrow\text{Module}[\{\}, \\ & \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\ \text{HoldPattern}\left[\mathcal{K}^{\ddag_1}_-{}^{\underline{\chi\underline{\beta\underline{\alpha}}}}\epsilon^{\parallel}_{\underline{\alpha\underline{\beta\underline{\chi}}}}\mathcal{K}^{\ddag_1}_+{}^\dagger\right]&\rightarrow\text{Module}[\{\}, \\ -& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\ \text{HoldPattern}\left[\mathcal{K}^{\ddag_1}_-{}^{\underline{\chi}}_{\underline{\beta\underline{\alpha}}}{}^{\underline{\alpha\underline{\beta}}}{}_{\underline{\chi}}\mathcal{K}^{\ddag_1}_+{}^\dagger\right]&\rightarrow\text{Module}[\{\}, \\ -& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\ \text{HoldPattern}\left[\mathcal{K}^{\ddag_1}_-{}^{\underline{\beta}}_{\underline{\chi}}{}^{\underline{\alpha}}_{\underline{\alpha}}{}^{\underline{\chi}}_{\underline{\beta}}\mathcal{K}^{\ddag_1}_+{}^\dagger\right]&\rightarrow\text{Module}[\{\}, \\ -& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\ \text{HoldPattern}\left[\mathcal{K}^{\ddag_1}_-{}^{\underline{\chi\underline{\beta}}}{}_{\underline{\alpha}}{}^{\underline{\alpha}}{}^{\underline{\chi}}_{\underline{\beta\underline{\chi}}}\mathcal{K}^{\ddag_1}_+{}^\dagger\right]&\rightarrow\text{Module}[\{\}, \\ -& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\ \text{HoldPattern}\left[\mathcal{K}^{\ddag_1}_-{}^{\underline{\chi\underline{\beta}}}{}_{\underline{\alpha}}{}^{\underline{\alpha}}{}^{\underline{\beta}}_{\underline{\alpha\underline{\chi}}}\mathcal{K}^{\ddag_1}_+{}^\dagger\right]&\rightarrow\text{Module}[\{\}, \\ -& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\ \text{HoldPattern}\left[\mathcal{K}^{\ddag_1}_-{}^{\underline{\beta\underline{\alpha}}}{}_{\underline{\chi}}{}^{\underline{\chi}}_{\underline{\alpha\underline{\beta}}}\mathcal{K}^{\ddag_1}_+{}^\dagger\right]&\rightarrow\text{Module}[\{\}, \end{aligned}$$

```

-- Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\chi \beta \alpha}} \epsilon^{\parallel \underline{\underline{\alpha \beta \chi}}} \mathcal{K}_{\ominus^+}^{\dagger_1} \dagger$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\beta \alpha \chi}}} \mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha \beta \chi}} \mathcal{K}_{\ominus^+}^{\dagger_1} \dagger$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\beta \alpha}}}_{\underline{\underline{\chi}}} \mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\chi}}_{\underline{\underline{\alpha \beta}}} \mathcal{K}_{\ominus^+}^{\dagger_1} \dagger$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha \chi}}}_{\underline{\underline{\beta}}} \mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta}}_{\underline{\underline{\alpha \chi}}} \mathcal{K}_{\ominus^+}^{\dagger_1} \dagger$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha}}}_{\underline{\underline{\beta \chi}}} \mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta \chi}}_{\underline{\underline{\alpha}}} \mathcal{K}_{\ominus^+}^{\dagger_1} \dagger$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\beta}}}_{\underline{\underline{\alpha \chi}}} \mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha \chi}}_{\underline{\underline{\beta}}} \mathcal{K}_{\ominus^+}^{\dagger_1} \dagger$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\beta \alpha}}}_{\underline{\underline{\chi}}} \mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha \beta}}_{\underline{\underline{\chi}}} \mathcal{K}_{\ominus^+}^{\dagger_1} \dagger$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta \alpha \chi}} \epsilon^{\parallel \underline{\underline{\beta \alpha \chi}}} \mathcal{K}_{\ominus^+}^{\dagger_1} \dagger$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta \alpha}}_{\underline{\underline{\chi}}} \epsilon^{\parallel \underline{\underline{\beta \alpha}}}_{\underline{\underline{\chi}}} \mathcal{K}_{\ominus^+}^{\dagger_1} \dagger$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta}}_{\underline{\underline{\alpha \chi}}} \epsilon^{\parallel \underline{\underline{\alpha \chi}}}_{\underline{\underline{\beta}}} \mathcal{K}_{\ominus^+}^{\dagger_1} \dagger$ ] := Module[{},

```

[illegible]

[illegible]

$\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \frac{\alpha}{\beta} \frac{\chi}{\chi}} \mathcal{K}_{\ominus^-}^{\dagger\dagger} \frac{\beta}{\alpha} \mathcal{K}_{\ominus^+}^{\dagger\dagger} \dagger\right] \rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \frac{\beta}{\alpha} \frac{\chi}{\chi}} \mathcal{K}_{\ominus^-}^{\dagger\dagger} \frac{\alpha}{\beta} \mathcal{K}_{\ominus^+}^{\dagger\dagger} \dagger\right] \rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \frac{\beta}{\alpha\chi}} \mathcal{K}_{\ominus^-}^{\dagger\dagger} \frac{\chi\alpha}{\beta} \mathcal{K}_{\ominus^+}^{\dagger\dagger} \dagger\right] \rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \frac{\chi}{\beta\alpha}} \mathcal{K}_{\ominus^-}^{\dagger\dagger} \frac{\alpha\beta}{\chi} \mathcal{K}_{\ominus^+}^{\dagger\dagger} \dagger\right] \rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \frac{\beta\alpha\chi}{\chi\beta\alpha}} \mathcal{K}_{\ominus^-}^{\dagger\dagger} \frac{\chi\alpha\beta}{\chi\beta\alpha} \mathcal{K}_{\ominus^+}^{\dagger\dagger} \dagger\right] \rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{K}_{\ominus^-}^{\dagger\dagger} \frac{\chi\beta\alpha}{\chi\beta\alpha} \epsilon^{\parallel \frac{\beta\alpha\chi}{\chi\beta\alpha}} \mathcal{K}_{\ominus^+}^{\dagger\dagger} \dagger\right] \rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{K}_{\ominus^-}^{\dagger\dagger} \frac{\chi}{\beta\alpha} \epsilon^{\parallel \frac{\beta\alpha}{\chi}} \mathcal{K}_{\ominus^+}^{\dagger\dagger} \dagger\right] \rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{K}_{\ominus^-}^{\dagger\dagger} \frac{\beta}{\chi} \epsilon^{\parallel \frac{\alpha\chi}{\beta}} \mathcal{K}_{\ominus^+}^{\dagger\dagger} \dagger\right] \rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{K}_{\ominus^-}^{\dagger\dagger} \frac{\chi\beta}{\alpha} \epsilon^{\parallel \frac{\alpha}{\beta} \frac{\chi}{\chi}} \mathcal{K}_{\ominus^+}^{\dagger\dagger} \dagger\right] \rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{K}_{\ominus^-}^{\dagger\dagger} \frac{\chi}{\beta} \epsilon^{\parallel \frac{\beta}{\alpha\chi}} \mathcal{K}_{\ominus^+}^{\dagger\dagger} \dagger\right] \rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{K}_{\ominus^-}^{\dagger\dagger} \frac{\beta\alpha}{\chi} \epsilon^{\parallel \frac{\chi}{\beta\alpha}} \mathcal{K}_{\ominus^+}^{\dagger\dagger} \dagger\right] \rightarrow \text{Module}[\{\},$

```

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\chi\beta\alpha}} \epsilon_{\underline{\underline{\beta\alpha\chi}}}^{\parallel} \mathcal{K}_{\ominus^+}^{\dagger_1}$ ] := Module[{ },
    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\alpha\chi\beta}}}^{\parallel} \mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha\beta\chi}} \mathcal{K}_{\ominus^+}^{\dagger_1}$ ] := Module[{ },
    -Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\chi}}_{\underline{\underline{\beta}}}}^{\parallel \frac{\alpha}{\chi}} \mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha\beta}}_{\underline{\underline{\chi}}}^{\frac{\chi}{\alpha}} \mathcal{K}_{\ominus^+}^{\dagger_1}$ ] := Module[{ },
    -Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\beta}}_{\underline{\underline{\alpha}}}}^{\parallel \frac{\alpha\chi}{\beta}} \mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha}}_{\underline{\underline{\chi}}}^{\frac{\beta}{\alpha}} \mathcal{K}_{\ominus^+}^{\dagger_1}$ ] := Module[{ },
    -Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\chi\beta}}_{\underline{\underline{\alpha}}}}^{\parallel \frac{\alpha}{\chi\beta}} \mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta\chi}}_{\underline{\underline{\alpha}}}^{\frac{\beta\chi}{\alpha}} \mathcal{K}_{\ominus^+}^{\dagger_1}$ ] := Module[{ },
    -Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\alpha\chi}}_{\underline{\underline{\beta}}}}^{\parallel \frac{\beta}{\alpha\chi}} \mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha}}_{\underline{\underline{\beta}}}^{\frac{\alpha}{\beta}} \underline{\underline{\chi}}^{\frac{\chi}{\alpha}}$ ] := Module[{ },
    -Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\alpha}}_{\underline{\underline{\beta}}}}^{\parallel \frac{\chi}{\alpha}} \mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha\beta}}_{\underline{\underline{\chi}}}^{\frac{\alpha\beta}{\chi}} \mathcal{K}_{\ominus^+}^{\dagger_1}$ ] := Module[{ },
    -Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\alpha\chi\beta}}}^{\parallel} \mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha\beta\chi}} \mathcal{K}_{\ominus^+}^{\dagger_1}$ ] := Module[{ },
    -Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta\alpha\chi}} \epsilon_{\underline{\underline{\alpha\chi\beta}}}^{\parallel} \mathcal{K}_{\ominus^+}^{\dagger_1}$ ] := Module[{ },
    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta\alpha}}_{\underline{\underline{\chi}}}^{\frac{\chi}{\beta\alpha}} \epsilon_{\underline{\underline{\alpha}}_{\underline{\underline{\chi}}}}^{\parallel \frac{\alpha}{\beta}} \mathcal{K}_{\ominus^+}^{\dagger_1}$ ] := Module[{ },
    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta\alpha\chi}} \epsilon_{\underline{\underline{\alpha\chi\beta}}}^{\parallel} \mathcal{K}_{\ominus^+}^{\dagger_1}$ ] := Module[{ },

```


[illegible]

```

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \underline{\underline{\alpha\chi\beta}} \epsilon \underline{\underline{\alpha\chi\beta}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha\chi\beta}} \mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \underline{\underline{\beta\chi\alpha}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha}} \underline{\underline{\beta}} \mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \underline{\underline{\chi}} \underline{\underline{\alpha}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha\chi}} \underline{\underline{\beta}} \mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \underline{\underline{\chi\alpha}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha}} \underline{\underline{\chi\beta}} \mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \underline{\underline{\beta\chi}} \underline{\underline{\alpha}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha\chi\beta}} \mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \underline{\underline{\beta\chi\alpha}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha\chi\beta}} \mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \underline{\underline{\chi\alpha\beta}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha}} \underline{\underline{\beta}} \mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \underline{\underline{\alpha\beta}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha\chi}} \underline{\underline{\beta}} \mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \underline{\underline{\chi\alpha}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1} \dagger$ ] :> Module[{}],

```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$\neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger]$,
 $\text{HoldPattern}\left[\mathcal{K}_{\ominus^-}^{\ddagger_1} \underline{\underline{\chi\beta\alpha}} \epsilon_{\underline{\underline{\beta\chi\alpha}}}^{\parallel} \mathcal{K}_{\ominus^+}^{\ddagger_1} \dagger\right] \rightarrow \text{Module}[\{\},$

$\neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger]$,
 $\text{HoldPattern}\left[\epsilon_{\underline{\underline{\chi\alpha\beta}}}^{\parallel} \mathcal{K}_{\ominus^-}^{\ddagger_1} \underline{\underline{\alpha\beta\chi}} \mathcal{K}_{\ominus^+}^{\ddagger_1} \dagger\right] \rightarrow \text{Module}[\{\},$

$\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger]$,
 $\text{HoldPattern}\left[\epsilon_{\underline{\underline{\chi}}}^{\parallel} \underline{\underline{\alpha\beta}} \mathcal{K}_{\ominus^-}^{\ddagger_1} \underline{\underline{\alpha\beta}}^{\chi} \mathcal{K}_{\ominus^+}^{\ddagger_1} \dagger\right] \rightarrow \text{Module}[\{\},$

$\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger]$,
 $\text{HoldPattern}\left[\epsilon_{\underline{\underline{\beta}}}^{\parallel} \underline{\underline{\chi\alpha}} \mathcal{K}_{\ominus^-}^{\ddagger_1} \underline{\underline{\alpha}}^{\beta} \underline{\underline{\beta}}^{\chi} \mathcal{K}_{\ominus^+}^{\ddagger_1} \dagger\right] \rightarrow \text{Module}[\{\},$

$\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger]$,
 $\text{HoldPattern}\left[\epsilon_{\underline{\underline{\chi}}}^{\parallel} \underline{\underline{\alpha}}^{\beta} \mathcal{K}_{\ominus^-}^{\ddagger_1} \underline{\underline{\beta\chi}}^{\alpha} \mathcal{K}_{\ominus^+}^{\ddagger_1} \dagger\right] \rightarrow \text{Module}[\{\},$

$\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger]$,
 $\text{HoldPattern}\left[\epsilon_{\underline{\underline{\alpha}}}^{\parallel} \underline{\underline{\chi}}^{\beta} \mathcal{K}_{\ominus^-}^{\ddagger_1} \underline{\underline{\beta\chi}}^{\alpha} \mathcal{K}_{\ominus^+}^{\ddagger_1} \dagger\right] \rightarrow \text{Module}[\{\},$

$\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger]$,
 $\text{HoldPattern}\left[\epsilon_{\underline{\underline{\chi\alpha\beta}}}^{\parallel} \mathcal{K}_{\ominus^-}^{\ddagger_1} \underline{\underline{\alpha\beta\chi}} \mathcal{K}_{\ominus^+}^{\ddagger_1} \dagger\right] \rightarrow \text{Module}[\{\},$

$\neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger]$,
 $\text{HoldPattern}\left[\mathcal{K}_{\ominus^-}^{\ddagger_1} \underline{\underline{\beta\alpha\chi}} \epsilon_{\underline{\underline{\chi\alpha\beta}}}^{\parallel} \mathcal{K}_{\ominus^+}^{\ddagger_1} \dagger\right] \rightarrow \text{Module}[\{\},$

$\neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger]$,
 $\text{HoldPattern}\left[\mathcal{K}_{\ominus^-}^{\ddagger_1} \underline{\underline{\beta\alpha}}^{\chi} \epsilon_{\underline{\underline{\chi}}}^{\parallel} \underline{\underline{\alpha\beta}} \mathcal{K}_{\ominus^+}^{\ddagger_1} \dagger\right] \rightarrow \text{Module}[\{\},$

$\neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger]$,
 $\text{HoldPattern}\left[\mathcal{K}_{\ominus^-}^{\ddagger_1} \underline{\underline{\alpha\chi}}^{\beta} \epsilon_{\underline{\underline{\beta}}}^{\parallel} \underline{\underline{\chi\alpha}} \mathcal{K}_{\ominus^+}^{\ddagger_1} \dagger\right] \rightarrow \text{Module}[\{\},$

[illegible]

[illegible]

```

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\theta^-}^{\dagger\dagger} \underline{\underline{\chi\beta\alpha}} \epsilon \parallel \underline{\underline{\chi\alpha\beta}} \mathcal{K}_{\theta^+}^{\dagger\dagger} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \parallel \underline{\underline{\chi\beta\alpha}} \mathcal{K}_{\theta^-}^{\dagger\dagger} \underline{\underline{\alpha\beta\chi}} \mathcal{K}_{\theta^+}^{\dagger\dagger} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \parallel \underline{\underline{\chi}} \underline{\underline{\beta\alpha}} \mathcal{K}_{\theta^-}^{\dagger\dagger} \underline{\underline{\alpha\beta}} \underline{\underline{\chi}} \mathcal{K}_{\theta^+}^{\dagger\dagger} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \parallel \underline{\underline{\chi}} \underline{\underline{\alpha}} \mathcal{K}_{\theta^-}^{\dagger\dagger} \underline{\underline{\beta}} \underline{\underline{\chi}} \mathcal{K}_{\theta^+}^{\dagger\dagger} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \parallel \underline{\underline{\chi\beta}} \underline{\underline{\alpha}} \mathcal{K}_{\theta^-}^{\dagger\dagger} \underline{\underline{\beta\chi}} \mathcal{K}_{\theta^+}^{\dagger\dagger} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \parallel \underline{\underline{\chi\beta}} \underline{\underline{\alpha}} \mathcal{K}_{\theta^-}^{\dagger\dagger} \underline{\underline{\alpha}} \underline{\underline{\beta\chi}} \mathcal{K}_{\theta^+}^{\dagger\dagger} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \parallel \underline{\underline{\chi}} \underline{\underline{\beta}} \underline{\underline{\alpha}} \mathcal{K}_{\theta^-}^{\dagger\dagger} \underline{\underline{\alpha}} \underline{\underline{\chi}} \mathcal{K}_{\theta^+}^{\dagger\dagger} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\theta^-}^{\dagger\dagger} \underline{\underline{\beta\alpha\chi}} \epsilon \parallel \underline{\underline{\chi\beta\alpha}} \mathcal{K}_{\theta^+}^{\dagger\dagger} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\theta^-}^{\dagger\dagger} \underline{\underline{\beta\alpha}} \underline{\underline{\chi}} \epsilon \parallel \underline{\underline{\chi}} \underline{\underline{\beta\alpha}} \mathcal{K}_{\theta^+}^{\dagger\dagger} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\theta^-}^{\dagger\dagger} \underline{\underline{\beta}} \underline{\underline{\alpha\chi}} \epsilon \parallel \underline{\underline{\chi}} \underline{\underline{\alpha}} \underline{\underline{\beta}} \mathcal{K}_{\theta^+}^{\dagger\dagger} \dagger$ ] :> Module[{}],

```

[illegible]

[illegible]

[illegible]

```

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\sharp_1 \chi \beta \alpha} \epsilon_{\underline{\chi \beta \alpha}}^{\parallel} \mathcal{K}_{\ominus^+}^{\sharp_1} \dagger$ ] := Module[{ },

    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\alpha \beta \chi}}^{\parallel} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\alpha \beta \chi} \mathcal{T}_{\ominus^+}^{\sharp_1} \dagger$ ] := Module[{ },

    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\alpha \beta}}^{\parallel} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\alpha \beta}^{\chi} \mathcal{T}_{\ominus^+}^{\sharp_1} \dagger$ ] := Module[{ },

    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\alpha}}^{\parallel} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\alpha}^{\beta} \mathcal{T}_{\ominus^+}^{\sharp_1} \dagger$ ] := Module[{ },

    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\alpha}}^{\parallel} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\alpha}^{\beta \chi} \mathcal{T}_{\ominus^+}^{\sharp_1} \dagger$ ] := Module[{ },

    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\alpha}}^{\parallel} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\alpha}^{\beta \chi} \mathcal{T}_{\ominus^+}^{\sharp_1} \dagger$ ] := Module[{ },

    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\alpha}}^{\parallel} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\alpha}^{\beta} \mathcal{T}_{\ominus^+}^{\sharp_1} \dagger$ ] := Module[{ },

    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\alpha \beta \chi}}^{\parallel} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\alpha \beta \chi} \mathcal{T}_{\ominus^+}^{\sharp_1} \dagger$ ] := Module[{ },

    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\beta \alpha \chi} \epsilon_{\underline{\alpha \beta \chi}}^{\parallel} \mathcal{T}_{\ominus^+}^{\sharp_1} \dagger$ ] := Module[{ },

    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\beta \alpha}^{\chi} \epsilon_{\underline{\alpha \beta}}^{\parallel} \mathcal{T}_{\ominus^+}^{\sharp_1} \dagger$ ] := Module[{ },

    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\beta}^{\alpha \chi} \epsilon_{\underline{\alpha \chi}}^{\parallel} \mathcal{T}_{\ominus^+}^{\sharp_1} \dagger$ ] := Module[{ },

```



```

--Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1 \underline{\beta}}_{\underline{\alpha}} \epsilon^{\parallel \underline{\alpha}}_{\underline{\beta X}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger$ ] := Module[{},

--Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1 \underline{\alpha}}_{\underline{\beta}} \epsilon^{\parallel \underline{\beta X}}_{\underline{\alpha}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger$ ] := Module[{},

--Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1 \underline{\alpha X}}_{\underline{\beta}} \epsilon^{\parallel \underline{\beta}}_{\underline{\alpha X}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger$ ] := Module[{},

--Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1 \underline{\beta \alpha}}_{\underline{X}} \epsilon^{\parallel \underline{X}}_{\underline{\alpha \beta}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger$ ] := Module[{},

--Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1 \underline{\beta \alpha X}}_{\underline{\alpha \beta X}} \epsilon^{\parallel \underline{\alpha \beta X}}_{\underline{\alpha \beta X}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger$ ] := Module[{},

--Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1 \underline{\alpha X \beta}}_{\underline{\alpha X \beta}} \epsilon^{\parallel \underline{\alpha \beta X}}_{\underline{\alpha \beta X}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger$ ] := Module[{},

--Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1 \underline{X}}_{\underline{\alpha}}_{\underline{\beta}} \epsilon^{\parallel \underline{\alpha \beta}}_{\underline{X}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger$ ] := Module[{},

--Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1 \underline{\beta}}_{\underline{\alpha X}} \epsilon^{\parallel \underline{\alpha}}_{\underline{\beta}} \frac{\underline{X}}{\underline{\beta}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger$ ] := Module[{},

--Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1 \underline{X \beta}}_{\underline{\alpha}} \epsilon^{\parallel \underline{\alpha}}_{\underline{\beta X}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger$ ] := Module[{},

--Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1 \underline{\alpha}}_{\underline{X \beta}} \epsilon^{\parallel \underline{\beta X}}_{\underline{\alpha}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger$ ] := Module[{},

--Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1 \underline{\alpha X}}_{\underline{\beta}} \epsilon^{\parallel \underline{\beta}}_{\underline{\alpha}} \frac{\underline{\beta}}{\underline{X}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger$ ] := Module[{},

--Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1 \underline{\alpha}}_{\underline{X}} \frac{\underline{\beta}}{\underline{X}} \epsilon^{\parallel \underline{X}}_{\underline{\alpha \beta}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger$ ] := Module[{}
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]


```

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\#1} \underline{\underline{\alpha\chi\beta}} \epsilon \underline{\underline{\alpha\chi\beta}} \mathcal{T}_{\ominus^+}^{\#1} \dagger$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha\chi\beta}} \mathcal{T}_{\ominus^-}^{\#1} \underline{\underline{\beta\chi\alpha}} \mathcal{T}_{\ominus^+}^{\#1} \dagger$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha}} \underline{\underline{\beta}} \mathcal{T}_{\ominus^-}^{\#1} \underline{\underline{\chi}} \underline{\underline{\alpha}} \mathcal{T}_{\ominus^+}^{\#1} \dagger$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha\chi}} \underline{\underline{\beta}} \mathcal{T}_{\ominus^-}^{\#1} \underline{\underline{\chi\alpha}} \mathcal{T}_{\ominus^+}^{\#1} \dagger$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha}} \underline{\underline{\chi\beta}} \mathcal{T}_{\ominus^-}^{\#1} \underline{\underline{\beta\chi}} \underline{\underline{\alpha}} \mathcal{T}_{\ominus^+}^{\#1} \dagger$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha\chi\beta}} \mathcal{T}_{\ominus^-}^{\#1} \underline{\underline{\beta\chi\alpha}} \mathcal{T}_{\ominus^+}^{\#1} \dagger$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha}} \underline{\underline{\chi}} \underline{\underline{\beta}} \mathcal{T}_{\ominus^-}^{\#1} \underline{\underline{\beta}} \underline{\underline{\alpha}} \mathcal{T}_{\ominus^+}^{\#1} \dagger$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha\chi\beta}} \mathcal{T}_{\ominus^-}^{\#1} \underline{\underline{\chi\alpha\beta}} \mathcal{T}_{\ominus^+}^{\#1} \dagger$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha}} \underline{\underline{\chi}} \underline{\underline{\beta}} \mathcal{T}_{\ominus^-}^{\#1} \underline{\underline{\alpha\beta}} \mathcal{T}_{\ominus^+}^{\#1} \dagger$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon \underline{\underline{\alpha\chi}} \underline{\underline{\beta}} \mathcal{T}_{\ominus^-}^{\#1} \underline{\underline{\chi\alpha}} \mathcal{T}_{\ominus^+}^{\#1} \dagger$ ] := Module[{},

```

$$\begin{aligned}
& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
\text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\chi\underline{\beta}}} \mathcal{T}^{\ddag_1}_-{}_{\underline{\alpha}}^{\underline{\chi}} =_{\underline{\alpha}}^{\underline{\beta}} \mathcal{T}^{\ddag_1}_+{}_{\dagger}\right] &\Rightarrow \text{Module}[\{\}, \\
& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
\text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\alpha}} \frac{\chi\beta}{=} \mathcal{T}^{\ddag_1}_-{}_{\underline{\chi}}^{\underline{\alpha}} =_{\underline{\beta}}^{\dagger} \mathcal{T}^{\ddag_1}_+{}_{\dagger}\right] &\Rightarrow \text{Module}[\{\}, \\
& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
\text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\alpha}\underline{\chi}} \frac{\beta}{=} \mathcal{T}^{\ddag_1}_-{}_{\underline{\alpha}}^{\underline{\chi}} =_{\underline{\beta}}^{\dagger} \mathcal{T}^{\ddag_1}_+{}_{\dagger}\right] &\Rightarrow \text{Module}[\{\}, \\
& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
\text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\alpha}} \frac{\underline{\chi}}{\underline{\beta}} \mathcal{T}^{\ddag_1}_-{}_{\underline{\alpha}}^{\underline{\chi}} =_{\underline{\chi}}^{\underline{\alpha}\beta} \mathcal{T}^{\ddag_1}_+{}_{\dagger}\right] &\Rightarrow \text{Module}[\{\}, \\
& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
\text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\alpha}\underline{\chi}\underline{\beta}} \mathcal{T}^{\ddag_1}_-{}_{\underline{\alpha}\underline{\chi}\underline{\beta}}^{\dagger} \mathcal{T}^{\ddag_1}_+{}_{\dagger}\right] &\Rightarrow \text{Module}[\{\}, \\
& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
\text{HoldPattern}\left[\mathcal{T}^{\ddag_1}_-{}_{\underline{\chi}\underline{\beta}\underline{\alpha}} \epsilon^{\parallel}_{\underline{\alpha}\underline{\chi}\underline{\beta}} \mathcal{T}^{\ddag_1}_+{}_{\dagger}\right] &\Rightarrow \text{Module}[\{\}, \\
& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
\text{HoldPattern}\left[\mathcal{T}^{\ddag_1}_-{}_{\underline{\beta}\underline{\alpha}} \frac{\underline{\chi}}{\underline{\underline{\alpha}}} \epsilon^{\parallel}_{\underline{\alpha}} \frac{\underline{\beta}}{\underline{\underline{\chi}}} \mathcal{T}^{\ddag_1}_+{}_{\dagger}\right] &\Rightarrow \text{Module}[\{\}, \\
& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
\text{HoldPattern}\left[\mathcal{T}^{\ddag_1}_-{}_{\underline{\chi}} \frac{\underline{\beta}}{\underline{\underline{\alpha}}} \epsilon^{\parallel}_{\underline{\alpha}\underline{\chi}} \frac{\underline{\alpha}\underline{\chi}}{\underline{\underline{\beta}}} \mathcal{T}^{\ddag_1}_+{}_{\dagger}\right] &\Rightarrow \text{Module}[\{\}, \\
& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
\text{HoldPattern}\left[\mathcal{T}^{\ddag_1}_-{}_{\underline{\chi}\underline{\beta}} \frac{\underline{\alpha}}{\underline{\underline{\alpha}}} \epsilon^{\parallel}_{\underline{\alpha}} \frac{\underline{\chi}\underline{\beta}}{\underline{\underline{\chi}}} \mathcal{T}^{\ddag_1}_+{}_{\dagger}\right] &\Rightarrow \text{Module}[\{\}, \\
& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
\text{HoldPattern}\left[\mathcal{T}^{\ddag_1}_-{}_{\underline{\beta}} \frac{\underline{\underline{\chi}}}{\underline{\underline{\beta}}} \epsilon^{\parallel}_{\underline{\alpha}\underline{\chi}} \frac{\underline{\beta}}{\underline{\underline{\alpha}}} \mathcal{T}^{\ddag_1}_+{}_{\dagger}\right] &\Rightarrow \text{Module}[\{\}, \\
& \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
\text{HoldPattern}\left[\mathcal{T}^{\ddag_1}_-{}_{\underline{\chi}} \frac{\underline{\beta}\underline{\alpha}}{\underline{\underline{\chi}}} \epsilon^{\parallel}_{\underline{\alpha}} \frac{\underline{\underline{\chi}}}{\underline{\underline{\beta}}} \mathcal{T}^{\ddag_1}_+{}_{\dagger}\right] &\Rightarrow \text{Module}[\{\},
\end{aligned}$$

```

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\chi \beta \alpha}} \epsilon_{\underline{\underline{\alpha \chi \beta}}}^{\parallel} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ]:> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\beta \chi \alpha}}}^{\parallel} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha \beta \chi}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ]:> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\chi}}}_{\underline{\underline{\beta}}}^{\parallel \underline{\underline{\beta}} \underline{\underline{\alpha}}} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha \beta}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ]:> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\beta}}}^{\parallel \underline{\underline{\chi \alpha}}} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha}}_{\underline{\underline{\chi}}}^{\underline{\underline{\beta}}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ]:> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\beta \chi}}}^{\parallel \underline{\underline{\alpha}}} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha}}_{\underline{\underline{\beta \chi}}}^{\underline{\underline{\beta \chi}}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ]:> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\beta \chi}}}^{\parallel \underline{\underline{\alpha}}} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha}}_{\underline{\underline{\beta \chi}}}^{\underline{\underline{\alpha}}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ]:> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\chi}}}^{\parallel \underline{\underline{\beta}}} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha \beta}}_{\underline{\underline{\chi}}}^{\underline{\underline{\alpha \beta}}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ]:> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon_{\underline{\underline{\beta \chi \alpha}}}^{\parallel \underline{\underline{\alpha}}} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha \beta \chi}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ]:> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta \alpha \chi}} \epsilon_{\underline{\underline{\beta \chi \alpha}}}^{\parallel \underline{\underline{\alpha}}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ]:> Module[{},

— Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta \alpha}}_{\underline{\underline{\beta \alpha}}}^{\underline{\underline{\chi}}} \epsilon_{\underline{\underline{\chi}}}^{\parallel \underline{\underline{\beta}} \underline{\underline{\alpha}}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ]:> Module[{},

— Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta}}_{\underline{\underline{\alpha \chi}}}^{\underline{\underline{\beta}}} \epsilon_{\underline{\underline{\beta}}}^{\parallel \underline{\underline{\chi \alpha}}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ]:> Module[{},

```

[illegible]


```

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\beta} \chi}_{\underline{\alpha}} \mathcal{T}^{\sharp_1 \chi}_{\ominus^-}{}_{\underline{\alpha}}^{\underline{\beta}} \mathcal{T}^{\sharp_1}_{\oplus^+} \dagger$ ] := Module[{ },
    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\beta} \chi}_{\underline{\alpha}} \mathcal{T}^{\sharp_1}_{\ominus^-}{}_{\underline{\chi}}^{\underline{\alpha}} \mathcal{T}^{\sharp_1}_{\oplus^+} \dagger$ ] := Module[{ },
    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\beta}}_{\underline{\chi} \underline{\alpha}} \mathcal{T}^{\sharp_1}_{\ominus^-}{}_{\underline{\beta}}^{\underline{\chi} \underline{\alpha}} \mathcal{T}^{\sharp_1}_{\oplus^+} \dagger$ ] := Module[{ },
    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\chi}}_{\underline{\beta}}{}_{\underline{\alpha}} \mathcal{T}^{\sharp_1}_{\ominus^-}{}_{\underline{\chi}}^{\underline{\alpha} \underline{\beta}} \mathcal{T}^{\sharp_1}_{\oplus^+} \dagger$ ] := Module[{ },
    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\beta} \chi \alpha}_{\underline{\alpha}} \mathcal{T}^{\sharp_1}_{\ominus^-}{}_{\underline{\beta}}^{\underline{\chi} \underline{\alpha} \underline{\beta}} \mathcal{T}^{\sharp_1}_{\oplus^+} \dagger$ ] := Module[{ },
    Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{\ominus^-}{}_{\underline{\chi} \underline{\beta} \underline{\alpha}} \epsilon^{\parallel \underline{\beta} \chi \alpha}_{\underline{\alpha}} \mathcal{T}^{\sharp_1}_{\oplus^+} \dagger$ ] := Module[{ },
    — Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{\ominus^-}{}_{\underline{\beta} \underline{\alpha}}^{\underline{\chi}} \epsilon^{\parallel \underline{\beta}}_{\underline{\chi}}{}_{\underline{\alpha}} \mathcal{T}^{\sharp_1}_{\oplus^+} \dagger$ ] := Module[{ },
    — Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{\ominus^-}{}_{\underline{\chi}}^{\underline{\beta}}{}_{\underline{\alpha}} \epsilon^{\parallel \underline{\chi} \underline{\alpha}}_{\underline{\beta}} \mathcal{T}^{\sharp_1}_{\oplus^+} \dagger$ ] := Module[{ },
    — Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{\ominus^-}{}_{\underline{\chi} \underline{\beta}}^{\underline{\alpha}} \epsilon^{\parallel \underline{\beta} \chi}_{\underline{\alpha}} \mathcal{T}^{\sharp_1}_{\oplus^+} \dagger$ ] := Module[{ },
    — Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{\ominus^-}{}_{\underline{\beta}}^{\underline{\chi} \underline{\alpha}} \epsilon^{\parallel \underline{\beta}}_{\underline{\chi} \underline{\alpha}} \mathcal{T}^{\sharp_1}_{\oplus^+} \dagger$ ] := Module[{ },
    — Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{\ominus^-}{}_{\underline{\chi}}^{\underline{\beta} \underline{\alpha}} \epsilon^{\parallel \underline{\chi}}_{\underline{\beta}}{}_{\underline{\alpha}} \mathcal{T}^{\sharp_1}_{\oplus^+} \dagger$ ] := Module[{ },

```

$\neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$

$$\text{HoldPattern}\left[\mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\underline{\chi\beta\alpha}} \epsilon_{\underline{\underline{\beta\chi\alpha}}}^{\parallel} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger\right] \rightarrow \text{Module}[\{\},$$

$\neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$

$$\text{HoldPattern}\left[\epsilon_{\underline{\underline{\chi\alpha\beta}}}^{\parallel} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\underline{\alpha\beta\chi}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger\right] \rightarrow \text{Module}[\{\},$$

$\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$

$$\text{HoldPattern}\left[\epsilon_{\underline{\underline{\chi}}}^{\parallel} \underline{\underline{\alpha\beta}} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\underline{\chi}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger\right] \rightarrow \text{Module}[\{\},$$

$\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$

$$\text{HoldPattern}\left[\epsilon_{\underline{\underline{\beta}}}^{\parallel} \underline{\underline{\chi\alpha}} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\underline{\beta}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger\right] \rightarrow \text{Module}[\{\},$$

$\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$

$$\text{HoldPattern}\left[\epsilon_{\underline{\underline{\chi}}}^{\parallel} \underline{\underline{\alpha}} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\underline{\beta\chi}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger\right] \rightarrow \text{Module}[\{\},$$

$\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$

$$\text{HoldPattern}\left[\epsilon_{\underline{\underline{\chi}}}^{\parallel} \underline{\underline{\alpha}} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\underline{\beta\chi}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger\right] \rightarrow \text{Module}[\{\},$$

$\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$

$$\text{HoldPattern}\left[\epsilon_{\underline{\underline{\chi\alpha}}}^{\parallel} \underline{\underline{\beta}} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\underline{\alpha}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger\right] \rightarrow \text{Module}[\{\},$$

$\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$

$$\text{HoldPattern}\left[\epsilon_{\underline{\underline{\alpha\beta}}}^{\parallel} \mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\underline{\alpha\beta}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger\right] \rightarrow \text{Module}[\{\},$$

$\neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$

$$\text{HoldPattern}\left[\mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\underline{\beta\alpha\chi}} \epsilon_{\underline{\underline{\chi\alpha\beta}}}^{\parallel} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger\right] \rightarrow \text{Module}[\{\},$$

$\neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$

$$\text{HoldPattern}\left[\mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\underline{\beta\alpha}} \epsilon_{\underline{\underline{\chi}}}^{\parallel} \underline{\underline{\alpha\beta}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger\right] \rightarrow \text{Module}[\{\},$$

$\neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$

$$\text{HoldPattern}\left[\mathcal{T}_{\ominus^-}^{\sharp_1} \underline{\underline{\beta}} \epsilon_{\underline{\underline{\alpha\chi}}}^{\parallel} \underline{\underline{\chi\alpha}} \mathcal{T}_{\oplus^+}^{\sharp_1} \dagger\right] \rightarrow \text{Module}[\{\},$$

[illegible]

[illegible]

[illegible]

```

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\chi \beta \alpha}} \epsilon^{\parallel \underline{\underline{\chi \alpha \beta}}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\chi \beta \alpha}}} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha \beta \chi}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\beta \alpha}}} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha \beta}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\chi}} \underline{\underline{\alpha}}} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\chi \beta}} \underline{\underline{\alpha}}} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta \chi}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\chi \beta}} \underline{\underline{\alpha}}} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha \beta \chi}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\beta}} \underline{\underline{\alpha}}} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\chi}} \underline{\underline{\beta \alpha}}} \mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha \beta}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta \alpha \chi}} \epsilon^{\parallel \underline{\underline{\chi \beta \alpha}}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\beta \alpha}} \epsilon^{\parallel \underline{\underline{\chi}} \underline{\underline{\beta \alpha}}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †],
HoldPattern[ $\mathcal{J}_{\ominus^-}^{\dagger_1} \underline{\underline{\alpha \chi}} \epsilon^{\parallel \underline{\underline{\chi}} \underline{\underline{\alpha}}} \mathcal{J}_{\ominus^+}^{\dagger_1} \dagger$ ] :> Module[{}],

```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \dagger_{\underline{\underline{\underline{\chi\beta\alpha}}}} \epsilon_{\underline{\underline{\underline{\beta\alpha\chi}}}}^{\parallel} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1}$ ] :> Module[{},
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\underline{\underline{\alpha\chi\beta}}}}^{\parallel} \mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \dagger_{\underline{\underline{\underline{\alpha\beta\chi}}}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1}$ ] :> Module[{},
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\underline{\underline{\alpha}}}}^{\parallel} \mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \dagger_{\underline{\underline{\underline{\chi}}}}^{\underline{\underline{\underline{\beta}}}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1}$ ] :> Module[{},
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\underline{\underline{\alpha\chi}}}}^{\parallel} \mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \dagger_{\underline{\underline{\underline{\alpha}}}}^{\underline{\underline{\underline{\beta}}}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1}$ ] :> Module[{},
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\underline{\underline{\alpha}}}}^{\parallel} \mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \dagger_{\underline{\underline{\underline{\chi\beta}}}}^{\underline{\underline{\underline{\alpha}}}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1}$ ] :> Module[{},
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\underline{\underline{\alpha\chi\beta}}}}^{\parallel} \mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \dagger_{\underline{\underline{\underline{\alpha\beta\chi}}}} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1}$ ] :> Module[{},
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \dagger_{\underline{\underline{\underline{\beta\alpha\chi}}}} \epsilon_{\underline{\underline{\underline{\alpha\chi\beta}}}}^{\parallel} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1}$ ] :> Module[{},
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \dagger_{\underline{\underline{\underline{\beta\alpha}}}}^{\underline{\underline{\underline{\chi}}}} \epsilon_{\underline{\underline{\underline{\alpha}}}}^{\parallel} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1}$ ] :> Module[{},
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\dagger\dagger_1} \dagger_{\underline{\underline{\underline{\alpha\chi}}}}^{\underline{\underline{\underline{\beta}}}} \epsilon_{\underline{\underline{\underline{\alpha\chi}}}}^{\parallel} \mathcal{K}_{\ominus^+}^{\dagger\dagger_1}$ ] :> Module[{}]
```


[illegible]

[illegible]

[illegible]

[illegible]

```

-- Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha\chi\beta}} \epsilon_{\underline{\underline{\beta\chi\alpha}}} \mathcal{K}_{\oplus^+}^{\sharp_1}$ ] := Module[{ },

-- Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\underline{\beta\chi\alpha}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta\chi\alpha}} \mathcal{K}_{\oplus^+}^{\sharp_1}$ ] := Module[{ },

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\underline{\beta}} \underline{\underline{\chi}} \underline{\underline{\alpha}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}} \underline{\underline{\beta}} \underline{\underline{\alpha}} \mathcal{K}_{\oplus^+}^{\sharp_1}$ ] := Module[{ },

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\underline{\beta}} \underline{\underline{\alpha}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta}} \underline{\underline{\chi\alpha}} \mathcal{K}_{\oplus^+}^{\sharp_1}$ ] := Module[{ },

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\underline{\beta\chi}} \underline{\underline{\alpha}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta\chi}} \underline{\underline{\alpha}} \mathcal{K}_{\oplus^+}^{\sharp_1}$ ] := Module[{ },

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\underline{\beta\chi}} \underline{\underline{\alpha}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta\chi}} \underline{\underline{\alpha}} \mathcal{K}_{\oplus^+}^{\sharp_1}$ ] := Module[{ },

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\underline{\beta}} \underline{\underline{\chi\alpha}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi\alpha}} \underline{\underline{\beta}} \mathcal{K}_{\oplus^+}^{\sharp_1}$ ] := Module[{ },

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\underline{\beta}} \underline{\underline{\alpha}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha\beta}} \underline{\underline{\chi}} \mathcal{K}_{\oplus^+}^{\sharp_1}$ ] := Module[{ },

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\underline{\beta}} \underline{\underline{\alpha}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi\alpha}} \underline{\underline{\beta}} \mathcal{K}_{\oplus^+}^{\sharp_1}$ ] := Module[{ },

```

[illegible]

```

-- Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi\beta\alpha}} \epsilon^{\parallel \underline{\underline{\beta\chi\alpha}}} \mathcal{K}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\chi\alpha\beta}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha\beta\chi}} \mathcal{K}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha\beta}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}} \mathcal{K}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\chi\alpha}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta}} \mathcal{K}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta\chi}} \mathcal{K}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\chi}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha\beta\chi}} \mathcal{K}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\chi\alpha\beta}}} \mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha\beta\chi}} \mathcal{K}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta\alpha\chi}} \epsilon^{\parallel \underline{\underline{\chi\alpha\beta}}} \mathcal{K}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta\alpha}} \epsilon^{\parallel \underline{\underline{\chi}}} \mathcal{K}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{K}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha\chi}} \epsilon^{\parallel \underline{\underline{\chi\alpha}}} \mathcal{K}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

```


[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```

    Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
    HoldPattern[ $\mathcal{J}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi\beta\alpha}} \epsilon \parallel \underline{\underline{\alpha\beta\chi}} \mathcal{J}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},
    Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
    HoldPattern[ $\epsilon \parallel \underline{\underline{\beta\alpha\chi}} \mathcal{J}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha\beta\chi}} \mathcal{J}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},
    Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
    HoldPattern[ $\epsilon \parallel \underline{\underline{\beta\alpha}} \mathcal{J}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}} \mathcal{J}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},
    Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
    HoldPattern[ $\epsilon \parallel \underline{\underline{\alpha\chi}} \mathcal{J}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta}} \mathcal{J}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},
    Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
    HoldPattern[ $\epsilon \parallel \underline{\underline{\alpha}} \mathcal{J}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta\chi}} \mathcal{J}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},
    Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
    HoldPattern[ $\epsilon \parallel \underline{\underline{\beta}} \mathcal{J}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha}} \mathcal{J}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},
    Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
    HoldPattern[ $\epsilon \parallel \underline{\underline{\beta}} \mathcal{J}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha\chi}} \mathcal{J}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},
    Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
    HoldPattern[ $\epsilon \parallel \underline{\underline{\beta\alpha\chi}} \mathcal{J}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha\beta\chi}} \mathcal{J}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},
    Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
    HoldPattern[ $\mathcal{J}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta\alpha\chi}} \epsilon \parallel \underline{\underline{\beta\alpha\chi}} \mathcal{J}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},
    Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
    HoldPattern[ $\mathcal{J}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta\alpha}} \epsilon \parallel \underline{\underline{\beta\alpha}} \mathcal{J}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},
    Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
    HoldPattern[ $\mathcal{J}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta}} \epsilon \parallel \underline{\underline{\alpha\chi}} \mathcal{J}_{\ominus^+}^{\sharp_1}$ ] :> Module[{}],

```

[illegible]

[illegible]

```

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \frac{\alpha}{\beta} \underline{\underline{\chi}}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}}^{\frac{\beta}{\alpha}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \frac{\beta}{\alpha} \underline{\underline{\chi}}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}}^{\frac{\alpha}{\beta}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \frac{\beta}{\alpha \chi}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}}^{\frac{\chi \alpha}{\beta}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \beta \underline{\underline{\alpha}}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}}^{\frac{\alpha \beta}{\chi}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \beta \underline{\underline{\alpha \chi}}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}}^{\frac{\chi \alpha \beta}{\chi}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}}^{\beta \alpha} \epsilon^{\parallel \frac{\beta \alpha \chi}{\chi}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}}^{\frac{\beta}{\alpha}} \epsilon^{\parallel \frac{\beta \alpha}{\chi}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}}^{\frac{\beta}{\alpha}} \epsilon^{\parallel \frac{\alpha \chi}{\beta}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}}^{\frac{\beta}{\alpha}} \epsilon^{\parallel \frac{\alpha}{\beta \chi}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}}^{\frac{\alpha}{\beta}} \epsilon^{\parallel \frac{\beta}{\alpha \chi}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] :> Module[{},

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi}}^{\frac{\beta \alpha}{\chi}} \epsilon^{\parallel \frac{\chi}{\beta \alpha}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] :> Module[{}],

```

```

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi\beta\alpha}} \epsilon^{\parallel \underline{\underline{\beta\alpha\chi}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},
  Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha\chi\beta}}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha\beta\chi}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},
  Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha}} \underline{\underline{\beta}}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha\beta}} \underline{\underline{\chi}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},
  Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha\chi}}} \underline{\underline{\beta}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha}} \underline{\underline{\beta}} \underline{\underline{\chi}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},
  Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha}} \underline{\underline{\chi\beta}}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha}} \underline{\underline{\beta\chi}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},
  Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha\chi\beta}}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha}} \underline{\underline{\beta\chi}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},
  Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha\chi\beta}}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha\beta\chi}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},
  Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta\alpha\chi}} \epsilon^{\parallel \underline{\underline{\alpha\chi\beta}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},
  Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\beta\alpha}} \epsilon^{\parallel \underline{\underline{\alpha}} \underline{\underline{\beta}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},
  Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\alpha\chi}} \epsilon^{\parallel \underline{\underline{\alpha\chi}}} \underline{\underline{\beta}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

```


[illegible]


```

-- Rank3AntisymmetricPara0mConstantSymbol ⊢ Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \frac{\alpha}{\chi \beta}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \frac{\chi}{\underline{\underline{\alpha}}} \frac{\beta}{\underline{\underline{\alpha}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol ⊢ Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \frac{\chi \beta}{\underline{\underline{\alpha}}}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \frac{\alpha}{\underline{\underline{\chi}}} \frac{\beta}{\underline{\underline{\beta}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol ⊢ Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \frac{\beta}{\underline{\underline{\alpha \chi}}}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \frac{\chi \alpha}{\underline{\underline{\beta}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol ⊢ Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \frac{\chi}{\underline{\underline{\alpha}}} \frac{\beta}{\underline{\underline{\beta}}}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \frac{\alpha \beta}{\underline{\underline{\chi}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol ⊢ Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha \chi \beta}}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \frac{\chi \alpha \beta}{\underline{\underline{\alpha \beta}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

-- Rank3AntisymmetricPara0mConstantSymbol ⊢ Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \underline{\underline{\chi \beta \alpha}} \epsilon^{\parallel \frac{\alpha \chi \beta}{\underline{\underline{\alpha \beta}}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol ⊢ Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \frac{\chi}{\underline{\underline{\beta \alpha}}} \epsilon^{\parallel \frac{\alpha}{\underline{\underline{\chi}}} \frac{\beta}{\underline{\underline{\chi}}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol ⊢ Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \frac{\beta}{\underline{\underline{\chi}}} \frac{\alpha}{\underline{\underline{\alpha}}} \epsilon^{\parallel \frac{\alpha \chi}{\underline{\underline{\beta}}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol ⊢ Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \frac{\chi \beta}{\underline{\underline{\alpha}}} \epsilon^{\parallel \frac{\alpha}{\underline{\underline{\chi}}} \frac{\beta}{\underline{\underline{\chi \beta}}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol ⊢ Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \frac{\alpha}{\underline{\underline{\chi \beta}}} \epsilon^{\parallel \frac{\chi \beta}{\underline{\underline{\alpha}}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol ⊢ Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \frac{\chi}{\underline{\underline{\beta}}} \frac{\alpha}{\underline{\underline{\beta}}} \epsilon^{\parallel \frac{\beta}{\underline{\underline{\alpha \chi}}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

Rank3AntisymmetricPara0mConstantSymbol ⊢ Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger \frac{\beta \alpha}{\underline{\underline{\chi}}} \epsilon^{\parallel \frac{\chi}{\underline{\underline{\alpha}}} \frac{\beta}{\underline{\underline{\beta}}}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{},

```


[illegible]


```

Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\beta} \underline{\chi}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger_{\underline{\alpha}}^{\underline{\chi} \underline{\beta}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{ },
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\beta} \underline{\chi}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger_{\underline{\chi} \underline{\beta}}^{\underline{\alpha}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{ },
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\beta}}_{\underline{\chi} \underline{\alpha}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger_{\underline{\beta}}^{\underline{\chi} \underline{\alpha}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{ },
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\beta}}_{\underline{\alpha}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger_{\underline{\chi}}^{\underline{\alpha} \underline{\beta}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{ },
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\epsilon^{\parallel \underline{\beta} \underline{\chi} \underline{\alpha}} \mathcal{T}_{\ominus^-}^{\sharp_1} \dagger_{\underline{\alpha} \underline{\beta}}^{\underline{\chi} \underline{\alpha} \underline{\beta}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{ },
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger_{\underline{\chi} \underline{\beta} \underline{\alpha}} \epsilon^{\parallel \underline{\beta} \underline{\chi} \underline{\alpha}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{ },
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger_{\underline{\beta} \underline{\alpha}}^{\underline{\chi}} \epsilon^{\parallel \underline{\beta}}_{\underline{\chi}}^{\underline{\alpha}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{ },
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger_{\underline{\chi}}^{\underline{\beta}} \epsilon^{\parallel \underline{\beta}}_{\underline{\alpha}}^{\underline{\chi} \underline{\alpha}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{ },
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger_{\underline{\alpha}}^{\underline{\chi} \underline{\beta}} \epsilon^{\parallel \underline{\beta} \underline{\chi}}_{\underline{\alpha}}^{\underline{\alpha}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{ },
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger_{\underline{\chi} \underline{\beta}}^{\underline{\alpha}} \epsilon^{\parallel \underline{\beta} \underline{\chi}}_{\underline{\alpha}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{ },
Rank3AntisymmetricPara0mConstantSymbol † Rank3AntisymmetricPara0pConstantSymbol],
HoldPattern[ $\mathcal{T}_{\ominus^-}^{\sharp_1} \dagger_{\underline{\chi}}^{\underline{\beta} \underline{\alpha}} \epsilon^{\parallel \underline{\beta}}_{\underline{\alpha}}^{\underline{\chi}} \mathcal{T}_{\ominus^+}^{\sharp_1}$ ] := Module[{ },

```


[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$\neg \text{Rank3AntisymmetricPara0mConstantSymbol} \vdash \text{Rank3AntisymmetricPara0pConstantSymbol}]$,
 $\text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\chi}\beta} \mathcal{T}_{\ominus^{-}}^{\sharp_1} + \frac{\chi}{\underline{\alpha}} \mathcal{T}_{\ominus^{+}}^{\sharp_1}\right] \Rightarrow \text{Module}[\{\},$
 $\neg \text{Rank3AntisymmetricPara0mConstantSymbol} \vdash \text{Rank3AntisymmetricPara0pConstantSymbol}]$,
 $\text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\alpha}} \mathcal{T}_{\ominus^{-}}^{\sharp_1} + \frac{\alpha}{\underline{\chi}\beta} \mathcal{T}_{\ominus^{+}}^{\sharp_1}\right] \Rightarrow \text{Module}[\{\},$
 $\neg \text{Rank3AntisymmetricPara0mConstantSymbol} \vdash \text{Rank3AntisymmetricPara0pConstantSymbol}]$,
 $\text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\chi}\underline{\alpha}} \mathcal{T}_{\ominus^{-}}^{\sharp_1} + \frac{\chi\alpha}{\underline{\beta}} \mathcal{T}_{\ominus^{+}}^{\sharp_1}\right] \Rightarrow \text{Module}[\{\},$
 $\neg \text{Rank3AntisymmetricPara0mConstantSymbol} \vdash \text{Rank3AntisymmetricPara0pConstantSymbol}]$,
 $\text{HoldPattern}\left[\epsilon^{\parallel}_{\underline{\beta}\alpha} \mathcal{T}_{\ominus^{-}}^{\sharp_1} + \frac{\alpha\beta}{\underline{\chi}} \mathcal{T}_{\ominus^{+}}^{\sharp_1}\right] \Rightarrow \text{Module}[\{\},$
 $\neg \text{Rank3AntisymmetricPara0mConstantSymbol} \vdash \text{Rank3AntisymmetricPara0pConstantSymbol}]$,
 $\text{HoldPattern}\left[\mathcal{T}_{\ominus^{-}}^{\sharp_1} + \frac{\chi}{\underline{\beta}\alpha} \epsilon^{\parallel}_{\underline{\chi}\beta\alpha} \mathcal{T}_{\ominus^{+}}^{\sharp_1}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol} \vdash \text{Rank3AntisymmetricPara0pConstantSymbol}]$,
 $\text{HoldPattern}\left[\mathcal{T}_{\ominus^{-}}^{\sharp_1} + \frac{\chi}{\underline{\beta}\alpha} \epsilon^{\parallel}_{\underline{\chi}} \frac{\beta\alpha}{\underline{\quad}} \mathcal{T}_{\ominus^{+}}^{\sharp_1}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol} \vdash \text{Rank3AntisymmetricPara0pConstantSymbol}]$,
 $\text{HoldPattern}\left[\mathcal{T}_{\ominus^{-}}^{\sharp_1} + \frac{\beta}{\underline{\chi}\underline{\alpha}} \epsilon^{\parallel}_{\underline{\quad}} \frac{\alpha}{\underline{\beta}} \mathcal{T}_{\ominus^{+}}^{\sharp_1}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol} \vdash \text{Rank3AntisymmetricPara0pConstantSymbol}]$,
 $\text{HoldPattern}\left[\mathcal{T}_{\ominus^{-}}^{\sharp_1} + \frac{\chi\beta}{\underline{\alpha}} \epsilon^{\parallel}_{\underline{\chi}\beta} \frac{\alpha}{\underline{\quad}} \mathcal{T}_{\ominus^{+}}^{\sharp_1}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol} \vdash \text{Rank3AntisymmetricPara0pConstantSymbol}]$,
 $\text{HoldPattern}\left[\mathcal{T}_{\ominus^{-}}^{\sharp_1} + \frac{\alpha}{\underline{\chi}\beta} \epsilon^{\parallel}_{\underline{\quad}} \frac{\chi\beta}{\underline{\alpha}} \mathcal{T}_{\ominus^{+}}^{\sharp_1}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol} \vdash \text{Rank3AntisymmetricPara0pConstantSymbol}]$,
 $\text{HoldPattern}\left[\mathcal{T}_{\ominus^{-}}^{\sharp_1} + \frac{\chi}{\underline{\beta}} \frac{\alpha}{\underline{\quad}} \epsilon^{\parallel}_{\underline{\chi}} \frac{\beta}{\underline{\alpha}} \mathcal{T}_{\ominus^{+}}^{\sharp_1}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPara0mConstantSymbol} \vdash \text{Rank3AntisymmetricPara0pConstantSymbol}]$,
 $\text{HoldPattern}\left[\mathcal{T}_{\ominus^{-}}^{\sharp_1} + \frac{\beta\alpha}{\underline{\chi}} \epsilon^{\parallel}_{\underline{\chi}} \frac{\alpha}{\underline{\beta}\alpha} \mathcal{T}_{\ominus^{+}}^{\sharp_1}\right] \Rightarrow \text{Module}[\{\},$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]


```

    - Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1+}^{\#1} \dagger_{\underline{\underline{\beta\alpha}}} \mathcal{K}_{1+}^{\#2} \underline{\underline{\alpha\beta}}$ ] :> Module[{},

    - Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1+}^{\#1} \dagger_{\underline{\underline{\beta}}} \mathcal{K}_{1+}^{\#2} \underline{\underline{\alpha}} \underline{\underline{\beta}}$ ] :> Module[{},

    - Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1+}^{\#1} \dagger_{\underline{\underline{\alpha}}} \mathcal{K}_{1+}^{\#2} \underline{\underline{\alpha}} \underline{\underline{\beta}}$ ] :> Module[{},

    - Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1+}^{\#1} \dagger_{\underline{\underline{\beta\alpha}}} \mathcal{K}_{1+}^{\#2} \underline{\underline{\alpha\beta}}$ ] :> Module[{},

    - Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1+}^{\#2} \underline{\underline{\beta\alpha}} \mathcal{K}_{1+}^{\#1} \dagger_{\underline{\underline{\beta\alpha}}}$ ] :> Module[{},

    Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1+}^{\#2} \underline{\underline{\beta}} \underline{\underline{\alpha}} \mathcal{K}_{1+}^{\#1} \dagger_{\underline{\underline{\beta}}} \underline{\underline{\alpha}}$ ] :> Module[{},

    Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1+}^{\#2} \underline{\underline{\alpha}} \underline{\underline{\beta}} \mathcal{K}_{1+}^{\#1} \dagger_{\underline{\underline{\alpha}}} \underline{\underline{\beta}}$ ] :> Module[{},

    Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1+}^{\#2} \underline{\underline{\beta\alpha}} \mathcal{K}_{1+}^{\#1} \dagger_{\underline{\underline{\beta\alpha}}}$ ] :> Module[{},

    Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1+}^{\#1} \underline{\underline{\alpha\beta}} \mathcal{K}_{1+}^{\#2} \dagger_{\underline{\underline{\alpha\beta}}}$ ] :> Module[{},

    Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{1+}^{\#1} \underline{\underline{\alpha}} \underline{\underline{\beta}} \mathcal{K}_{1+}^{\#2} \dagger_{\underline{\underline{\alpha}}} \underline{\underline{\beta}}$ ] :> Module[{},

    Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{1+}^{\#1} \underline{\underline{\beta}} \underline{\underline{\alpha}} \mathcal{K}_{1+}^{\#2} \dagger_{\underline{\underline{\beta}}} \underline{\underline{\alpha}}$ ] :> Module[{},

    Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{K}_{1+}^{\#1} \underline{\underline{\alpha\beta}} \mathcal{K}_{1+}^{\#2} \dagger_{\underline{\underline{\alpha\beta}}}$ ] :> Module[{},

    Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],

```



```

Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{1^+}^{\#1} \underline{\underline{\alpha\beta}} \mathcal{T}_{1^+}^{\#2} \dagger \underline{\underline{\alpha\beta}}$ ] :> Module[{},
Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \dagger \underline{\underline{\beta\alpha}} \mathcal{T}_{1^+}^{\#1} \underline{\underline{\alpha\beta}}$ ] :> Module[{},
Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \dagger \underline{\underline{\beta}} \underline{\underline{\alpha}} \mathcal{T}_{1^+}^{\#1} \underline{\underline{\alpha}} \underline{\underline{\beta}}$ ] :> Module[{},
Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \dagger \underline{\underline{\alpha}} \underline{\underline{\beta}} \mathcal{T}_{1^+}^{\#1} \underline{\underline{\beta}} \underline{\underline{\alpha}}$ ] :> Module[{},
Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \dagger \underline{\underline{\beta\alpha}} \mathcal{T}_{1^+}^{\#1} \underline{\underline{\alpha\beta}}$ ] :> Module[{},
Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{1^+}^{\#1} \underline{\underline{\beta\alpha}} \mathcal{T}_{1^+}^{\#2} \dagger \underline{\underline{\alpha\beta}}$ ] :> Module[{},
Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{1^+}^{\#1} \underline{\underline{\beta}} \underline{\underline{\alpha}} \mathcal{T}_{1^+}^{\#2} \dagger \underline{\underline{\alpha}} \underline{\underline{\beta}}$ ] :> Module[{},
Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{1^+}^{\#1} \underline{\underline{\beta}} \underline{\underline{\alpha}} \mathcal{T}_{1^+}^{\#2} \dagger \underline{\underline{\alpha\beta}}$ ] :> Module[{},
Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \dagger \underline{\underline{\beta\alpha}} \mathcal{T}_{1^+}^{\#1} \underline{\underline{\beta\alpha}}$ ] :> Module[{},
Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \dagger \underline{\underline{\beta}} \underline{\underline{\alpha}} \mathcal{T}_{1^+}^{\#1} \underline{\underline{\beta}} \underline{\underline{\alpha}}$ ] :> Module[{},
Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \dagger \underline{\underline{\alpha}} \underline{\underline{\beta}} \mathcal{T}_{1^+}^{\#1} \underline{\underline{\beta}} \underline{\underline{\alpha}}$ ] :> Module[{}

```



```

-- Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{J}_{1^{+}}^{\dagger_2} \dagger_{\underline{\underline{\beta \alpha}}} \mathcal{J}_{1^{+}}^{\dagger_2 \underline{\underline{\beta \alpha}}}$ ] := Module[{ },
    Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{J}_{1^{+}}^{\dagger_2} \dagger_{\underline{\underline{\alpha}}}^{\beta} \mathcal{J}_{1^{+}}^{\dagger_2 \underline{\underline{\alpha}}}^{\underline{\underline{\alpha}}}$ ] := Module[{ },
    Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{J}_{1^{+}}^{\dagger_2} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\alpha}}} \mathcal{J}_{1^{+}}^{\dagger_2 \underline{\underline{\beta}}}^{\underline{\underline{\alpha}}}$ ] := Module[{ },
    Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\mathcal{J}_{1^{+}}^{\dagger_2} \dagger_{\underline{\underline{\alpha}}}^{\underline{\underline{\beta \alpha}}} \mathcal{J}_{1^{+}}^{\dagger_2 \underline{\underline{\alpha}}}^{\underline{\underline{\beta \alpha}}}$ ] := Module[{ },
    Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha \beta \chi}}} \mathcal{K}_{1^{-}}^{\dagger_1 \underline{\underline{\alpha}}} \mathcal{K}_{1^{+}}^{\dagger_1} \dagger_{\underline{\underline{\alpha}}}^{\underline{\underline{\beta \chi}}}$ ] := Module[{ },
    Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha \beta}}}^{\chi} \mathcal{K}_{1^{-}}^{\dagger_1 \underline{\underline{\alpha}}} \mathcal{K}_{1^{+}}^{\dagger_1} \dagger_{\underline{\underline{\chi}}}^{\underline{\underline{\beta}}}$ ] := Module[{ },
    Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha}}}^{\underline{\underline{\beta}}} \mathcal{K}_{1^{-}}^{\dagger_1 \underline{\underline{\alpha}}} \mathcal{K}_{1^{+}}^{\dagger_1} \dagger_{\underline{\underline{\chi}}}^{\underline{\underline{\beta \chi}}}$ ] := Module[{ },
    Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha \beta \chi}}} \mathcal{K}_{1^{-}}^{\dagger_1 \underline{\underline{\alpha}}} \mathcal{K}_{1^{+}}^{\dagger_1} \dagger_{\underline{\underline{\alpha}}}^{\underline{\underline{\beta \chi}}}$ ] := Module[{ },
    Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha \beta}}}^{\underline{\underline{\chi}}} \mathcal{K}_{1^{-}}^{\dagger_1 \underline{\underline{\alpha}}} \mathcal{K}_{1^{+}}^{\dagger_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\chi}}}$ ] := Module[{ },
    Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1pConstantSymbol †],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha \beta}}}^{\underline{\underline{\alpha \chi}}} \mathcal{K}_{1^{-}}^{\dagger_1 \underline{\underline{\alpha}}} \mathcal{K}_{1^{+}}^{\dagger_1} \dagger_{\underline{\underline{\beta \chi}}}^{\underline{\underline{\alpha}}}$ ] := Module[{ },
    Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1pConstantSymbol †],

```


[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]


```

HoldPattern[ $\mathcal{T}_{1^+}^{\#1} \underline{\underline{\chi\beta}} \epsilon \underline{\underline{\chi\beta}}^{\underline{\underline{\alpha}}} \mathcal{T}_{1^-}^{\#1} \dagger_{\underline{\underline{\alpha}}}$ ] := Module[{},
  Rank3AntisymmetricParaImConstantSymbol  $\dagger$  Rank3AntisymmetricParaIpConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#1} \underline{\underline{\beta}} \epsilon \underline{\underline{\chi}}^{\underline{\underline{\alpha}}} \mathcal{T}_{1^-}^{\#1} \dagger_{\underline{\underline{\alpha}}}$ ] := Module[{},
  Rank3AntisymmetricParaImConstantSymbol  $\dagger$  Rank3AntisymmetricParaIpConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#1} \underline{\underline{\chi}} \epsilon \underline{\underline{\beta\alpha}} \mathcal{T}_{1^-}^{\#1} \dagger_{\underline{\underline{\alpha}}}$ ] := Module[{},
  Rank3AntisymmetricParaImConstantSymbol  $\dagger$  Rank3AntisymmetricParaIpConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#1} \underline{\underline{\chi\beta}} \epsilon \underline{\underline{\chi\beta\alpha}} \mathcal{T}_{1^-}^{\#1} \dagger_{\underline{\underline{\alpha}}}$ ] := Module[{},
  Rank3AntisymmetricParaImConstantSymbol  $\dagger$  Rank3AntisymmetricParaIpConstantSymbol],
HoldPattern[ $\epsilon \underline{\underline{\alpha\beta\chi}} \mathcal{T}_{1^-}^{\#1} \dagger_{\underline{\underline{\alpha}}} \mathcal{T}_{1^+}^{\#2} \underline{\underline{\beta\chi}}$ ] := Module[{},
  Rank3AntisymmetricParaImConstantSymbol  $\dagger$  Rank3AntisymmetricPerpIpConstantSymbol],
HoldPattern[ $\epsilon \underline{\underline{\alpha\beta}}^{\underline{\underline{\chi}}} \mathcal{T}_{1^-}^{\#1} \dagger_{\underline{\underline{\alpha}}} \mathcal{T}_{1^+}^{\#2} \underline{\underline{\beta}}_{\underline{\underline{\chi}}}$ ] := Module[{},
  Rank3AntisymmetricParaImConstantSymbol  $\dagger$  Rank3AntisymmetricPerpIpConstantSymbol],
HoldPattern[ $\epsilon \underline{\underline{\alpha}}^{\underline{\underline{\beta}}} \mathcal{T}_{1^-}^{\#1} \dagger_{\underline{\underline{\alpha}}} \mathcal{T}_{1^+}^{\#2} \underline{\underline{\chi}}_{\underline{\underline{\beta}}}$ ] := Module[{},
  Rank3AntisymmetricParaImConstantSymbol  $\dagger$  Rank3AntisymmetricPerpIpConstantSymbol],
HoldPattern[ $\epsilon \underline{\underline{\alpha}}^{\underline{\underline{\beta\chi}}} \mathcal{T}_{1^-}^{\#1} \dagger_{\underline{\underline{\alpha}}} \mathcal{T}_{1^+}^{\#2} \underline{\underline{\beta\chi}}$ ] := Module[{},
  Rank3AntisymmetricParaImConstantSymbol  $\dagger$  Rank3AntisymmetricPerpIpConstantSymbol],
HoldPattern[ $\epsilon \underline{\underline{\alpha}}^{\underline{\underline{\beta}}} \mathcal{T}_{1^-}^{\#1} \dagger_{\underline{\underline{\alpha}}} \mathcal{T}_{1^+}^{\#2} \underline{\underline{\beta}}_{\underline{\underline{\chi}}}$ ] := Module[{},
  Rank3AntisymmetricParaImConstantSymbol  $\dagger$  Rank3AntisymmetricPerpIpConstantSymbol],
HoldPattern[ $\epsilon \underline{\underline{\alpha\beta}}^{\underline{\underline{\chi}}} \mathcal{T}_{1^-}^{\#1} \dagger_{\underline{\underline{\alpha}}} \mathcal{T}_{1^+}^{\#2} \underline{\underline{\chi}}_{\underline{\underline{\beta}}}$ ] := Module[{},
  Rank3AntisymmetricParaImConstantSymbol  $\dagger$  Rank3AntisymmetricPerpIpConstantSymbol],
HoldPattern[ $\epsilon \underline{\underline{\alpha\beta\chi}} \mathcal{T}_{1^-}^{\#1} \dagger_{\underline{\underline{\alpha}}} \mathcal{T}_{1^+}^{\#2} \underline{\underline{\beta\chi}}$ ] := Module[{},
  Rank3AntisymmetricParaImConstantSymbol  $\dagger$  Rank3AntisymmetricPerpIpConstantSymbol],

```


[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```

HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_2 \chi \beta} \epsilon_{\underline{\chi \beta}}^{\underline{\alpha}} \mathcal{J}_{1^-}^{\sharp_2} \dagger_{\underline{\alpha}}$ ] := Module[{},
  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_2} \underline{\chi}^{\underline{\beta}} \epsilon_{\underline{\beta}}^{\underline{\chi \alpha}} \mathcal{J}_{1^-}^{\sharp_2} \dagger_{\underline{\alpha}}$ ] := Module[{},
  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_2} \underline{\chi}^{\underline{\beta}} \epsilon_{\underline{\beta}}^{\underline{\chi \alpha}} \mathcal{J}_{1^-}^{\sharp_2} \dagger_{\underline{\alpha}}$ ] := Module[{},
  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_2} \underline{\chi}^{\underline{\beta}} \epsilon_{\underline{\beta}}^{\underline{\chi \alpha}} \mathcal{J}_{1^-}^{\sharp_2} \dagger_{\underline{\alpha}}$ ] := Module[{},
  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_2} \underline{\chi}^{\underline{\beta}} \epsilon_{\underline{\beta}}^{\underline{\chi \alpha}} \mathcal{J}_{1^-}^{\sharp_2} \dagger_{\underline{\alpha}}$ ] := Module[{},
  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1^-}^{\sharp_1 \underline{\alpha}} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}}$ ] := Module[{},
  Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{K}_{1^-}^{\sharp_1 \underline{\alpha}} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}}$ ] := Module[{},
  Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}} \mathcal{K}_{1^-}^{\sharp_2 \underline{\alpha}}$ ] := Module[{},
  Rank3AntisymmetricPara1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1mConstantSymbol],
HoldPattern[ $\mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}} \mathcal{K}_{1^-}^{\sharp_2 \underline{\alpha}}$ ] := Module[{},
  Rank3AntisymmetricPara1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1mConstantSymbol],
HoldPattern[ $\mathcal{K}_{1^-}^{\sharp_1 \underline{\alpha}} \mathcal{K}_{1^-}^{\sharp_2} \dagger_{\underline{\alpha}}$ ] := Module[{},
  Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{K}_{1^-}^{\sharp_1 \underline{\alpha}} \mathcal{K}_{1^-}^{\sharp_2} \dagger_{\underline{\alpha}}$ ] := Module[{},
  Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{K}_{1^-}^{\sharp_2 \underline{\alpha}} \mathcal{K}_{1^-}^{\sharp_2} \dagger_{\underline{\alpha}}$ ] := Module[{},
  Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^-}^{\sharp_1 \underline{\alpha}} \mathcal{J}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}}$ ] := Module[{},
  Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^-}^{\sharp_1 \underline{\alpha}} \mathcal{J}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}}$ ] := Module[{},

```


[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\begin{aligned} & \neg\text{Rank3AntisymmetricPara2mConstantSymbol} \vdash \text{Rank3AntisymmetricPara2pConstantSymbol}, \\ \text{HoldPattern}\left[\mathcal{K}_{2^{-}}^{\sharp_1} + \frac{\delta}{X} \underline{\beta} \epsilon^{\parallel X}_{\underline{\delta\alpha}} \mathcal{K}_{2^{+}}^{\sharp_1} \underline{\alpha}\right] &\Rightarrow \text{Module}[\{\}, \\ & \neg\text{Rank3AntisymmetricPara2mConstantSymbol} \vdash \text{Rank3AntisymmetricPara2pConstantSymbol}, \\ \text{HoldPattern}\left[\mathcal{K}_{2^{-}}^{\sharp_1} + \frac{\beta}{\underline{\delta X}} \epsilon^{\parallel X\delta}_{\underline{\alpha}} \mathcal{K}_{2^{+}}^{\sharp_1} \underline{\alpha}\right] &\Rightarrow \text{Module}[\{\}, \\ & \neg\text{Rank3AntisymmetricPara2mConstantSymbol} \vdash \text{Rank3AntisymmetricPara2pConstantSymbol}, \\ \text{HoldPattern}\left[\mathcal{K}_{2^{-}}^{\sharp_1} + \frac{\delta X}{\underline{\beta}} \epsilon^{\parallel X\delta\alpha}_{\underline{\alpha}} \mathcal{K}_{2^{+}}^{\sharp_1} \underline{\beta\alpha}\right] &\Rightarrow \text{Module}[\{\}, \\ & \neg\text{Rank3AntisymmetricPara2mConstantSymbol} \vdash \text{Rank3AntisymmetricPara2pConstantSymbol}, \\ \text{HoldPattern}\left[\mathcal{K}_{2^{-}}^{\sharp_1} + \frac{X}{\underline{\delta}} \underline{\beta} \epsilon^{\parallel X}_{\underline{\alpha}} \mathcal{K}_{2^{+}}^{\sharp_1} \underline{\beta\alpha}\right] &\Rightarrow \text{Module}[\{\}, \\ & \neg\text{Rank3AntisymmetricPara2mConstantSymbol} \vdash \text{Rank3AntisymmetricPara2pConstantSymbol}, \\ \text{HoldPattern}\left[\mathcal{K}_{2^{-}}^{\sharp_1} + \frac{\delta}{X\underline{\beta}} \epsilon^{\parallel X}_{\underline{\delta\alpha}} \mathcal{K}_{2^{+}}^{\sharp_1} \underline{\beta\alpha}\right] &\Rightarrow \text{Module}[\{\}, \\ & \neg\text{Rank3AntisymmetricPara2mConstantSymbol} \vdash \text{Rank3AntisymmetricPara2pConstantSymbol}, \\ \text{HoldPattern}\left[\mathcal{K}_{2^{-}}^{\sharp_1} + \frac{\delta X\beta}{\underline{\alpha}} \epsilon^{\parallel X\delta}_{\underline{\alpha}} \mathcal{K}_{2^{+}}^{\sharp_1} \underline{\beta\alpha}\right] &\Rightarrow \text{Module}[\{\}, \\ & \neg\text{Rank3AntisymmetricPara2mConstantSymbol} \vdash \text{Rank3AntisymmetricPara2pConstantSymbol}, \\ \text{HoldPattern}\left[\epsilon^{\parallel \underline{\alpha}}_{\underline{\underline{X}}} \mathcal{K}_{2^{-}}^{\sharp_1} + \frac{X\delta\beta}{\underline{\alpha}} \mathcal{K}_{2^{+}}^{\sharp_1} \underline{\alpha\beta}\right] &\Rightarrow \text{Module}[\{\}, \\ \text{Rank3AntisymmetricPara2mConstantSymbol} \vdash \text{Rank3AntisymmetricPara2pConstantSymbol}, \\ \text{HoldPattern}\left[\epsilon^{\parallel \underline{\delta\alpha}}_{\underline{\underline{X}}} \mathcal{K}_{2^{-}}^{\sharp_1} + \frac{X}{\underline{\delta}} \underline{\beta} \mathcal{K}_{2^{+}}^{\sharp_1} \underline{\alpha\beta}\right] &\Rightarrow \text{Module}[\{\}, \\ \text{Rank3AntisymmetricPara2mConstantSymbol} \vdash \text{Rank3AntisymmetricPara2pConstantSymbol}, \\ \text{HoldPattern}\left[\epsilon^{\parallel \underline{\alpha X}}_{\underline{\underline{\delta}}} \mathcal{K}_{2^{-}}^{\sharp_1} + \frac{\delta\beta}{X} \mathcal{K}_{2^{+}}^{\sharp_1} \underline{\alpha\beta}\right] &\Rightarrow \text{Module}[\{\}, \\ \text{Rank3AntisymmetricPara2mConstantSymbol} \vdash \text{Rank3AntisymmetricPara2pConstantSymbol}, \\ \text{HoldPattern}\left[\epsilon^{\parallel \underline{\delta\alpha X}}_{\underline{\underline{\delta}}} \mathcal{K}_{2^{-}}^{\sharp_1} + \frac{\beta}{X\underline{\delta}} \mathcal{K}_{2^{+}}^{\sharp_1} \underline{\alpha\beta}\right] &\Rightarrow \text{Module}[\{\}, \\ \text{Rank3AntisymmetricPara2mConstantSymbol} \vdash \text{Rank3AntisymmetricPara2pConstantSymbol}, \\ \text{HoldPattern}\left[\epsilon^{\parallel \underline{\alpha}}_{\underline{\underline{\delta}} \underline{X}} \mathcal{K}_{2^{-}}^{\sharp_1} + \frac{X\delta}{\underline{\beta}} \mathcal{K}_{2^{+}}^{\sharp_1} \underline{\beta}\right] &\Rightarrow \text{Module}[\{\}, \\ \text{Rank3AntisymmetricPara2mConstantSymbol} \vdash \text{Rank3AntisymmetricPara2pConstantSymbol}, \end{aligned}$$


```

HoldPattern[ $\mathcal{K}_{2^{-}}^{\sharp_1} \dagger_{\underline{\chi}\beta}^\delta \epsilon_{\underline{\alpha}}^{\parallel \underline{\delta\alpha}} \mathcal{K}_{2^{+}}^{\sharp_1 \underline{\beta\alpha}}$ ] := Module[{ },
  -Rank3AntisymmetricPara2mConstantSymbol \(\dagger Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\mathcal{K}_{2^{-}}^{\sharp_1} \dagger_{\underline{\delta\chi\beta}}^\delta \epsilon_{\underline{\alpha}}^{\parallel \underline{\chi}} \mathcal{K}_{2^{+}}^{\sharp_1 \underline{\beta\alpha}}$ ] := Module[{ },
  -Rank3AntisymmetricPara2mConstantSymbol \(\dagger Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\delta\chi}}^{\parallel \underline{\alpha}} \mathcal{K}_{2^{-}}^{\sharp_1} \dagger_{\underline{\alpha}\beta}^{\underline{\chi\delta\beta}} \mathcal{K}_{2^{+}}^{\sharp_1 \underline{\alpha\beta}}$ ] := Module[{ },
  -Rank3AntisymmetricPara2mConstantSymbol \(\dagger Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\chi}}^{\parallel \underline{\delta\alpha}} \mathcal{K}_{2^{-}}^{\sharp_1} \dagger_{\underline{\underline{\delta}}}^{\underline{\chi}} \mathcal{K}_{2^{+}}^{\sharp_1 \underline{\alpha\beta}}$ ] := Module[{ },
  -Rank3AntisymmetricPara2mConstantSymbol \(\dagger Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\delta}}^{\parallel \underline{\chi\alpha}} \mathcal{K}_{2^{-}}^{\sharp_1} \dagger_{\underline{\chi}}^{\underline{\delta\beta}} \mathcal{K}_{2^{+}}^{\sharp_1 \underline{\alpha\beta}}$ ] := Module[{ },
  -Rank3AntisymmetricPara2mConstantSymbol \(\dagger Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\delta\chi\alpha}}^{\parallel \underline{\underline{\delta\chi\alpha}}} \mathcal{K}_{2^{-}}^{\sharp_1} \dagger_{\underline{\chi\underline{\delta}}}^{\underline{\beta}} \mathcal{K}_{2^{+}}^{\sharp_1 \underline{\alpha\beta}}$ ] := Module[{ },
  -Rank3AntisymmetricPara2mConstantSymbol \(\dagger Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\delta\chi}}^{\parallel \underline{\alpha}} \mathcal{K}_{2^{-}}^{\sharp_1} \dagger_{\underline{\underline{\delta\beta}}}^{\underline{\chi\delta}} \mathcal{K}_{2^{+}}^{\sharp_1 \underline{\beta}}_{\underline{\alpha}}$ ] := Module[{ },
  -Rank3AntisymmetricPara2mConstantSymbol \(\dagger Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\chi}}^{\parallel \underline{\delta\alpha}} \mathcal{K}_{2^{-}}^{\sharp_1} \dagger_{\underline{\underline{\delta\beta}}}^{\underline{\chi}} \mathcal{K}_{2^{+}}^{\sharp_1 \underline{\beta}}_{\underline{\alpha}}$ ] := Module[{ },
  -Rank3AntisymmetricPara2mConstantSymbol \(\dagger Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\delta}}^{\parallel \underline{\chi\alpha}} \mathcal{K}_{2^{-}}^{\sharp_1} \dagger_{\underline{\underline{\chi}}}^{\underline{\delta}} \mathcal{K}_{2^{+}}^{\sharp_1 \underline{\beta}}_{\underline{\alpha}}$ ] := Module[{ },
  -Rank3AntisymmetricPara2mConstantSymbol \(\dagger Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\delta\chi\alpha}}^{\parallel \underline{\underline{\delta\chi\alpha}}} \mathcal{K}_{2^{-}}^{\sharp_1} \dagger_{\underline{\underline{\chi\delta\beta}}}^{\underline{\beta}}$ ] := Module[{ },
  -Rank3AntisymmetricPara2mConstantSymbol \(\dagger Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\delta\chi\alpha}}^{\parallel \underline{\underline{\delta\chi\alpha}}} \mathcal{K}_{2^{-}}^{\sharp_1} \dagger_{\underline{\underline{\alpha\beta}}}^{\underline{\chi\delta\beta}} \mathcal{K}_{2^{+}}^{\sharp_1 \underline{\alpha}}_{\underline{\underline{\beta}}}$ ] := Module[{ },
  -Rank3AntisymmetricPara2mConstantSymbol \(\dagger Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\epsilon_{\underline{\chi\alpha}}^{\parallel \underline{\delta}} \mathcal{K}_{2^{-}}^{\sharp_1} \dagger_{\underline{\underline{\delta}}}^{\underline{\chi}} \mathcal{K}_{2^{+}}^{\sharp_1 \underline{\alpha}}_{\underline{\underline{\beta}}}$ ] := Module[{ },

```


[illegible]

[illegible]

[illegible]

[illegible]


```

Rank3AntisymmetricPara2mConstantSymbol † Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\mathcal{T}_{2^-}^{\dagger\dagger_1} + \frac{\chi\beta}{\underline{\delta}} \epsilon^{\underline{\alpha}}_{\underline{\chi}} \frac{\delta}{\underline{\delta}} \mathcal{T}_{2^+}^{\dagger\dagger_1} \beta_{\underline{\alpha}}$ ] := Module[{}],

Rank3AntisymmetricPara2mConstantSymbol † Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\mathcal{T}_{2^-}^{\dagger\dagger_1} + \frac{\delta}{\underline{\chi}} \frac{\beta}{\underline{\delta}} \epsilon^{\underline{\alpha}\chi}_{\underline{\delta}} \mathcal{T}_{2^+}^{\dagger\dagger_1} \beta_{\underline{\alpha}}$ ] := Module[{}],

Rank3AntisymmetricPara2mConstantSymbol † Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\mathcal{T}_{2^-}^{\dagger\dagger_1} + \frac{\beta}{\underline{\delta}\chi} \epsilon^{\underline{\alpha}\chi\delta}_{\underline{\delta}} \mathcal{T}_{2^+}^{\dagger\dagger_1} \beta_{\underline{\alpha}}$ ] := Module[{}],

Rank3AntisymmetricPara2mConstantSymbol † Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\mathcal{T}_{2^-}^{\dagger\dagger_1} + \frac{\delta\chi}{\underline{\beta}} \epsilon^{\underline{\alpha}}_{\underline{\chi}\delta} \mathcal{T}_{2^+}^{\dagger\dagger_1} \frac{\beta}{\underline{\alpha}}$ ] := Module[{}],

Rank3AntisymmetricPara2mConstantSymbol † Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\mathcal{T}_{2^-}^{\dagger\dagger_1} + \frac{\chi}{\underline{\delta}} \frac{\beta}{\underline{\delta}} \epsilon^{\underline{\alpha}}_{\underline{\chi}} \mathcal{T}_{2^+}^{\dagger\dagger_1} \frac{\beta}{\underline{\alpha}}$ ] := Module[{}],

Rank3AntisymmetricPara2mConstantSymbol † Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\mathcal{T}_{2^-}^{\dagger\dagger_1} + \frac{\delta}{\underline{\chi}\beta} \epsilon^{\underline{\alpha}\chi}_{\underline{\delta}} \mathcal{T}_{2^+}^{\dagger\dagger_1} \frac{\beta}{\underline{\alpha}}$ ] := Module[{}],

Rank3AntisymmetricPara2mConstantSymbol † Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\mathcal{T}_{2^-}^{\dagger\dagger_1} + \frac{\delta\chi\beta}{\underline{\delta}} \epsilon^{\underline{\alpha}\chi\delta}_{\underline{\delta}} \mathcal{T}_{2^+}^{\dagger\dagger_1} \frac{\beta}{\underline{\alpha}}$ ] := Module[{}],

Rank3AntisymmetricPara2mConstantSymbol † Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\mathcal{T}_{2^-}^{\dagger\dagger_1} + \frac{\delta\chi\beta}{\underline{\delta}} \epsilon^{\underline{\alpha}\chi\delta}_{\underline{\delta}} \mathcal{T}_{2^+}^{\dagger\dagger_1} \frac{\alpha}{\underline{\beta}}$ ] := Module[{}],

Rank3AntisymmetricPara2mConstantSymbol † Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\mathcal{T}_{2^-}^{\dagger\dagger_1} + \frac{\chi\beta}{\underline{\delta}} \epsilon^{\underline{\alpha}}_{\underline{\alpha}\chi} \mathcal{T}_{2^+}^{\dagger\dagger_1} \frac{\alpha}{\underline{\beta}}$ ] := Module[{}],

Rank3AntisymmetricPara2mConstantSymbol † Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\mathcal{T}_{2^-}^{\dagger\dagger_1} + \frac{\delta}{\underline{\chi}} \frac{\beta}{\underline{\delta}} \epsilon^{\underline{\alpha}}_{\underline{\alpha}} \frac{\chi}{\underline{\delta}} \mathcal{T}_{2^+}^{\dagger\dagger_1} \frac{\alpha}{\underline{\beta}}$ ] := Module[{}],

Rank3AntisymmetricPara2mConstantSymbol † Rank3AntisymmetricPara2pConstantSymbol],
HoldPattern[ $\mathcal{T}_{2^-}^{\dagger\dagger_1} + \frac{\beta}{\underline{\delta}\chi} \epsilon^{\underline{\alpha}}_{\underline{\alpha}} \frac{\chi\delta}{\underline{\delta}} \mathcal{T}_{2^+}^{\dagger\dagger_1} \frac{\alpha}{\underline{\beta}}$ ] := Module[{}],

Rank3AntisymmetricPara2mConstantSymbol † Rank3AntisymmetricPara2pConstantSymbol],

```


[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]


```

HoldPattern[ $\mathcal{J}_{2^-}^{\#1} \dagger_{\underline{\underline{\beta}}}^{\underline{\alpha}} \underline{\underline{\chi}} \mathcal{J}_{2^-}^{\#1} \underline{\underline{\beta\chi}}$ ] :> Module[{},
  -Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol †],
HoldPattern[ $\mathcal{J}_{2^-}^{\#1} \dagger_{\underline{\underline{\beta}}}^{\underline{\alpha\chi}} \mathcal{J}_{2^-}^{\#1} \underline{\underline{\beta}}_{\underline{\underline{\alpha}} \underline{\underline{\chi}}}$ ] :> Module[{},
  -Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol †],
HoldPattern[ $\mathcal{J}_{2^-}^{\#1} \dagger_{\underline{\underline{\chi}}}^{\underline{\beta\alpha}} \mathcal{J}_{2^-}^{\#1} \underline{\underline{\alpha\beta}}_{\underline{\underline{\chi}}}$ ] :> Module[{},
  -Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol †],
HoldPattern[ $\mathcal{J}_{2^-}^{\#1} \dagger_{\underline{\underline{\alpha\chi}}}^{\underline{\beta\alpha\chi}} \mathcal{J}_{2^-}^{\#1} \underline{\underline{\alpha\beta\chi}}$ ] :> Module[{},
  -Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol †],
HoldPattern[ $\mathcal{J}_{2^-}^{\#1} \underline{\underline{\beta\alpha\chi}}_{\underline{\underline{\alpha\beta\chi}}} \mathcal{J}_{2^-}^{\#1} \dagger_{\underline{\underline{\alpha\beta\chi}}}$ ] :> Module[{},
  -Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol †],
HoldPattern[ $\mathcal{J}_{2^-}^{\#1} \underline{\underline{\beta\alpha}}_{\underline{\underline{\chi}}} \mathcal{J}_{2^-}^{\#1} \dagger_{\underline{\underline{\alpha\beta}}}^{\underline{\chi}}$ ] :> Module[{},
  -Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol †],
HoldPattern[ $\mathcal{J}_{2^-}^{\#1} \underline{\underline{\alpha\chi}}_{\underline{\underline{\beta}}} \mathcal{J}_{2^-}^{\#1} \dagger_{\underline{\underline{\alpha}} \underline{\underline{\chi}}}^{\underline{\beta}}$ ] :> Module[{},
  -Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol †],
HoldPattern[ $\mathcal{J}_{2^-}^{\#1} \underline{\underline{\alpha}}_{\underline{\underline{\beta}} \underline{\underline{\chi}}} \mathcal{J}_{2^-}^{\#1} \dagger_{\underline{\underline{\alpha}}}^{\underline{\beta\chi}}$ ] :> Module[{},
  -Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol †],
HoldPattern[ $\mathcal{J}_{2^-}^{\#1} \underline{\underline{\beta}}_{\underline{\underline{\alpha}}}^{\underline{\chi}} \mathcal{J}_{2^-}^{\#1} \dagger_{\underline{\underline{\beta\chi}}}^{\underline{\alpha}}$ ] :> Module[{},
  -Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol †],
HoldPattern[ $\mathcal{J}_{2^-}^{\#1} \underline{\underline{\beta}}_{\underline{\underline{\alpha\chi}}}^{\underline{\chi}} \mathcal{J}_{2^-}^{\#1} \dagger_{\underline{\underline{\beta}}}^{\underline{\alpha}} \underline{\underline{\chi}}$ ] :> Module[{},
  -Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol †],
HoldPattern[ $\mathcal{J}_{2^-}^{\#1} \underline{\underline{\beta\alpha}}_{\underline{\underline{\chi}}}^{\underline{\chi}} \mathcal{J}_{2^-}^{\#1} \dagger_{\underline{\underline{\alpha\beta}}}^{\underline{\alpha\beta\chi}}$ ] :> Module[{},
  -Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol †],
HoldPattern[ $\mathcal{J}_{2^-}^{\#1} \underline{\underline{\beta\alpha\chi}}_{\underline{\underline{\alpha\beta\chi}}} \mathcal{J}_{2^-}^{\#1} \dagger_{\underline{\underline{\alpha\beta\chi}}}$ ] :> Module[{}

```



```

HoldPattern[ $\mathcal{J}_{2^{-}}^{\#1} \dagger_{\underline{\underline{\beta\alpha\chi}}} \mathcal{J}_{2^{-}}^{\#1} \underline{\underline{\beta\alpha\chi}}$ ] :> Module[{ },
  Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{2^{-}}^{\#1} \dagger_{\underline{\underline{\beta\alpha}}}^{\underline{\underline{\chi}}} \mathcal{J}_{2^{-}}^{\#1} \underline{\underline{\beta\alpha}}_{\underline{\underline{\chi}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{2^{-}}^{\#1} \dagger_{\underline{\underline{\alpha\chi}}}^{\underline{\underline{\beta}}} \mathcal{J}_{2^{-}}^{\#1} \underline{\underline{\beta}}_{\underline{\underline{\alpha\chi}}}^{\underline{\underline{\alpha\chi}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{2^{-}}^{\#1} \dagger_{\underline{\underline{\alpha}}}^{\underline{\underline{\beta}}} \underline{\underline{\chi}} \mathcal{J}_{2^{-}}^{\#1} \underline{\underline{\beta}}_{\underline{\underline{\chi}}}^{\underline{\underline{\alpha}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{2^{-}}^{\#1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\alpha}}} \underline{\underline{\chi}} \mathcal{J}_{2^{-}}^{\#1} \underline{\underline{\beta}}_{\underline{\underline{\chi}}}^{\underline{\underline{\alpha\chi}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{2^{-}}^{\#1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\alpha\chi}}} \mathcal{J}_{2^{-}}^{\#1} \underline{\underline{\beta}}_{\underline{\underline{\alpha\chi}}}^{\underline{\underline{\beta}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{2^{-}}^{\#1} \dagger_{\underline{\underline{\chi}}}^{\underline{\underline{\beta\alpha}}} \mathcal{J}_{2^{-}}^{\#1} \underline{\underline{\beta\alpha}}_{\underline{\underline{\chi}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{2^{-}}^{\#1} \dagger_{\underline{\underline{\beta\alpha\chi}}}^{\underline{\underline{\beta\alpha\chi}}} \mathcal{J}_{2^{-}}^{\#1} \underline{\underline{\beta\alpha\chi}}$ ] :> Module[{ },
  Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol  $\dagger$ }]

```

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`SymbolicLagrangian\$103292 will now be printed.

$$- \frac{\binom{2}{1} \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol} \dagger + \binom{2}{1} \text{Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol} \dagger}{3} +$$

$$- \frac{\binom{2}{1} \text{Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1mConstantSymbol} \dagger}{2}$$

$$+ \binom{2}{1} \text{Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol} \dagger + \frac{\binom{2}{1} 8 \text{Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol} \dagger}{9}$$

$$+ \binom{2}{1} \text{Rank3AntisymmetricPara2pConstantSymbol Rank3AntisymmetricPara2pConstantSymbol} \dagger +$$

$$\binom{2}{1} \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1mConstantSymbol} \dagger -$$

$$\binom{2}{1} \text{Rank3AntisymmetricPara1pConstantSymbol} \dagger \text{Rank3AntisymmetricPerp1pConstantSymbol} -$$

$$\binom{2}{1} \text{Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol} \dagger -$$

$$- \frac{4}{3} \binom{2}{1} \mathcal{K}_{0^-}^{\dagger 1} \dagger_{\alpha\beta\chi} \mathcal{K}_{2^-}^{\dagger 1} \alpha\beta\chi - \frac{4}{3} \binom{2}{1} \mathcal{K}_{0^-}^{\dagger 1} \alpha\beta\chi \mathcal{K}_{2^-}^{\dagger 1} \dagger_{\alpha\beta\chi}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`SpinParityConstantSymbols\$103292 will now be printed.

<| xAct`PSALter`A23Field`Rank3Antisymmetric →

<| 0 → <| xAct`PSALter`Private`Even → {Rank3AntisymmetricPara0pConstantSymbol},
xAct`PSALter`Private`Odd → {Rank3AntisymmetricPara0mConstantSymbol} |> ,

1 → <| xAct`PSALter`Private`Even → {Rank3AntisymmetricPara1pConstantSymbol,
Rank3AntisymmetricPerp1pConstantSymbol},
xAct`PSALter`Private`Odd → {Rank3AntisymmetricPara1mConstantSymbol,
Rank3AntisymmetricPerp1mConstantSymbol} |> ,

2 → <| xAct`PSALter`Private`Even → {Rank3AntisymmetricPara2pConstantSymbol},
xAct`PSALter`Private`Odd → {Rank3AntisymmetricPara2mConstantSymbol} |> |> |>

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`SpinParityRescalingSymbols\$103292 will now be printed.

```
<| xAct`PSALter`A23Field`Rank3Antisymmetric →
<| 0 → <| xAct`PSALter`Private`Even → {Rank3AntisymmetricPara0pRescalingSymbol},
xAct`PSALter`Private`Odd → {Rank3AntisymmetricPara0mRescalingSymbol} |>,
1 → <| xAct`PSALter`Private`Even → {Rank3AntisymmetricPara1pRescalingSymbol,
Rank3AntisymmetricPerp1pRescalingSymbol},
xAct`PSALter`Private`Odd → {Rank3AntisymmetricPara1mRescalingSymbol,
Rank3AntisymmetricPerp1mRescalingSymbol} |>,
2 → <| xAct`PSALter`Private`Even → {Rank3AntisymmetricPara2pRescalingSymbol},
xAct`PSALter`Private`Odd → {Rank3AntisymmetricPara2mRescalingSymbol} |> |> |>
```

****** xAct`PSALter`Private`Diagnostic: the value of the
variable xAct`PSALter`Private`Symbols\$103292 will now be printed.

```
<| |>
```

****** xAct`PSALter`Private`Diagnostic: the value of the
variable xAct`PSALter`Private`Symbols\$103292 will now be printed.

```
<| 0 → {Rank3AntisymmetricPara0pConstantSymbol,
Rank3AntisymmetricPara0mConstantSymbol, Rank3AntisymmetricPara0pConstantSymbol†,
Rank3AntisymmetricPara0mConstantSymbol†},
1 → {Rank3AntisymmetricPara1pConstantSymbol, Rank3AntisymmetricPerp1pConstantSymbol,
Rank3AntisymmetricPara1mConstantSymbol, Rank3AntisymmetricPerp1mConstantSymbol,
Rank3AntisymmetricPara1pConstantSymbol†,
Rank3AntisymmetricPerp1pConstantSymbol†, Rank3AntisymmetricPara1mConstantSymbol†,
Rank3AntisymmetricPerp1mConstantSymbol†},
2 → {Rank3AntisymmetricPara2pConstantSymbol, Rank3AntisymmetricPara2mConstantSymbol,
Rank3AntisymmetricPara2pConstantSymbol†,
Rank3AntisymmetricPara2mConstantSymbol†} |>
```

****** xAct`PSALter`Private`Diagnostic: the value of the
variable xAct`PSALter`Private`DummySectors\$103292 will now be printed.

$\langle | \rightarrow \langle | \text{ xAct`PSALter`Private`Even} \rightarrow$

Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol $\dagger$,
 xAct`PSALter`Private`Odd $\rightarrow$ Rank3AntisymmetricPara0mConstantSymbol
 Rank3AntisymmetricPara0mConstantSymbol $\dagger | \rangle$,

1 $\rightarrow \langle | \text{ xAct`PSALter`Private`Even} \rightarrow$

Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol $\dagger +$
 Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol $\dagger$,
 xAct`PSALter`Private`Odd $\rightarrow$ Rank3AntisymmetricPara1mConstantSymbol
 Rank3AntisymmetricPara1mConstantSymbol $\dagger +$
 Rank3AntisymmetricPerp1mConstantSymbol
 Rank3AntisymmetricPerp1mConstantSymbol $\dagger | \rangle$,

2 $\rightarrow \langle | \text{ xAct`PSALter`Private`Even} \rightarrow$ Rank3AntisymmetricPara2pConstantSymbol

Rank3AntisymmetricPara2pConstantSymbol $\dagger$,
 xAct`PSALter`Private`Odd $\rightarrow$ Rank3AntisymmetricPara2mConstantSymbol
 Rank3AntisymmetricPara2mConstantSymbol $\dagger | \rangle | \rangle$

****** xAct`PSALter`Private`Diagnostic: the value of
 the variable xAct`PSALter`Private`Rescalings will now be printed.

$\langle | \rightarrow \left\{ \frac{1}{\text{Rank3AntisymmetricPara0pRescalingSymbol}}, \right.$

$\left. \frac{1}{\text{Rank3AntisymmetricPara0mRescalingSymbol}} \right\},$

1 $\rightarrow \left\{ \frac{1}{\text{Rank3AntisymmetricPara1pRescalingSymbol}}, \right.$

$\frac{1}{\text{Rank3AntisymmetricPerp1pRescalingSymbol}}, \frac{1}{\text{Rank3AntisymmetricPara1mRescalingSymbol}},$

$\left. \frac{1}{\text{Rank3AntisymmetricPerp1mRescalingSymbol}} \right\},$

2 $\rightarrow \left\{ \frac{1}{\text{Rank3AntisymmetricPara2pRescalingSymbol}}, \right.$

$\left. \frac{1}{\text{Rank3AntisymmetricPara2mRescalingSymbol}} \right\} | \rangle$

****** xAct`PSALter`Private`Diagnostic: the value of the
 variable xAct`PSALter`Private`RaisedIndexSources will now be printed.

$$\langle | \rightarrow \{ \mathcal{T}_{\theta^+}^{\#1}, \mathcal{T}_{\theta^-}^{\#1} \alpha \beta \chi \},$$

$$1 \rightarrow \{ \mathcal{T}_{1^+}^{\#1} \alpha \beta, \mathcal{T}_{1^+}^{\#2} \alpha \beta, \mathcal{T}_{1^-}^{\#1} \alpha, \mathcal{T}_{1^-}^{\#2} \alpha \}, 2 \rightarrow \{ \mathcal{T}_{2^+}^{\#1} \alpha \beta, \mathcal{T}_{2^-}^{\#1} \alpha \beta \chi \} | \rangle$$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable xAct`PSALTER`Private`LoweredIndexSources will now be printed.

$$\langle | \rightarrow \{ \mathcal{T}_{\theta^+}^{\#1}, \mathcal{T}_{\theta^-}^{\#1} \alpha \beta \chi \},$$

$$1 \rightarrow \{ \mathcal{T}_{1^+}^{\#1} \alpha \beta, \mathcal{T}_{1^+}^{\#2} \alpha \beta, \mathcal{T}_{1^-}^{\#1} \alpha, \mathcal{T}_{1^-}^{\#2} \alpha \}, 2 \rightarrow \{ \mathcal{T}_{2^+}^{\#1} \alpha \beta, \mathcal{T}_{2^-}^{\#1} \alpha \beta \chi \} | \rangle$$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable xAct`PSALTER`Private`RaisedIndexFields\$103292 will now be printed.

$$\langle | \rightarrow \{ \mathcal{K}_{\theta^+}^{\#1}, \mathcal{K}_{\theta^-}^{\#1} \alpha \beta \chi \},$$

$$1 \rightarrow \{ \mathcal{K}_{1^+}^{\#1} \alpha \beta, \mathcal{K}_{1^+}^{\#2} \alpha \beta, \mathcal{K}_{1^-}^{\#1} \alpha, \mathcal{K}_{1^-}^{\#2} \alpha \}, 2 \rightarrow \{ \mathcal{K}_{2^+}^{\#1} \alpha \beta, \mathcal{K}_{2^-}^{\#1} \alpha \beta \chi \} | \rangle$$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable xAct`PSALTER`Private`LoweredIndexFields\$103292 will now be printed.

$$\langle | \rightarrow \{ \mathcal{K}_{\theta^+}^{\#1}, \mathcal{K}_{\theta^-}^{\#1} \alpha \beta \chi \},$$

$$1 \rightarrow \{ \mathcal{K}_{1^+}^{\#1} \alpha \beta, \mathcal{K}_{1^+}^{\#2} \alpha \beta, \mathcal{K}_{1^-}^{\#1} \alpha, \mathcal{K}_{1^-}^{\#2} \alpha \}, 2 \rightarrow \{ \mathcal{K}_{2^+}^{\#1} \alpha \beta, \mathcal{K}_{2^-}^{\#1} \alpha \beta \chi \} | \rangle$$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable xAct`PSALTER`Private`\$Neglect will now be printed.

$$\langle | \rightarrow \langle | \text{ xAct`PSALTER`Private`Even} \rightarrow \text{False}, \text{ xAct`PSALTER`Private`Odd} \rightarrow \text{False} | \rangle ,$$

$$1 \rightarrow \langle | \text{ xAct`PSALTER`Private`Even} \rightarrow \text{False}, \text{ xAct`PSALTER`Private`Odd} \rightarrow \text{False} | \rangle ,$$

$$2 \rightarrow \langle | \text{ xAct`PSALTER`Private`Even} \rightarrow \text{False}, \text{ xAct`PSALTER`Private`Odd} \rightarrow \text{False} | \rangle | \rangle$$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable xAct`PSALTER`Private`SymbolicLagrangian\$103292 will now be printed.

$$\frac{-K_1^{(2)} \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol} \dagger + K_1^{(2)} \text{Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol} \dagger}{3} +$$

$$\frac{K_1^{(2)} \text{Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1mConstantSymbol} \dagger}{2} -$$

$$K_1^{(2)} \text{Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol} \dagger + \frac{8 K_1^{(2)} \text{Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol} \dagger}{9}$$

$$+ K_1^{(2)} \text{Rank3AntisymmetricPara2pConstantSymbol Rank3AntisymmetricPara2pConstantSymbol} \dagger +$$

$$K_1^{(2)} \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1mConstantSymbol} \dagger -$$

$$K_1^{(2)} \text{Rank3AntisymmetricPara1pConstantSymbol} \dagger \text{Rank3AntisymmetricPerp1pConstantSymbol} -$$

$$K_1^{(2)} \text{Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol} \dagger -$$

$$-\frac{4}{3} K_1^{(2)} \mathcal{K}_{0^-}^{\dagger 1} \dagger_{\alpha\beta\chi} \mathcal{K}_{2^-}^{\dagger 1} \alpha\beta\chi - \frac{4}{3} K_1^{(2)} \mathcal{K}_{0^-}^{\dagger 1} \alpha\beta\chi \mathcal{K}_{2^-}^{\dagger 1} \dagger_{\alpha\beta\chi}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`MatrixLagrangian will now be printed.

$$\left\langle \left| 0 \rightarrow \text{SparseArray} \left[\begin{array}{c} \text{Specified elements: 2} \\ \text{Dimensions: \{2, 2\}} \end{array} \right], \right. \right.$$

$$1 \rightarrow \text{SparseArray} \left[\begin{array}{c} \text{Specified elements: 5} \\ \text{Dimensions: \{4, 4\}} \end{array} \right], 2 \rightarrow \text{SparseArray} \left[\begin{array}{c} \text{Specified elements: 2} \\ \text{Dimensions: \{2, 2\}} \end{array} \right] \left| \right\rangle$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`MatrixLagrangian will now be printed.

$\langle \mid \rightarrow \text{SparseArray} \left[\begin{array}{c} \text{Specified elements: 2} \\ \text{Dimensions: \{2, 2\}} \end{array} \right],$

$1 \rightarrow \text{SparseArray} \left[\begin{array}{c} \text{Specified elements: 5} \\ \text{Dimensions: \{4, 4\}} \end{array} \right], 2 \rightarrow \text{SparseArray} \left[\begin{array}{c} \text{Specified elements: 2} \\ \text{Dimensions: \{2, 2\}} \end{array} \right] \mid \rangle$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
 xAct`PSALter`Private`Class\$103292[xAct`PSALter`Private`InverseRescalingMatrix]
 will now be printed.

$\langle \mid \rightarrow \left\{ \left\{ \frac{1}{\text{Rank3AntisymmetricPara0pRescalingSymbol}^2}, \right. \right.$

$$\left. \frac{i \sqrt{6}}{\text{Rank3AntisymmetricPara0mRescalingSymbol Rank3AntisymmetricPara0pRescalingSymbol}} \right\},$$

$\left\{ -\frac{i \sqrt{6}}{\text{Rank3AntisymmetricPara0mRescalingSymbol Rank3AntisymmetricPara0pRescalingSymbol}} \right.$

$$\left. , \frac{1}{\text{Rank3AntisymmetricPara0mRescalingSymbol}^2} \right\} \Bigg\},$$

$1 \rightarrow \left\{ \left\{ \frac{1}{\text{Rank3AntisymmetricPara1pRescalingSymbol}^2}, \right. \right.$

$$\left. \frac{1}{\text{Rank3AntisymmetricPara1pRescalingSymbol Rank3AntisymmetricPerp1pRescalingSymbol}} \right\},$$

$$\frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara1mRescalingSymbol Rank3AntisymmetricPara1pRescalingSymbol}},$$

$\left. \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara1pRescalingSymbol Rank3AntisymmetricPerp1mRescalingSymbol}} \right\},$

$$\begin{aligned}
& \left\{ \frac{1}{\text{Rank3AntisymmetricPara1pRescalingSymbol Rank3AntisymmetricPerp1pRescalingSymbol}}, \right. \\
& \frac{1}{\text{Rank3AntisymmetricPerp1pRescalingSymbol}^2}, \\
& \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara1mRescalingSymbol Rank3AntisymmetricPerp1pRescalingSymbol}}, \\
& \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPerp1mRescalingSymbol Rank3AntisymmetricPerp1pRescalingSymbol}} \Bigg\}, \\
& \left\{ \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara1mRescalingSymbol Rank3AntisymmetricPara1pRescalingSymbol}}, \right. \\
& \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara1mRescalingSymbol Rank3AntisymmetricPerp1pRescalingSymbol}}, \\
& \frac{1}{\text{Rank3AntisymmetricPara1mRescalingSymbol}^2}, \\
& \left. \frac{1}{\text{Rank3AntisymmetricPara1mRescalingSymbol Rank3AntisymmetricPerp1mRescalingSymbol}} \right\}, \\
& \left\{ \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara1pRescalingSymbol Rank3AntisymmetricPerp1mRescalingSymbol}}, \right. \\
& \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPerp1mRescalingSymbol Rank3AntisymmetricPerp1pRescalingSymbol}}, \\
& \left. \right\}
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{\text{Rank3AntisymmetricPara1mRescalingSymbol Rank3AntisymmetricPerp1mRescalingSymbol}}, \\
& \frac{1}{\text{Rank3AntisymmetricPerp1mRescalingSymbol}^2} \Bigg\} \Bigg\}, \\
2 \rightarrow & \left\{ \left\{ \frac{1}{\text{Rank3AntisymmetricPara2pRescalingSymbol}^2}, \right. \right. \\
& \left. \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara2mRescalingSymbol Rank3AntisymmetricPara2pRescalingSymbol}} \right\}, \\
& \left\{ - \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara2mRescalingSymbol Rank3AntisymmetricPara2pRescalingSymbol}} \right. \\
& \left. , \frac{1}{\text{Rank3AntisymmetricPara2mRescalingSymbol}^2} \right\} \Bigg\} \Bigg| \Bigg\rangle
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`MatrixLagrangian will now be printed.

$$\begin{aligned}
& \langle | \rightarrow \left\{ \left\{ \frac{{}^{(2)}K_1}{3 \text{ Rank3AntisymmetricPara0pRescalingSymbol}^2}, 0 \right\}, \right. \\
& \quad \left. \left\{ 0, -\frac{{}^{(2)}K_1}{\text{Rank3AntisymmetricPara0mRescalingSymbol}^2} \right\} \right\}, \\
& \quad 1 \rightarrow \left\{ \left\{ -\frac{{}^{(2)}K_1}{\text{Rank3AntisymmetricPara1pRescalingSymbol}^2}, \right. \right. \\
& \quad \quad \left. \left. \frac{{}^{(2)}K_1}{\text{Rank3AntisymmetricPara1pRescalingSymbol} \text{ Rank3AntisymmetricPerp1pRescalingSymbol}}, 0, 0 \right\}, \right. \\
& \quad \left. \left\{ -\frac{{}^{(2)}K_1}{\text{Rank3AntisymmetricPara1pRescalingSymbol} \text{ Rank3AntisymmetricPerp1pRescalingSymbol}}, 0, 0, 0 \right\}, \right. \\
& \quad \left. \left\{ 0, 0, \frac{{}^{(2)}K_1}{2 \text{ Rank3AntisymmetricPara1mRescalingSymbol}^2}, 0 \right\}, \right. \\
& \quad \left. \left\{ 0, 0, 0, \frac{{}^{(2)}K_1}{\text{Rank3AntisymmetricPerp1mRescalingSymbol}^2} \right\} \right\}, \\
& \quad 2 \rightarrow \left\{ \left\{ \frac{{}^{(2)}K_1}{\text{Rank3AntisymmetricPara2pRescalingSymbol}^2}, 0 \right\}, \right. \\
& \quad \left. \left\{ 0, \frac{{}^{(2)}K_1}{9 \text{ Rank3AntisymmetricPara2mRescalingSymbol}^2} \right\} \right\} | \rangle
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`MatrixLagrangian will now be printed.

$$\begin{aligned}
\langle | \rightarrow & \left\{ \left\{ \frac{\binom{2}{K_1}}{\sqrt[3]{\frac{2}{3}}}, 0 \right\}, \left\{ 0, -\frac{\binom{2}{K_1}}{1^2} \right\} \right\}, \\
1 \rightarrow & \left\{ \left\{ -\frac{\binom{2}{K_1}}{\sqrt{2^2}}, -\frac{\binom{2}{K_1}}{\sqrt{2}}, 0, 0 \right\}, \left\{ -\frac{\binom{2}{K_1}}{\sqrt{2}}, 0, 0, 0 \right\}, \left\{ 0, 0, \frac{\binom{2}{K_1}}{2 \times 1^2}, 0 \right\}, \right. \\
& \left. \left\{ 0, 0, 0, \frac{\binom{2}{K_1}}{\sqrt{2^2}} \right\} \right\}, 2 \rightarrow \left\{ \left\{ \frac{\binom{2}{K_1}}{\sqrt{2^2}}, 0 \right\}, \left\{ 0, \frac{\binom{2}{K_1}}{9 \left(\frac{4}{3}\right)^2} \right\} \right\} | \rangle
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`MatrixLagrangian will now be printed.

$$\begin{aligned}
\langle | \rightarrow & \left\{ \left\{ \frac{\binom{2}{K_1}}{2}, 0 \right\}, \left\{ 0, -\frac{\binom{2}{K_1}}{1} \right\} \right\}, \\
1 \rightarrow & \left\{ \left\{ -\frac{\binom{2}{K_1}}{2}, -\frac{\binom{2}{K_1}}{\sqrt{2}}, 0, 0 \right\}, \left\{ -\frac{\binom{2}{K_1}}{\sqrt{2}}, 0, 0, 0 \right\}, \left\{ 0, 0, \frac{\binom{2}{K_1}}{2}, 0 \right\}, \left\{ 0, 0, 0, \frac{\binom{2}{K_1}}{2} \right\} \right\}, \\
2 \rightarrow & \left\{ \left\{ \frac{\binom{2}{K_1}}{2}, 0 \right\}, \left\{ 0, \frac{\binom{2}{K_1}}{2} \right\} \right\} | \rangle
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`AntiMaskMatrixValue will now be printed.

$$\begin{aligned}
\langle | \rightarrow & \langle | \text{xAct`PSALter`Private`Even} \rightarrow \{\{1, 0\}, \{0, 0\}\}, \\
& \text{xAct`PSALter`Private`Odd} \rightarrow \{\{0, 0\}, \{0, 1\}\} | \rangle, \\
1 \rightarrow & \langle | \text{xAct`PSALter`Private`Even} \rightarrow \\
& \{\{1, 1, 0, 0\}, \{1, 1, 0, 0\}, \{0, 0, 0, 0\}, \{0, 0, 0, 0\}\}, \text{xAct`PSALter`Private`Odd} \rightarrow \\
& \{\{0, 0, 0, 0\}, \{0, 0, 0, 0\}, \{0, 0, 1, 1\}, \{0, 0, 1, 1\}\} | \rangle, \\
2 \rightarrow & \langle | \text{xAct`PSALter`Private`Even} \rightarrow \{\{1, 0\}, \{0, 0\}\}, \\
& \text{xAct`PSALter`Private`Odd} \rightarrow \{\{0, 0\}, \{0, 1\}\} | \rangle | \rangle
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`\$BMatricesValues will now be printed.

$$\begin{aligned}
& \langle | \rightarrow \left\{ \left\{ \left\{ \frac{\binom{(2)}{K_1}}{2}, 0 \right\}, \{0, 0\} \right\}, \left\{ \{0, 0\}, \left\{ 0, -\binom{(2)}{K_1} \right\} \right\} \right\}, \\
& {}_1 \rightarrow \left\{ \left\{ \left\{ -\frac{\binom{(2)}{K_1}}{2}, -\frac{\binom{(2)}{K_1}}{\sqrt{2}}, 0, 0 \right\}, \left\{ -\frac{\binom{(2)}{K_1}}{\sqrt{2}}, 0, 0, 0 \right\}, \{0, 0, 0, 0\}, \{0, 0, 0, 0\} \right\}, \right. \\
& \quad \left. \left\{ \{0, 0, 0, 0\}, \{0, 0, 0, 0\}, \left\{ 0, 0, \frac{\binom{(2)}{K_1}}{2}, 0 \right\}, \left\{ 0, 0, 0, \frac{\binom{(2)}{K_1}}{2} \right\} \right\} \right\}, \\
& {}_2 \rightarrow \left\{ \left\{ \left\{ \frac{\binom{(2)}{K_1}}{2}, 0 \right\}, \{0, 0\} \right\}, \left\{ \{0, 0\}, \left\{ 0, \frac{\binom{(2)}{K_1}}{2} \right\} \right\} \right\} | \rangle
\end{aligned}$$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable xAct`PSALTER`Private`\$ValuesAllMatrices will now be printed.

$$\begin{aligned}
& \left\{ \left\{ \left\{ \frac{\binom{(2)}{K_1}}{2}, 0 \right\}, \left\{ 0, -\binom{(2)}{K_1} \right\} \right\}, \right. \\
& \quad \left\{ \left\{ -\frac{\binom{(2)}{K_1}}{2}, -\frac{\binom{(2)}{K_1}}{\sqrt{2}}, 0, 0 \right\}, \left\{ -\frac{\binom{(2)}{K_1}}{\sqrt{2}}, 0, 0, 0 \right\}, \left\{ 0, 0, \frac{\binom{(2)}{K_1}}{2}, 0 \right\}, \left\{ 0, 0, 0, \frac{\binom{(2)}{K_1}}{2} \right\} \right\}, \\
& \quad \left. \left\{ \left\{ \frac{\binom{(2)}{K_1}}{2}, 0 \right\}, \left\{ 0, \frac{\binom{(2)}{K_1}}{2} \right\} \right\} \right\}
\end{aligned}$$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable xAct`PSALTER`Private`EmptyDeterminants\$103292 will now be printed.

$$\{\}, \{\}, \{\}$$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable xAct`PSALTER`Private`CrossingRules\$167183 will now be printed.

$$\left\{ \begin{array}{l} \text{HoldPattern}\left[\partial^u \partial^\omega \mathcal{K}^{\alpha\beta\chi} \partial^\psi \partial^\pi \mathcal{K}^{\xi\phi\zeta}\right] \rightarrow \\ \quad \text{Module}[\{\}, k^\pi k^\omega k^u k^\psi \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^{\dagger\alpha\beta\chi}], \\ \text{HoldPattern}\left[\partial^\omega \mathcal{K}^{\alpha\beta\chi} \partial^\psi \partial^\pi \mathcal{K}^{\xi\phi\zeta}\right] \rightarrow \\ \quad \text{Module}[\{\}, -i k^\pi k^\omega k^\psi \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^{\dagger\alpha\beta\chi}], \\ \text{HoldPattern}\left[\partial^\pi \mathcal{K}^{\xi\phi\zeta} \partial^u \partial^\omega \mathcal{K}^{\alpha\beta\chi}\right] \rightarrow \\ \quad \text{Module}[\{\}, i k^\pi k^\omega k^u \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^{\dagger\alpha\beta\chi}], \\ \text{HoldPattern}\left[\partial^\pi \mathcal{K}^{\xi\phi\zeta} \partial^\omega \mathcal{K}^{\alpha\beta\chi}\right] \rightarrow \text{Module}[\{\}, k^\pi k^\omega \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^{\dagger\alpha\beta\chi}], \\ \text{HoldPattern}\left[\mathcal{K}^{\xi\phi\zeta} \partial^\omega \partial^\pi \mathcal{K}^{\alpha\beta\chi}\right] \rightarrow \text{Module}[\{\}, -k^\pi k^\omega \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^{\dagger\alpha\beta\chi}], \\ \text{HoldPattern}\left[\mathcal{K}^{\alpha\beta\chi} \partial^\omega \partial^\pi \mathcal{K}^{\xi\phi\zeta}\right] \rightarrow \text{Module}[\{\}, -k^\pi k^\omega \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^{\dagger\alpha\beta\chi}], \\ \text{HoldPattern}\left[\mathcal{K}^{\alpha\beta\chi} \partial^\pi \mathcal{K}^{\xi\phi\zeta}\right] \rightarrow \text{Module}[\{\}, -i k^\pi \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^{\dagger\alpha\beta\chi}], \\ \text{HoldPattern}\left[\mathcal{K}^{\alpha\beta\chi} \mathcal{K}^{\xi\phi\zeta}\right] \rightarrow \text{Module}[\{\}, \mathcal{K}^{\xi\phi\zeta} \mathcal{K}^{\dagger\alpha\beta\chi}] \end{array} \right\}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`ToMomentumExpr\$167183 will now be printed.

$$\begin{aligned} & \overset{(2)}{K}_1 \mathcal{K}^{\alpha\beta\chi} \mathcal{K}^{\dagger}_{\alpha\beta\chi} + \overset{(4)}{K}_3 k_\beta k_\delta \mathcal{K}^{\alpha\beta\chi} \mathcal{K}^{\dagger}_{\alpha\chi}{}^\delta + \\ & \overset{(2)}{K}_3 \mathcal{K}^\chi{}_{\beta\chi} \mathcal{K}^{\dagger\alpha}{}_{\alpha}{}^\beta + \overset{(4)}{K}_2 k_\chi k^\chi \mathcal{K}^\delta{}_{\beta\delta} \mathcal{K}^{\dagger\alpha}{}_{\alpha}{}^\beta + \\ & \overset{(4)}{K}_1 k_\beta k^\chi \mathcal{K}^\delta{}_{\chi\delta} \mathcal{K}^{\dagger\alpha}{}_{\alpha}{}^\beta + \overset{(4)}{K}_{15} k_\delta k^\delta \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\dagger\alpha\beta\chi} + \\ & \overset{(2)}{K}_2 \mathcal{K}_{\beta\alpha\chi} \mathcal{K}^{\dagger\alpha\beta\chi} + \overset{(4)}{K}_{16} k_\delta k^\delta \mathcal{K}_{\beta\alpha\chi} \mathcal{K}^{\dagger\alpha\beta\chi} + \\ & \overset{(4)}{K}_4 k_\alpha k_\delta \mathcal{K}^{\alpha\beta\chi} \mathcal{K}^{\dagger}_{\beta\chi}{}^\delta + \overset{(4)}{K}_6 k_\beta k_\delta \mathcal{K}^{\alpha\beta\chi} \mathcal{K}^{\dagger}_{\chi\alpha}{}^\delta + \\ & \overset{(4)}{K}_7 k^\chi k_\delta \mathcal{K}^\alpha{}_{\alpha}{}^\beta \mathcal{K}^{\dagger}_{\chi\beta}{}^\delta + \overset{(4)}{K}_9 k_\alpha k_\delta \mathcal{K}^{\alpha\beta\chi} \mathcal{K}^{\dagger}_{\beta\chi}{}^\delta \end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`ToMomentumExpr\$167183 will now be printed.

$$\begin{aligned}
& \overset{(2)}{K}_1 \mathcal{K}^{\alpha\beta\chi} \mathcal{K}^\dagger_{\alpha\beta\chi} + \overset{(4)}{K}_3 k_\beta k_\delta \mathcal{K}^{\alpha\beta\chi} \mathcal{K}^\dagger_{\alpha\chi}{}^\delta + \\
& \overset{(2)}{K}_3 \mathcal{K}^\chi_{\beta\chi} \mathcal{K}^\dagger_{\alpha}{}^{\beta} + \overset{(4)}{K}_2 k_\chi k^\chi \mathcal{K}^\delta_{\beta\delta} \mathcal{K}^\dagger_{\alpha}{}^{\beta} + \\
& \overset{(4)}{K}_1 k_\beta k^\chi \mathcal{K}^\delta_{\chi\delta} \mathcal{K}^\dagger_{\alpha}{}^{\beta} + \overset{(4)}{K}_{15} k_\delta k^\delta \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^\dagger^{\alpha\beta\chi} + \\
& \overset{(2)}{K}_2 \mathcal{K}_{\beta\alpha\chi} \mathcal{K}^\dagger^{\alpha\beta\chi} + \overset{(4)}{K}_{16} k_\delta k^\delta \mathcal{K}_{\beta\alpha\chi} \mathcal{K}^\dagger^{\alpha\beta\chi} + \\
& \overset{(4)}{K}_4 k_\alpha k_\delta \mathcal{K}^{\alpha\beta\chi} \mathcal{K}^\dagger_{\beta\chi}{}^\delta + \overset{(4)}{K}_6 k_\beta k_\delta \mathcal{K}^{\alpha\beta\chi} \mathcal{K}^\dagger_{\chi\alpha}{}^\delta + \\
& \overset{(4)}{K}_7 k^\chi k_\delta \mathcal{K}^\alpha_{\alpha}{}^\beta \mathcal{K}^\dagger_{\chi\beta}{}^\delta + \overset{(4)}{K}_9 k_\alpha k_\delta \mathcal{K}^{\alpha\beta\chi} \mathcal{K}^\dagger_{\beta\chi}{}^\delta
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`ToMomentumExpr\$167183 will now be printed.

$$\begin{aligned}
& \overset{(2)}{K}_1 \mathcal{K}^{\alpha\beta\chi} \mathcal{K}^\dagger_{\alpha\beta\chi} - \overset{(4)}{K}_7 k^\alpha k^\beta \mathcal{K}^\chi_{\chi}{}^\delta \mathcal{K}^\dagger_{\alpha\beta\delta} + \\
& \overset{(2)}{K}_2 \mathcal{K}^{\alpha\beta\chi} \mathcal{K}^\dagger_{\beta\alpha\chi} + \overset{(4)}{K}_9 k^\alpha k^\beta \mathcal{K}^\chi_{\alpha}{}^\delta \mathcal{K}^\dagger_{\beta\chi\delta} + \\
& \overset{(4)}{K}_{15} k_\alpha k^\alpha \mathcal{K}^{\beta\chi\delta} \mathcal{K}^\dagger_{\beta\chi\delta} - \overset{(4)}{K}_4 k^\alpha k^\beta \mathcal{K}^\chi_{\alpha}{}^\delta \mathcal{K}^\dagger_{\chi\beta\delta} + \\
& \overset{(4)}{K}_{16} k_\alpha k^\alpha \mathcal{K}^{\beta\chi\delta} \mathcal{K}^\dagger_{\chi\beta\delta} - \overset{(4)}{K}_3 k^\alpha k^\beta \mathcal{K}^\chi_{\alpha}{}^\delta \mathcal{K}^\dagger_{\chi\beta\delta} + \\
& \overset{(2)}{K}_3 \mathcal{K}^\alpha_{\alpha}{}^\beta \mathcal{K}^\dagger_{\beta\chi}{}^\chi - \overset{(4)}{K}_6 k^\alpha k^\beta \mathcal{K}^\chi_{\alpha}{}^\delta \mathcal{K}^\dagger_{\delta\beta\chi} - \\
& \overset{(4)}{K}_1 k^\alpha k^\beta \mathcal{K}^\chi_{\alpha\chi} \mathcal{K}^\dagger_{\beta\delta}{}^\delta + \overset{(4)}{K}_2 k_\alpha k^\alpha \mathcal{K}^\beta_{\beta}{}^\chi \mathcal{K}^\dagger_{\chi\delta}{}^\delta
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`ToMomentumExpr\$167183 will now be printed.

$$\begin{aligned}
& \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger\dagger 1} \alpha \beta \chi \mathcal{K}_{\theta^-}^{\dagger\dagger 1} \dagger \alpha \beta \chi + \\
& \frac{1}{3} \left(-k^2 \left(\binom{4}{3K_1} - \binom{4}{2K_{15}} - \binom{4}{K_{16}} + \binom{4}{3K_2} + \binom{4}{K_3} + \binom{4}{K_6} \right) + \binom{2}{2K_1} + \binom{2}{K_2} - \binom{2}{3K_3} \right) \mathcal{K}_{\theta^+}^{\dagger\dagger 1} \mathcal{K}_{\theta^+}^{\dagger\dagger 1} \dagger + \\
& \frac{1}{2} \left(k^2 \left(\binom{4}{2K_{15}} + \binom{4}{K_{16}} - \binom{4}{2K_2} \right) + \binom{2}{2K_1} + \binom{2}{K_2} - \binom{2}{2K_3} \right) \mathcal{K}_{1^-}^{\dagger\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger\dagger 1} \dagger \alpha + \\
& \left(k^2 \left(\binom{4}{2K_{15}} - \binom{4}{K_{16}} - \binom{4}{K_3} + \binom{4}{K_6} \right) + \binom{2}{2K_1} - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger\dagger 1} \dagger \alpha \beta + \\
& \frac{4}{3} \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger\dagger 1} \dagger \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger\dagger 1} \alpha \beta \chi + \\
& \frac{8}{9} \left(k^2 \left(\binom{4}{2K_{15}} + \binom{4}{K_{16}} \right) + \binom{2}{2K_1} + \binom{2}{K_2} \right) \mathcal{K}_{2^-}^{\dagger\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger\dagger 1} \dagger \alpha \beta \chi + \\
& \frac{4}{3} \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger\dagger 1} \dagger \alpha \beta \chi + \\
& \left(k^2 \left(\binom{4}{2K_{15}} + \binom{4}{K_{16}} - \binom{4}{K_3} - \binom{4}{K_6} \right) + \binom{2}{2K_1} + \binom{2}{K_2} \right) \mathcal{K}_{2^+}^{\dagger\dagger 1} \alpha \beta \mathcal{K}_{2^+}^{\dagger\dagger 1} \dagger \alpha \beta + \\
& \left(k^2 \binom{4}{K_2} + \binom{2}{K_3} \right) \mathcal{K}_{1^-}^{\dagger\dagger 1} \dagger \alpha \mathcal{K}_{1^-}^{\dagger\dagger 2} \alpha + \left(k^2 \left(\binom{4}{K_2} + \binom{4}{K_7} \right) + \binom{2}{K_3} \right) \mathcal{K}_{1^-}^{\dagger\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger\dagger 2} \dagger \alpha + \\
& \left(k^2 \left(\binom{4}{2K_{15}} + \binom{4}{K_{16}} - \binom{4}{K_2} - \binom{4}{K_3} - \binom{4}{K_4} - \binom{4}{K_7} + \binom{4}{2K_9} \right) + \binom{2}{2K_1} + \binom{2}{K_2} - \binom{2}{K_3} \right) \\
& \mathcal{K}_{1^-}^{\dagger\dagger 2} \alpha \mathcal{K}_{1^-}^{\dagger\dagger 2} \dagger \alpha + \left(k^2 \left(-\binom{4}{K_{16}} + \binom{4}{K_4} \right) - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger\dagger 1} \dagger \alpha \beta \mathcal{K}_{1^+}^{\dagger\dagger 2} \alpha \beta + \\
& \left(-k^2 \binom{4}{K_{16}} - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger\dagger 2} \dagger \alpha \beta + \left(k^2 \left(\binom{4}{K_{15}} + \binom{4}{K_9} \right) + \binom{2}{K_1} \right) \mathcal{K}_{1^+}^{\dagger\dagger 2} \alpha \beta \mathcal{K}_{1^+}^{\dagger\dagger 2} \dagger \alpha \beta
\end{aligned}$$

****** xAct`PSALTer`Private`Diagnostic: the value of the variable
xAct`PSALTer`Private`ToMomentumExpr\$167183 will now be printed.

$$\begin{aligned}
& \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger\dagger 1} \alpha \beta \chi \mathcal{K}_{\theta^-}^{\dagger\dagger 1} \dagger \alpha \beta \chi + \\
& \frac{1}{3} \left(-k^2 \left(\binom{4}{3K_1} - \binom{4}{2K_{15}} - \binom{4}{K_{16}} + \binom{4}{3K_2} + \binom{4}{K_3} + \binom{4}{K_6} \right) + \binom{2}{2K_1} + \binom{2}{K_2} - \binom{2}{3K_3} \right) \mathcal{K}_{\theta^+}^{\dagger\dagger 1} \mathcal{K}_{\theta^+}^{\dagger\dagger 1} \dagger + \\
& \frac{1}{2} \left(k^2 \left(\binom{4}{2K_{15}} + \binom{4}{K_{16}} - \binom{4}{2K_2} \right) + \binom{2}{2K_1} + \binom{2}{K_2} - \binom{2}{2K_3} \right) \mathcal{K}_{1^-}^{\dagger\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger\dagger 1} \dagger \alpha + \\
& \left(k^2 \left(\binom{4}{2K_{15}} - \binom{4}{K_{16}} - \binom{4}{K_3} + \binom{4}{K_6} \right) + \binom{2}{2K_1} - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger\dagger 1} \dagger \alpha \beta + \\
& \frac{4}{3} \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger\dagger 1} \dagger \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger\dagger 1} \alpha \beta \chi + \\
& \frac{8}{9} \left(k^2 \left(\binom{4}{2K_{15}} + \binom{4}{K_{16}} \right) + \binom{2}{2K_1} + \binom{2}{K_2} \right) \mathcal{K}_{2^-}^{\dagger\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger\dagger 1} \dagger \alpha \beta \chi + \\
& \frac{4}{3} \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger\dagger 1} \dagger \alpha \beta \chi + \\
& \left(k^2 \left(\binom{4}{2K_{15}} + \binom{4}{K_{16}} - \binom{4}{K_3} - \binom{4}{K_6} \right) + \binom{2}{2K_1} + \binom{2}{K_2} \right) \mathcal{K}_{2^+}^{\dagger\dagger 1} \alpha \beta \mathcal{K}_{2^+}^{\dagger\dagger 1} \dagger \alpha \beta + \\
& \left(k^2 \binom{4}{K_2} + \binom{2}{K_3} \right) \mathcal{K}_{1^-}^{\dagger\dagger 1} \dagger \alpha \mathcal{K}_{1^-}^{\dagger\dagger 2} \alpha + \left(k^2 \left(\binom{4}{K_2} + \binom{4}{K_7} \right) + \binom{2}{K_3} \right) \mathcal{K}_{1^-}^{\dagger\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger\dagger 2} \dagger \alpha + \\
& \left(k^2 \left(\binom{4}{2K_{15}} + \binom{4}{K_{16}} - \binom{4}{K_2} - \binom{4}{K_3} - \binom{4}{K_4} - \binom{4}{K_7} + \binom{4}{2K_9} \right) + \binom{2}{2K_1} + \binom{2}{K_2} - \binom{2}{K_3} \right) \\
& \mathcal{K}_{1^-}^{\dagger\dagger 2} \alpha \mathcal{K}_{1^-}^{\dagger\dagger 2} \dagger \alpha + \left(k^2 \left(-\binom{4}{K_{16}} + \binom{4}{K_4} \right) - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger\dagger 1} \dagger \alpha \beta \mathcal{K}_{1^+}^{\dagger\dagger 2} \alpha \beta + \\
& \left(-k^2 \binom{4}{K_{16}} - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger\dagger 2} \dagger \alpha \beta + \left(k^2 \left(\binom{4}{K_{15}} + \binom{4}{K_9} \right) + \binom{2}{K_1} \right) \mathcal{K}_{1^+}^{\dagger\dagger 2} \alpha \beta \mathcal{K}_{1^+}^{\dagger\dagger 2} \dagger \alpha \beta
\end{aligned}$$

****** xAct`PSALTer`Private`Diagnostic: the value of the variable
xAct`PSALTer`Private`ToMomentumExpr\$167183 will now be printed.

$$\begin{aligned}
& \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger\dagger 1} \alpha \beta \chi \mathcal{K}_{\theta^-}^{\dagger\dagger 1} \dagger \alpha \beta \chi + \\
& \frac{1}{3} \left(-k^2 \left(\binom{4}{3K_1} - \binom{4}{2K_{15}} - \binom{4}{K_{16}} + \binom{4}{3K_2} + \binom{4}{K_3} + \binom{4}{K_6} \right) + \binom{2}{2K_1} + \binom{2}{K_2} - \binom{2}{3K_3} \right) \mathcal{K}_{\theta^+}^{\dagger\dagger 1} \mathcal{K}_{\theta^+}^{\dagger\dagger 1} \dagger + \\
& \frac{1}{2} \left(k^2 \left(\binom{4}{2K_{15}} + \binom{4}{K_{16}} - \binom{4}{2K_2} \right) + \binom{2}{2K_1} + \binom{2}{K_2} - \binom{2}{2K_3} \right) \mathcal{K}_{1^-}^{\dagger\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger\dagger 1} \dagger \alpha + \\
& \left(k^2 \left(\binom{4}{2K_{15}} - \binom{4}{K_{16}} - \binom{4}{K_3} + \binom{4}{K_6} \right) + \binom{2}{2K_1} - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger\dagger 1} \dagger \alpha \beta + \\
& \frac{4}{3} \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger\dagger 1} \dagger \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger\dagger 1} \alpha \beta \chi + \\
& \frac{8}{9} \left(k^2 \left(\binom{4}{2K_{15}} + \binom{4}{K_{16}} \right) + \binom{2}{2K_1} + \binom{2}{K_2} \right) \mathcal{K}_{2^-}^{\dagger\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger\dagger 1} \dagger \alpha \beta \chi + \\
& \frac{4}{3} \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger\dagger 1} \dagger \alpha \beta \chi + \\
& \left(k^2 \left(\binom{4}{2K_{15}} + \binom{4}{K_{16}} - \binom{4}{K_3} - \binom{4}{K_6} \right) + \binom{2}{2K_1} + \binom{2}{K_2} \right) \mathcal{K}_{2^+}^{\dagger\dagger 1} \alpha \beta \mathcal{K}_{2^+}^{\dagger\dagger 1} \dagger \alpha \beta + \\
& \left(k^2 \binom{4}{K_2} + \binom{2}{K_3} \right) \mathcal{K}_{1^-}^{\dagger\dagger 1} \dagger \alpha \mathcal{K}_{1^-}^{\dagger\dagger 2} \alpha + \left(k^2 \left(\binom{4}{K_2} + \binom{4}{K_7} \right) + \binom{2}{K_3} \right) \mathcal{K}_{1^-}^{\dagger\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger\dagger 2} \dagger \alpha + \\
& \left(k^2 \left(\binom{4}{2K_{15}} + \binom{4}{K_{16}} - \binom{4}{K_2} - \binom{4}{K_3} - \binom{4}{K_4} - \binom{4}{K_7} + \binom{4}{2K_9} \right) + \binom{2}{2K_1} + \binom{2}{K_2} - \binom{2}{K_3} \right) \\
& \mathcal{K}_{1^-}^{\dagger\dagger 2} \alpha \mathcal{K}_{1^-}^{\dagger\dagger 2} \dagger \alpha + \left(k^2 \left(-\binom{4}{K_{16}} + \binom{4}{K_4} \right) - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger\dagger 1} \dagger \alpha \beta \mathcal{K}_{1^+}^{\dagger\dagger 2} \alpha \beta + \\
& \left(-k^2 \binom{4}{K_{16}} - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger\dagger 2} \dagger \alpha \beta + \left(k^2 \left(\binom{4}{K_{15}} + \binom{4}{K_9} \right) + \binom{2}{K_1} \right) \mathcal{K}_{1^+}^{\dagger\dagger 2} \alpha \beta \mathcal{K}_{1^+}^{\dagger\dagger 2} \dagger \alpha \beta
\end{aligned}$$

****** xAct`PSALTer`Private`Diagnostic: the value of the variable xAct`PSALTer`Private`\$ParityTranslate will now be printed.

$\langle \mid 1 \rightarrow \text{xAct`PSALTer`Private`Even}, -1 \rightarrow \text{xAct`PSALTer`Private`Odd} \mid \rangle$

****** xAct`PSALTer`Private`Diagnostic:
the value of the variable $\{\}$ will now be printed.

$\{\}$

****** xAct`PSALTer`Private`Diagnostic: the value of the variable xAct`PSALTer`Private`OmittedTensors\$167183 will now be printed.

}

** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`OmittedTensors\$167183 will now be printed.

}

** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`OmittedTensors\$167183 will now be printed.

}

** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`ToMomentumExpr\$167183 will now be printed.

$$\begin{aligned}
& \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{\theta^-}^{\dagger 1} \dagger_{\alpha \beta \chi} + \\
& - \frac{1}{3} \left(-k^2 \left(\binom{4}{3 K_1} - \binom{4}{2 K_{15}} - \binom{4}{K_{16}} + \binom{4}{3 K_2} + \binom{4}{K_3} + \binom{4}{K_6} \right) + \binom{2}{2 K_1} + \binom{2}{K_2} - \binom{2}{3 K_3} \right) \mathcal{K}_{\theta^+}^{\dagger 1} \mathcal{K}_{\theta^+}^{\dagger 1} \dagger + \\
& - \frac{1}{2} \left(k^2 \left(\binom{4}{2 K_{15}} + \binom{4}{K_{16}} - \binom{4}{2 K_2} \right) + \binom{2}{2 K_1} + \binom{2}{K_2} - \binom{2}{2 K_3} \right) \mathcal{K}_{1^-}^{\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger 1} \dagger_{\alpha} + \\
& \left(k^2 \left(\binom{4}{2 K_{15}} - \binom{4}{K_{16}} - \binom{4}{K_3} + \binom{4}{K_6} \right) + \binom{2}{2 K_1} - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 1} \dagger_{\alpha \beta} + \\
& - \frac{4}{3} \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger 1} \dagger_{\alpha \beta \chi} \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi + \\
& - \frac{8}{9} \left(k^2 \left(\binom{4}{2 K_{15}} + \binom{4}{K_{16}} \right) + \binom{2}{2 K_1} + \binom{2}{K_2} \right) \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger_{\alpha \beta \chi} + \\
& - \frac{4}{3} \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger_{\alpha \beta \chi} + \\
& \left(k^2 \left(\binom{4}{2 K_{15}} + \binom{4}{K_{16}} - \binom{4}{K_3} - \binom{4}{K_6} \right) + \binom{2}{2 K_1} + \binom{2}{K_2} \right) \mathcal{K}_{2^+}^{\dagger 1} \alpha \beta \mathcal{K}_{2^+}^{\dagger 1} \dagger_{\alpha \beta} + \\
& \left(k^2 \binom{4}{K_2} + \binom{2}{K_3} \right) \mathcal{K}_{1^-}^{\dagger 1} \dagger_{\alpha} \mathcal{K}_{1^-}^{\dagger 2} \alpha + \left(k^2 \left(\binom{4}{K_2} + \binom{4}{K_7} \right) + \binom{2}{K_3} \right) \mathcal{K}_{1^-}^{\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger 2} \dagger_{\alpha} + \\
& \left(k^2 \left(\binom{4}{2 K_{15}} + \binom{4}{K_{16}} - \binom{4}{K_2} - \binom{4}{K_3} - \binom{4}{K_4} - \binom{4}{K_7} + \binom{4}{2 K_9} \right) + \binom{2}{2 K_1} + \binom{2}{K_2} - \binom{2}{K_3} \right) \\
& \mathcal{K}_{1^-}^{\dagger 2} \alpha \mathcal{K}_{1^-}^{\dagger 2} \dagger_{\alpha} + \left(k^2 \left(-\binom{4}{K_{16}} + \binom{4}{K_4} \right) - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger 1} \dagger_{\alpha \beta} \mathcal{K}_{1^+}^{\dagger 2} \alpha \beta + \\
& \left(-k^2 \binom{4}{K_{16}} - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 2} \dagger_{\alpha \beta} + \left(k^2 \left(\binom{4}{K_{15}} + \binom{4}{K_9} \right) + \binom{2}{K_1} \right) \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 2} \dagger_{\alpha \beta}
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`DecomposeFieldsdLagrangian will now be printed.

$$\begin{aligned}
& \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{\theta^-}^{\dagger 1} \dagger \alpha \beta \chi + \\
& \frac{1}{3} \left(-k^2 \left(\binom{4}{3 K_1} - \binom{4}{2 K_{15}} - \binom{4}{K_{16}} + \binom{4}{3 K_2} + \binom{4}{K_3} + \binom{4}{K_6} \right) + \binom{2}{2 K_1} + \binom{2}{K_2} - \binom{2}{3 K_3} \right) \mathcal{K}_{\theta^+}^{\dagger 1} \mathcal{K}_{\theta^+}^{\dagger 1} \dagger + \\
& \frac{1}{2} \left(k^2 \left(\binom{4}{2 K_{15}} + \binom{4}{K_{16}} - \binom{4}{2 K_2} \right) + \binom{2}{2 K_1} + \binom{2}{K_2} - \binom{2}{2 K_3} \right) \mathcal{K}_{1^-}^{\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger 1} \dagger \alpha + \\
& \left(k^2 \left(\binom{4}{2 K_{15}} - \binom{4}{K_{16}} - \binom{4}{K_3} + \binom{4}{K_6} \right) + \binom{2}{2 K_1} - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 1} \dagger \alpha \beta + \\
& \frac{4}{3} \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger 1} \dagger \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi + \\
& \frac{8}{9} \left(k^2 \left(\binom{4}{2 K_{15}} + \binom{4}{K_{16}} \right) + \binom{2}{2 K_1} + \binom{2}{K_2} \right) \mathcal{K}_{2^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger \alpha \beta \chi + \\
& \frac{4}{3} \left(k^2 \left(\binom{4}{K_{15}} - \binom{4}{K_{16}} \right) + \binom{2}{K_1} - \binom{2}{K_2} \right) \mathcal{K}_{\theta^-}^{\dagger 1} \alpha \beta \chi \mathcal{K}_{2^-}^{\dagger 1} \dagger \alpha \beta \chi + \\
& \left(k^2 \left(\binom{4}{2 K_{15}} + \binom{4}{K_{16}} - \binom{4}{K_3} - \binom{4}{K_6} \right) + \binom{2}{2 K_1} + \binom{2}{K_2} \right) \mathcal{K}_{2^+}^{\dagger 1} \alpha \beta \mathcal{K}_{2^+}^{\dagger 1} \dagger \alpha \beta + \\
& \left(k^2 \binom{4}{K_2} + \binom{2}{K_3} \right) \mathcal{K}_{1^-}^{\dagger 1} \dagger \alpha \mathcal{K}_{1^-}^{\dagger 2} \alpha + \left(k^2 \left(\binom{4}{K_2} + \binom{4}{K_7} \right) + \binom{2}{K_3} \right) \mathcal{K}_{1^-}^{\dagger 1} \alpha \mathcal{K}_{1^-}^{\dagger 2} \dagger \alpha + \\
& \left(k^2 \left(\binom{4}{2 K_{15}} + \binom{4}{K_{16}} - \binom{4}{K_2} - \binom{4}{K_3} - \binom{4}{K_4} - \binom{4}{K_7} + \binom{4}{2 K_9} \right) + \binom{2}{2 K_1} + \binom{2}{K_2} - \binom{2}{K_3} \right) \\
& \mathcal{K}_{1^-}^{\dagger 2} \alpha \mathcal{K}_{1^-}^{\dagger 2} \dagger \alpha + \left(k^2 \left(-\binom{4}{K_{16}} + \binom{4}{K_4} \right) - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger 1} \dagger \alpha \beta \mathcal{K}_{1^+}^{\dagger 2} \alpha \beta + \\
& \left(-k^2 \binom{4}{K_{16}} - \binom{2}{K_2} \right) \mathcal{K}_{1^+}^{\dagger 1} \alpha \beta \mathcal{K}_{1^+}^{\dagger 2} \dagger \alpha \beta + \left(k^2 \left(\binom{4}{K_{15}} + \binom{4}{K_9} \right) + \binom{2}{K_1} \right) \mathcal{K}_{1^+}^{\dagger 2} \alpha \beta \mathcal{K}_{1^+}^{\dagger 2} \dagger \alpha \beta
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the
variable xAct`PSALter`Private`CouplingAssumptions will now be printed.

$$\left\{ \binom{4}{K_1} \in \mathbb{R}, \binom{4}{K_{15}} \in \mathbb{R}, \binom{4}{K_{16}} \in \mathbb{R}, \binom{4}{K_2} \in \mathbb{R}, \binom{4}{K_3} \in \mathbb{R}, \right. \\
\left. \binom{4}{K_4} \in \mathbb{R}, \binom{4}{K_6} \in \mathbb{R}, \binom{4}{K_7} \in \mathbb{R}, \binom{4}{K_9} \in \mathbb{R}, \binom{2}{K_1} \in \mathbb{R}, \binom{2}{K_2} \in \mathbb{R}, \binom{2}{K_3} \in \mathbb{R} \right\}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable

xAct`PSALter`Private`Class\$173389[xAct`PSALter`Private`InvariantToConstantRules]
will now be printed.

$$\left\{ \text{HoldPattern}\left[\mathcal{K}_{\theta^+}^{\dagger\dagger_1} \mathcal{K}_{\theta^+}^{\dagger\dagger_1} \dagger\right] \Rightarrow \text{Module}\left[\{\},\right.$$

$$\quad \text{Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$$

$$\text{HoldPattern}\left[\mathcal{J}_{\theta^+}^{\dagger\dagger_1} \mathcal{J}_{\theta^+}^{\dagger\dagger_1} \dagger\right] \Rightarrow \text{Module}\left[\{\},\right.$$

$$\quad \text{Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$$

$$\text{HoldPattern}\left[\epsilon^{\parallel \underline{\underline{\alpha\beta\chi}}} \mathcal{K}_{\theta^-}^{\dagger\dagger_1} \underline{\underline{\alpha\beta\chi}} \mathcal{K}_{\theta^+}^{\dagger\dagger_1} \dagger\right] \Rightarrow \text{Module}\left[\{\},\right.$$

$$\quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$$

$$\text{HoldPattern}\left[\epsilon^{\parallel \underline{\underline{\alpha\beta}}} \mathcal{K}_{\theta^-}^{\dagger\dagger_1} \underline{\underline{\chi}} \mathcal{K}_{\theta^+}^{\dagger\dagger_1} \dagger\right] \Rightarrow \text{Module}\left[\{\},\right.$$

$$\quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$$

$$\text{HoldPattern}\left[\epsilon^{\parallel \underline{\underline{\alpha}}} \mathcal{K}_{\theta^-}^{\dagger\dagger_1} \underline{\underline{\beta}} \mathcal{K}_{\theta^+}^{\dagger\dagger_1} \dagger\right] \Rightarrow \text{Module}\left[\{\},\right.$$

$$\quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$$

$$\text{HoldPattern}\left[\epsilon^{\parallel \underline{\underline{\alpha}}} \mathcal{K}_{\theta^-}^{\dagger\dagger_1} \underline{\underline{\beta\chi}} \mathcal{K}_{\theta^+}^{\dagger\dagger_1} \dagger\right] \Rightarrow \text{Module}\left[\{\},\right.$$

$$\quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$$

$$\text{HoldPattern}\left[\epsilon^{\parallel \underline{\underline{\alpha}}} \mathcal{K}_{\theta^-}^{\dagger\dagger_1} \underline{\underline{\beta\chi}} \mathcal{K}_{\theta^+}^{\dagger\dagger_1} \dagger\right] \Rightarrow \text{Module}\left[\{\},\right.$$

$$\quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$$

$$\text{HoldPattern}\left[\epsilon^{\parallel \underline{\underline{\alpha}}} \mathcal{K}_{\theta^-}^{\dagger\dagger_1} \underline{\underline{\beta\chi}} \mathcal{K}_{\theta^+}^{\dagger\dagger_1} \dagger\right] \Rightarrow \text{Module}\left[\{\},\right.$$

$$\quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$$

$$\text{HoldPattern}\left[\epsilon^{\parallel \underline{\underline{\alpha\beta\chi}}} \mathcal{K}_{\theta^-}^{\dagger\dagger_1} \underline{\underline{\alpha\beta\chi}} \mathcal{K}_{\theta^+}^{\dagger\dagger_1} \dagger\right] \Rightarrow \text{Module}\left[\{\},\right.$$

$$\quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$$

$$\text{HoldPattern}\left[\mathcal{K}_{\theta^-}^{\dagger\dagger_1} \underline{\underline{\beta\alpha\chi}} \epsilon^{\parallel \underline{\underline{\alpha\beta\chi}}} \mathcal{K}_{\theta^+}^{\dagger\dagger_1} \dagger\right] \Rightarrow \text{Module}\left[\{\},\right.$$

$$\quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],$$

[illegible]

[illegible]

$$\begin{aligned}
& \text{HoldPattern}\left[\mathcal{K}_{0^{-}}^{\dagger\dagger_1}\underline{\beta}_\alpha^\chi \epsilon^{\parallel\underline{\beta}}_{\underline{\alpha}\chi}\mathcal{K}_{0^{+}}^{\dagger\dagger_1}\right] \Rightarrow \text{Module}[\{\}, \\
& \quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\mathcal{K}_{0^{-}}^{\dagger\dagger_1}\underline{\beta}_\chi^\alpha \epsilon^{\parallel\underline{\beta}}_{\underline{\alpha}}\mathcal{K}_{0^{+}}^{\dagger\dagger_1}\right] \Rightarrow \text{Module}[\{\}, \\
& \quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\mathcal{K}_{0^{-}}^{\dagger\dagger_1}\underline{\chi}\beta_\alpha \epsilon^{\parallel\underline{\beta}}_{\underline{\alpha}\chi}\mathcal{K}_{0^{+}}^{\dagger\dagger_1}\right] \Rightarrow \text{Module}[\{\}, \\
& \quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\epsilon^{\parallel\underline{\alpha}\chi}_\beta \mathcal{K}_{0^{-}}^{\dagger\dagger_1}\underline{\alpha}\beta_\chi \mathcal{K}_{0^{+}}^{\dagger\dagger_1}\right] \Rightarrow \text{Module}[\{\}, \\
& \quad \neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\epsilon^{\parallel\underline{\alpha}}_\chi^\beta \mathcal{K}_{0^{-}}^{\dagger\dagger_1}\underline{\alpha}\beta_\chi \mathcal{K}_{0^{+}}^{\dagger\dagger_1}\right] \Rightarrow \text{Module}[\{\}, \\
& \quad \neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\epsilon^{\parallel\underline{\alpha}\chi}_\beta \mathcal{K}_{0^{-}}^{\dagger\dagger_1}\underline{\beta}_\alpha^\chi \mathcal{K}_{0^{+}}^{\dagger\dagger_1}\right] \Rightarrow \text{Module}[\{\}, \\
& \quad \neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\epsilon^{\parallel\underline{\alpha}}_\chi^\beta \mathcal{K}_{0^{-}}^{\dagger\dagger_1}\underline{\beta}_\chi^\alpha \mathcal{K}_{0^{+}}^{\dagger\dagger_1}\right] \Rightarrow \text{Module}[\{\}, \\
& \quad \neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\epsilon^{\parallel\underline{\chi}\beta}_\alpha \mathcal{K}_{0^{-}}^{\dagger\dagger_1}\underline{\alpha}_\beta^\chi \mathcal{K}_{0^{+}}^{\dagger\dagger_1}\right] \Rightarrow \text{Module}[\{\}, \\
& \quad \neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\epsilon^{\parallel\underline{\alpha}\chi}_\beta \mathcal{K}_{0^{-}}^{\dagger\dagger_1}\underline{\alpha}_\beta^\chi \mathcal{K}_{0^{+}}^{\dagger\dagger_1}\right] \Rightarrow \text{Module}[\{\}, \\
& \quad \neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\epsilon^{\parallel\underline{\chi}}_\alpha^\beta \mathcal{K}_{0^{-}}^{\dagger\dagger_1}\underline{\alpha}\beta_\chi \mathcal{K}_{0^{+}}^{\dagger\dagger_1}\right] \Rightarrow \text{Module}[\{\}, \\
& \quad \neg \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\mathcal{K}_{0^{-}}^{\dagger\dagger_1}\underline{\beta}_\alpha^\chi \epsilon^{\parallel\underline{\alpha}\chi}_\beta \mathcal{K}_{0^{+}}^{\dagger\dagger_1}\right] \Rightarrow \text{Module}[\{\}, \\
& \quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0pConstantSymbol } \dagger],
\end{aligned}$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\begin{aligned}
& \text{HoldPattern}\left[\mathcal{K}_{\ominus^{-}}^{\sharp_1} +_{\underline{\beta}}^{\underline{\alpha X}} \mathcal{K}_{\ominus^{-}}^{\sharp_1} =_{\underline{\underline{\alpha X}}}^{\beta}\right] := \text{Module}[\{\}, \\
& \quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\mathcal{K}_{\ominus^{-}}^{\sharp_1} +_{\underline{\underline{X}}}^{\frac{\alpha}{X}} \frac{\beta}{\underline{\underline{X}}} \mathcal{K}_{\ominus^{-}}^{\sharp_1} =_{\underline{\underline{\beta \alpha}}}^{\underline{X}}\right] := \text{Module}[\{\}, \\
& \quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\mathcal{K}_{\ominus^{-}}^{\sharp_1} +_{\underline{\underline{\alpha X \beta}}}^{\underline{\alpha X \beta}} \mathcal{K}_{\ominus^{-}}^{\sharp_1} =_{\underline{\underline{\beta \alpha X}}}^{\underline{\beta \alpha X}}\right] := \text{Module}[\{\}, \\
& \quad \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\mathcal{K}_{\ominus^{-}}^{\sharp_1} =_{\underline{\underline{\beta \alpha X}}}^{\underline{\beta \alpha X}} \mathcal{K}_{\ominus^{-}}^{\sharp_1} +_{\underline{\underline{\beta X \alpha}}}^{\underline{\beta X \alpha}}\right] := \text{Module}[\{\}, \\
& \quad - \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\mathcal{K}_{\ominus^{-}}^{\sharp_1} =_{\underline{\underline{X}}}^{\underline{\beta \alpha}} \mathcal{K}_{\ominus^{-}}^{\sharp_1} +_{\underline{\underline{\beta}}^{\underline{\underline{X}}}}^{\underline{\underline{X}}}\right] := \text{Module}[\{\}, \\
& \quad - \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\mathcal{K}_{\ominus^{-}}^{\sharp_1} =_{\underline{\underline{\beta}}^{\underline{\underline{\alpha X}}}}^{\underline{\underline{\alpha X}}} \mathcal{K}_{\ominus^{-}}^{\sharp_1} +_{\underline{\underline{\underline{X \alpha}}}}^{\underline{\underline{\beta}}}\right] := \text{Module}[\{\}, \\
& \quad - \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\mathcal{K}_{\ominus^{-}}^{\sharp_1} =_{\underline{\underline{\beta}}^{\underline{\underline{\alpha}}}}^{\underline{\underline{\alpha}}} \mathcal{K}_{\ominus^{-}}^{\sharp_1} +_{\underline{\underline{\underline{\beta X}}}}^{\underline{\underline{\beta X}}}\right] := \text{Module}[\{\}, \\
& \quad - \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\mathcal{K}_{\ominus^{-}}^{\sharp_1} =_{\underline{\underline{\underline{\alpha X}}}}^{\underline{\underline{\beta}}} \mathcal{K}_{\ominus^{-}}^{\sharp_1} +_{\underline{\underline{\underline{\beta}}}}^{\underline{\underline{\underline{X \alpha}}}}\right] := \text{Module}[\{\}, \\
& \quad - \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\mathcal{K}_{\ominus^{-}}^{\sharp_1} =_{\underline{\underline{\underline{\beta \alpha}}}}^{\underline{\underline{X}}} \mathcal{K}_{\ominus^{-}}^{\sharp_1} +_{\underline{\underline{\underline{X}}}}^{\underline{\underline{\underline{\beta}}}} \frac{\underline{\underline{\alpha}}}{\underline{\underline{\underline{X}}}}\right] := \text{Module}[\{\}, \\
& \quad - \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\mathcal{K}_{\ominus^{-}}^{\sharp_1} =_{\underline{\underline{\underline{\beta \alpha X}}}}^{\underline{\underline{\beta \alpha X}}} \mathcal{K}_{\ominus^{-}}^{\sharp_1} +_{\underline{\underline{\underline{\beta X \alpha}}}}^{\underline{\underline{\beta X \alpha}}}\right] := \text{Module}[\{\}, \\
& \quad - \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol } \dagger], \\
& \text{HoldPattern}\left[\mathcal{K}_{\ominus^{-}}^{\sharp_1} =_{\underline{\underline{\underline{\beta \alpha X}}}}^{\underline{\underline{\beta \alpha X}}} \mathcal{K}_{\ominus^{-}}^{\sharp_1} +_{\underline{\underline{\underline{X \alpha \beta}}}}^{\underline{\underline{X \alpha \beta}}}\right] := \text{Module}[\{\}, \\
& \quad - \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol } \dagger]
\end{aligned}$$

[illegible]

[illegible]

[illegible]

[illegible]


```

HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\alpha\chi}}} \mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\beta}}_{\underline{\underline{\alpha\chi}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi}}}^{\underline{\underline{\alpha}} \underline{\underline{\beta}}} \mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\beta\alpha}}_{\underline{\underline{\chi}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\alpha\chi\beta}}} \mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\beta\alpha\chi}}$ ] :> Module[{ },
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\beta\alpha\chi}}_{\underline{\underline{\beta}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta\chi\alpha}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\beta\alpha}}_{\underline{\underline{\chi}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\chi}} \underline{\underline{\alpha}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\beta}}_{\underline{\underline{\alpha}}}^{\underline{\underline{\chi}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta\chi\alpha}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\beta}}_{\underline{\underline{\alpha}}}^{\underline{\underline{\chi}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\chi\alpha}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\beta\alpha}}_{\underline{\underline{\alpha}}}^{\underline{\underline{\chi}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta\chi\alpha}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\beta\alpha\chi}}_{\underline{\underline{\alpha}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi\alpha\beta}}}$ ] :> Module[{ },
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],

```



```

HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\alpha\chi}}} \mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\alpha\chi}}_{\underline{\underline{\beta}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi}}}^{\underline{\underline{\alpha}} \underline{\underline{\beta}}} \mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\chi}}_{\underline{\underline{\alpha}} \underline{\underline{\beta}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\alpha\chi\beta}}}^{\underline{\underline{\alpha\chi\beta}}} \mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\alpha\chi\beta}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\alpha\chi\beta}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta\chi\alpha}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\alpha}}_{\underline{\underline{\chi\beta}}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\chi}} \underline{\underline{\alpha}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\alpha\chi}}_{\underline{\underline{\beta}}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi\alpha}}}^{\underline{\underline{\beta}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\alpha}}_{\underline{\underline{\chi\beta}}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\alpha}}}^{\underline{\underline{\beta\chi}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\chi\beta}}_{\underline{\underline{\alpha}}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta\chi}}}^{\underline{\underline{\alpha}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\alpha\chi}}_{\underline{\underline{\beta}}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\chi\alpha}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\chi}}_{\underline{\underline{\alpha}} \underline{\underline{\beta}}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi}}}^{\underline{\underline{\beta}} \underline{\underline{\alpha}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\alpha\chi\beta}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi\alpha\beta}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],

```



```

HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\alpha\chi}}} \mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\chi\alpha}}^{\underline{\underline{\beta}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi}}}^{\underline{\underline{\alpha}} \underline{\underline{\beta}}} \mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\alpha\beta}}^{\underline{\underline{\chi}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\alpha\chi\beta}}}^{\underline{\underline{\alpha\chi\beta}}} \mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\chi\alpha\beta}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\chi\alpha\beta}}^{\underline{\underline{\chi\alpha\beta}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta\chi\alpha}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\chi}}^{\underline{\underline{\alpha\beta}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\chi}} \underline{\underline{\alpha}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\chi\alpha}}^{\underline{\underline{\chi\alpha}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi\alpha}}}^{\underline{\underline{\beta}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\chi}}^{\underline{\underline{\alpha}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\alpha}}}^{\underline{\underline{\beta\chi}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\chi}}^{\underline{\underline{\beta}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta\chi}}}^{\underline{\underline{\alpha}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\chi\alpha}}^{\underline{\underline{\beta}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\chi\alpha}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\chi\alpha\beta}}^{\underline{\underline{\chi\alpha\beta}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi\alpha\beta}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{\theta^-}^{\sharp_1} \underline{\underline{\chi\alpha\beta}}^{\underline{\underline{\chi\alpha\beta}}} \mathcal{J}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi\alpha\beta}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],

```



```

HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\alpha\chi}}} \mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\chi}}_{\underline{\underline{\alpha}}}^{\underline{\underline{\beta}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi}}}^{\underline{\underline{\alpha}}} \underline{\underline{\beta}} \mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\chi}}_{\underline{\underline{\beta\alpha}}}^{\underline{\underline{\chi}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\alpha}}}^{\underline{\underline{\alpha\chi\beta}}} \mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\chi\beta\alpha}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\chi\beta\alpha}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta\chi\alpha}}}^{\underline{\underline{\beta}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\chi}}_{\underline{\underline{\beta}}}^{\underline{\underline{\beta\alpha}}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\underline{\chi}}} \underline{\underline{\alpha}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\chi}}_{\underline{\underline{\beta}}}^{\underline{\underline{\alpha}}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi\alpha}}}^{\underline{\underline{\beta}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\chi\beta}}_{\underline{\underline{\alpha}}}^{\underline{\underline{\alpha}}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\alpha}}}^{\underline{\underline{\beta\chi}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\chi\beta}}_{\underline{\underline{\alpha}}}^{\underline{\underline{\chi\beta}}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\beta\chi}}}^{\underline{\underline{\alpha}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\chi}}_{\underline{\underline{\beta\alpha}}}^{\underline{\underline{\beta}}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi}}}^{\underline{\underline{\beta}}} \underline{\underline{\alpha}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{T}_{\theta^-}^{\sharp_1} \underline{\underline{\chi\beta\alpha}} \mathcal{T}_{\theta^-}^{\sharp_1} \dagger_{\underline{\underline{\chi\alpha\beta}}}^{\underline{\underline{\beta\chi\alpha}}}$ ] :> Module[{},
  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol  $\dagger$ ],

```



```

HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_1 \underline{\beta} \underline{\alpha}} \mathcal{J}_{1^+}^{\sharp_1} \dagger_{\underline{\alpha} \underline{\beta}}$ ] := Module[{} ,
   $\neg$  Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_1} \underline{\underline{\beta}}^{\underline{\alpha}} \mathcal{J}_{1^+}^{\sharp_1} \dagger_{\underline{\alpha}}^{\underline{\beta}}$ ] := Module[{} ,
   $\neg$  Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_1} \underline{\underline{\beta}}_{\underline{\alpha}} \mathcal{J}_{1^+}^{\sharp_1} \dagger^{\underline{\alpha}}_{\underline{\beta}}$ ] := Module[{} ,
   $\neg$  Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_1} \underline{\underline{\beta} \underline{\alpha}} \mathcal{J}_{1^+}^{\sharp_1} \dagger^{\underline{\alpha} \underline{\beta}}$ ] := Module[{} ,
   $\neg$  Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_1} \dagger_{\underline{\underline{\beta} \underline{\alpha}}} \mathcal{J}_{1^+}^{\sharp_1} \underline{\underline{\beta} \underline{\alpha}}$ ] := Module[{} ,
  Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\beta}} \mathcal{J}_{1^+}^{\sharp_1} \underline{\underline{\alpha}}_{\underline{\beta}}$ ] := Module[{} ,
  Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\alpha}} \mathcal{J}_{1^+}^{\sharp_1} \underline{\underline{\beta}}_{\underline{\alpha}}$ ] := Module[{} ,
  Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_1} \dagger_{\underline{\underline{\beta} \underline{\alpha}}}^{\underline{\beta} \underline{\alpha}} \mathcal{J}_{1^+}^{\sharp_1} \underline{\underline{\beta} \underline{\alpha}}$ ] := Module[{} ,
  Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\alpha} \underline{\beta}} \mathcal{J}_{1^+}^{\sharp_2} \underline{\underline{\alpha} \underline{\beta}}$ ] := Module[{} ,
  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_1} \dagger_{\underline{\underline{\beta}}}^{\underline{\alpha}} \mathcal{J}_{1^+}^{\sharp_2} \underline{\underline{\beta}}_{\underline{\alpha}}$ ] := Module[{} ,
  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_1} \dagger_{\underline{\underline{\alpha}}}^{\underline{\beta}} \mathcal{J}_{1^+}^{\sharp_2} \underline{\underline{\alpha}}_{\underline{\beta}}$ ] := Module[{} ,
  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_2} \underline{\underline{\beta} \underline{\alpha}} \mathcal{J}_{1^+}^{\sharp_1} \dagger^{\underline{\alpha} \underline{\beta}}$ ] := Module[{} ,

```

```

    -Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \underline{\underline{\beta}} \underline{\underline{\alpha}} \mathcal{T}_{1^+}^{\#1} \dagger \underline{\underline{\alpha}} \underline{\underline{\beta}}$ ] :> Module[{},
    -Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \underline{\underline{\alpha}} \underline{\underline{\beta}} \mathcal{T}_{1^+}^{\#1} \dagger \underline{\underline{\beta}} \underline{\underline{\alpha}}$ ] :> Module[{},
    -Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \underline{\underline{\beta \alpha}} \mathcal{T}_{1^+}^{\#1} \dagger \underline{\underline{\alpha \beta}}$ ] :> Module[{},
    -Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#1} \dagger \underline{\underline{\beta \alpha}} \mathcal{T}_{1^+}^{\#2} \underline{\underline{\alpha \beta}}$ ] :> Module[{},
    -Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#1} \dagger \underline{\underline{\beta}} \underline{\underline{\alpha}} \mathcal{T}_{1^+}^{\#2} \underline{\underline{\beta}} \underline{\underline{\alpha}}$ ] :> Module[{},
    -Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#1} \dagger \underline{\underline{\beta}} \underline{\underline{\alpha}} \mathcal{T}_{1^+}^{\#2} \underline{\underline{\alpha}} \underline{\underline{\beta}}$ ] :> Module[{},
    -Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \underline{\underline{\beta \alpha}} \mathcal{T}_{1^+}^{\#1} \dagger \underline{\underline{\beta \alpha}}$ ] :> Module[{},
    Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \underline{\underline{\beta}} \underline{\underline{\alpha}} \mathcal{T}_{1^+}^{\#1} \dagger \underline{\underline{\beta}} \underline{\underline{\alpha}}$ ] :> Module[{},
    Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \underline{\underline{\alpha}} \underline{\underline{\beta}} \mathcal{T}_{1^+}^{\#1} \dagger \underline{\underline{\alpha}} \underline{\underline{\beta}}$ ] :> Module[{},
    Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#2} \underline{\underline{\beta \alpha}} \mathcal{T}_{1^+}^{\#1} \dagger \underline{\underline{\beta \alpha}}$ ] :> Module[{},
    Rank3AntisymmetricPara1pConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}_{1^+}^{\#1} \underline{\underline{\alpha \beta}} \mathcal{T}_{1^+}^{\#2} \dagger \underline{\underline{\alpha \beta}}$ ] :> Module[{},
    Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol †],

```


[illegible]

```

HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_2 \underline{\beta}} \mathcal{J}_{1^+}^{\sharp_2} \dagger \underline{\underline{\alpha}}_{\underline{\beta}}$ ] := Module[{} ,
   $\neg$ Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_2} \underline{\underline{\beta \alpha}} \mathcal{J}_{1^+}^{\sharp_2} \dagger \underline{\underline{\alpha \beta}}$ ] := Module[{} ,
   $\neg$ Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_2} \dagger \underline{\underline{\beta \alpha}} \mathcal{J}_{1^+}^{\sharp_2} \underline{\underline{\beta \alpha}}$ ] := Module[{} ,
  Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_2} \dagger \underline{\underline{\beta}}_{\underline{\alpha}} \mathcal{J}_{1^+}^{\sharp_2} \underline{\underline{\alpha}}_{\underline{\beta}}$ ] := Module[{} ,
  Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_2} \dagger \underline{\underline{\alpha}}_{\underline{\beta}} \mathcal{J}_{1^+}^{\sharp_2} \underline{\underline{\beta}}_{\underline{\alpha}}$ ] := Module[{} ,
  Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\mathcal{J}_{1^+}^{\sharp_2} \dagger \underline{\underline{\beta \alpha}} \mathcal{J}_{1^+}^{\sharp_2} \underline{\underline{\beta \alpha}}$ ] := Module[{} ,
  Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha \beta \chi}}} \mathcal{K}_{1^-}^{\sharp_1 \underline{\alpha}} \mathcal{K}_{1^+}^{\sharp_1} \dagger \underline{\underline{\beta \chi}}$ ] := Module[{} ,
  Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha \beta}}} \mathcal{K}_{1^-}^{\sharp_1 \underline{\alpha}} \mathcal{K}_{1^+}^{\sharp_1} \dagger \underline{\underline{\beta \chi}}$ ] := Module[{} ,
  Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha}}_{\underline{\chi}}} \mathcal{K}_{1^-}^{\sharp_1 \underline{\alpha}} \mathcal{K}_{1^+}^{\sharp_1} \dagger \underline{\underline{\beta \chi}}$ ] := Module[{} ,
  Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\beta \chi}}} \mathcal{K}_{1^-}^{\sharp_1 \underline{\alpha}} \mathcal{K}_{1^+}^{\sharp_1} \dagger \underline{\underline{\beta \chi}}$ ] := Module[{} ,
  Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha}}_{\underline{\beta \chi}}} \mathcal{K}_{1^-}^{\sharp_1} \mathcal{K}_{1^+}^{\sharp_1} \dagger \underline{\underline{\beta \chi}}$ ] := Module[{} ,
  Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],
HoldPattern[ $\epsilon^{\parallel \underline{\underline{\alpha}}_{\underline{\beta}}} \mathcal{K}_{1^-}^{\sharp_1} \mathcal{K}_{1^+}^{\sharp_1} \dagger \underline{\underline{\beta \chi}}$ ] := Module[{} ,
  Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1pConstantSymbol  $\dagger$ ],

```


[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$\text{HoldPattern}\left[\epsilon^{\parallel \underline{\beta} \underline{\alpha}}_{\underline{\chi}} \mathcal{K}^{\sharp_2}_{1^-} \underline{\alpha} \mathcal{K}^{\sharp_2}_{1^+} \dagger_{\underline{\beta}}^{\underline{\chi}}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \underline{\beta \chi \alpha}} \mathcal{K}^{\sharp_2}_{1^-} \underline{\alpha} \mathcal{K}^{\sharp_2}_{1^+} \dagger_{\underline{\beta \chi}}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{K}^{\sharp_2}_{1^+} \dagger_{\underline{\beta}}^{\underline{\chi}} \epsilon^{\parallel \underline{\beta \chi \alpha}} \mathcal{K}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{K}^{\sharp_2}_{1^+} \dagger_{\underline{\chi}}^{\underline{\beta}} \epsilon^{\parallel \underline{\beta}}_{\underline{\alpha}} \mathcal{K}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{K}^{\sharp_2}_{1^+} \dagger_{\underline{\beta}}^{\underline{\chi}} \epsilon^{\parallel \underline{\beta}}_{\underline{\chi \alpha}} \mathcal{K}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{K}^{\sharp_2}_{1^+} \dagger_{\underline{\chi \beta}} \epsilon^{\parallel \underline{\beta \chi}}_{\underline{\alpha}} \mathcal{K}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{K}^{\sharp_2}_{1^+} \dagger_{\underline{\chi}}^{\underline{\beta}} \epsilon^{\parallel \underline{\beta \chi \alpha}} \mathcal{K}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{K}^{\sharp_2}_{1^+} \dagger_{\underline{\beta}}^{\underline{\chi}} \epsilon^{\parallel \underline{\beta}}_{\underline{\chi}} \mathcal{K}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{K}^{\sharp_2}_{1^+} \dagger_{\underline{\chi \beta}} \epsilon^{\parallel \underline{\beta \chi \alpha}} \mathcal{K}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \underline{\chi \alpha \beta}} \mathcal{K}^{\sharp_2}_{1^-} \underline{\alpha} \mathcal{K}^{\sharp_2}_{1^+} \dagger_{\underline{\beta}}^{\underline{\chi}}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \underline{\chi}}_{\underline{\alpha \beta}} \mathcal{K}^{\sharp_2}_{1^-} \underline{\alpha} \mathcal{K}^{\sharp_2}_{1^+} \dagger_{\underline{\chi}}^{\underline{\beta}}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$

[illegible]

$\text{HoldPattern}\left[\epsilon^{\parallel \underline{\beta} \underline{\alpha}}_{\underline{\chi}} \mathcal{J}^{\sharp_1}_{1^-} \underline{\alpha} \mathcal{J}^{\sharp_1}_{1^+} \dagger_{\underline{\beta}}^{\underline{\chi}}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricParaImConstantSymbol Rank3AntisymmetricParaIpConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \underline{\beta \chi \alpha}} \mathcal{J}^{\sharp_1}_{1^-} \underline{\alpha} \mathcal{J}^{\sharp_1}_{1^+} \dagger_{\underline{\beta \chi}}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricParaImConstantSymbol Rank3AntisymmetricParaIpConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_1}_{1^+} \dagger_{\underline{\chi}}^{\underline{\beta}} \epsilon^{\parallel \underline{\beta \chi \alpha}} \mathcal{J}^{\sharp_1}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricParaImConstantSymbol Rank3AntisymmetricParaIpConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_1}_{1^+} \dagger_{\underline{\chi}}^{\underline{\beta}} \epsilon^{\parallel \underline{\chi}}_{\underline{\beta}} \underline{\alpha} \mathcal{J}^{\sharp_1}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricParaImConstantSymbol Rank3AntisymmetricParaIpConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_1}_{1^+} \dagger_{\underline{\beta}}^{\underline{\chi}} \epsilon^{\parallel \underline{\beta}}_{\underline{\chi \alpha}} \mathcal{J}^{\sharp_1}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricParaImConstantSymbol Rank3AntisymmetricParaIpConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_1}_{1^+} \dagger_{\underline{\chi \beta}} \epsilon^{\parallel \underline{\beta \chi}}_{\underline{\alpha}} \mathcal{J}^{\sharp_1}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricParaImConstantSymbol Rank3AntisymmetricParaIpConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_1}_{1^+} \dagger_{\underline{\chi}}^{\underline{\beta}} \epsilon^{\parallel \underline{\chi \alpha}}_{\underline{\beta}} \mathcal{J}^{\sharp_1}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricParaImConstantSymbol Rank3AntisymmetricParaIpConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_1}_{1^+} \dagger_{\underline{\beta}}^{\underline{\chi}} \epsilon^{\parallel \underline{\beta}}_{\underline{\chi}} \underline{\alpha} \mathcal{J}^{\sharp_1}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricParaImConstantSymbol Rank3AntisymmetricParaIpConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_1}_{1^+} \dagger_{\underline{\chi \beta}} \epsilon^{\parallel \underline{\beta \chi \alpha}} \mathcal{J}^{\sharp_1}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricParaImConstantSymbol Rank3AntisymmetricParaIpConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \underline{\chi \alpha \beta}} \mathcal{J}^{\sharp_1}_{1^-} \underline{\alpha} \mathcal{J}^{\sharp_1}_{1^+} \dagger_{\underline{\beta \chi}}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricParaImConstantSymbol Rank3AntisymmetricParaIpConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \underline{\chi}}_{\underline{\alpha \beta}} \mathcal{J}^{\sharp_1}_{1^-} \underline{\alpha} \mathcal{J}^{\sharp_1}_{1^+} \dagger_{\underline{\beta}}^{\underline{\chi}}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricParaImConstantSymbol Rank3AntisymmetricParaIpConstantSymbol } \dagger],$

[illegible]

[illegible]


```

HoldPattern[ $\epsilon^{\parallel}_{\underline{\chi}\underline{\alpha}}^{\underline{\beta}} \mathcal{T}^{\sharp_1}_{1^+} \dagger^{\underline{\chi}}_{\underline{\beta}} \mathcal{T}^{\sharp_2}_{1^-} \underline{\alpha}$ ] := Module[{},
  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1mConstantSymbol],
HoldPattern[ $\epsilon^{\parallel}_{\underline{\alpha}}^{\underline{\beta}} \mathcal{T}^{\sharp_1}_{1^+} \dagger^{\underline{\chi}}_{\underline{\beta}} \mathcal{T}^{\sharp_2}_{1^-} \underline{\alpha}$ ] := Module[{},
  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1mConstantSymbol],
HoldPattern[ $\epsilon^{\parallel}_{\underline{\chi}}^{\underline{\alpha}}_{\underline{\beta}} \mathcal{T}^{\sharp_1}_{1^+} \dagger^{\underline{\beta}\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \underline{\alpha}$ ] := Module[{},
  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1mConstantSymbol],
HoldPattern[ $\epsilon^{\parallel}_{\underline{\beta}}^{\underline{\chi}\underline{\alpha}} \mathcal{T}^{\sharp_1}_{1^+} \dagger^{\underline{\beta}}_{\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \underline{\alpha}$ ] := Module[{},
  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1mConstantSymbol],
HoldPattern[ $\epsilon^{\parallel}_{\underline{\chi}}^{\underline{\alpha}\underline{\beta}} \mathcal{T}^{\sharp_1}_{1^+} \dagger^{\underline{\chi}}_{\underline{\beta}} \mathcal{T}^{\sharp_2}_{1^-} \underline{\alpha}$ ] := Module[{},
  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1mConstantSymbol],
HoldPattern[ $\epsilon^{\parallel}_{\underline{\beta}}^{\underline{\chi}\underline{\alpha}\underline{\beta}} \mathcal{T}^{\sharp_1}_{1^+} \dagger^{\underline{\beta}\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \underline{\alpha}$ ] := Module[{},
  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1mConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{1^+} \dagger^{\underline{\chi}\underline{\beta}} \epsilon^{\parallel}_{\underline{\chi}\underline{\alpha}\underline{\beta}} \mathcal{T}^{\sharp_2}_{1^-} \underline{\alpha}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1mConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{1^+} \dagger^{\underline{\beta}}_{\underline{\chi}} \epsilon^{\parallel}_{\underline{\alpha}\underline{\beta}}^{\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \underline{\alpha}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1mConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{1^+} \dagger^{\underline{\chi}}_{\underline{\beta}} \epsilon^{\parallel}_{\underline{\chi}\underline{\alpha}}^{\underline{\beta}} \mathcal{T}^{\sharp_2}_{1^-} \underline{\alpha}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1mConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{1^+} \dagger^{\underline{\chi}\underline{\beta}} \epsilon^{\parallel}_{\underline{\alpha}}^{\underline{\chi}}_{\underline{\beta}} \mathcal{T}^{\sharp_2}_{1^-} \underline{\alpha}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1mConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{1^+} \dagger^{\underline{\chi}\underline{\beta}} \epsilon^{\parallel}_{\underline{\chi}}^{\underline{\alpha}}_{\underline{\beta}} \mathcal{T}^{\sharp_2}_{1^-} \underline{\alpha}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricPara1pConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1mConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{1^+} \dagger^{\underline{\beta}}_{\underline{\chi}} \epsilon^{\parallel}_{\underline{\beta}}^{\underline{\chi}\underline{\alpha}} \mathcal{T}^{\sharp_2}_{1^-} \underline{\alpha}$ ] := Module[{}

```

[illegible]

$$\begin{aligned} & \text{HoldPattern}\left[\mathcal{T}_{1^{+}}^{\dagger_2} +_{\underline{\beta}}^{\chi} \epsilon_{\underline{\alpha}\underline{\chi}}^{\parallel\beta} \mathcal{T}_{1^{-}}^{\dagger_2\alpha}\right] \Rightarrow \text{Module}[\{\}, \\ & \quad \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger], \\ & \text{HoldPattern}\left[\mathcal{T}_{1^{+}}^{\dagger_2} +_{\underline{\chi}\underline{\beta}}^{\parallel} \epsilon_{\underline{\alpha}}^{\parallel\frac{\chi\beta}{\underline{\alpha}}} \mathcal{T}_{1^{-}}^{\dagger_2\alpha}\right] \Rightarrow \text{Module}[\{\}, \\ & \quad \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger], \\ & \text{HoldPattern}\left[\mathcal{T}_{1^{+}}^{\dagger_2} +_{\underline{\beta}}^{\chi\beta} \epsilon_{\underline{\chi}\underline{\beta}}^{\parallel\alpha} \mathcal{T}_{1^{-}}^{\dagger_2\alpha}\right] \Rightarrow \text{Module}[\{\}, \\ & \quad \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger], \\ & \text{HoldPattern}\left[\mathcal{T}_{1^{+}}^{\dagger_2} +_{\underline{\chi}}^{\beta} \epsilon_{\underline{\beta}}^{\parallel\frac{\alpha\chi}{\underline{\beta}}} \mathcal{T}_{1^{-}}^{\dagger_2\alpha}\right] \Rightarrow \text{Module}[\{\}, \\ & \quad \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger], \\ & \text{HoldPattern}\left[\mathcal{T}_{1^{+}}^{\dagger_2} +_{\underline{\beta}}^{\chi} \epsilon_{\underline{\chi}}^{\parallel\frac{\alpha}{\underline{\chi}}} \mathcal{T}_{1^{-}}^{\dagger_2\alpha}\right] \Rightarrow \text{Module}[\{\}, \\ & \quad \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger], \\ & \text{HoldPattern}\left[\mathcal{T}_{1^{+}}^{\dagger_2} +_{\underline{\chi}\underline{\beta}}^{\parallel} \epsilon_{\underline{\alpha}}^{\parallel\frac{\alpha\chi\beta}{\underline{\alpha}}} \mathcal{T}_{1^{-}}^{\dagger_2\alpha}\right] \Rightarrow \text{Module}[\{\}, \\ & \quad \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger], \\ & \text{HoldPattern}\left[\epsilon_{\underline{\beta}\underline{\chi}\underline{\alpha}}^{\parallel} \mathcal{T}_{1^{-}}^{\dagger_2\alpha} \mathcal{T}_{1^{+}}^{\dagger_2} +_{\underline{\beta}}^{\beta\chi}\right] \Rightarrow \text{Module}[\{\}, \\ & \quad \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger], \\ & \text{HoldPattern}\left[\epsilon_{\underline{\beta}}^{\parallel\frac{\chi}{\underline{\beta}}} \mathcal{T}_{1^{-}}^{\dagger_2\alpha} \mathcal{T}_{1^{+}}^{\dagger_2} +_{\underline{\chi}}^{\beta}\right] \Rightarrow \text{Module}[\{\}, \\ & \quad \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger], \\ & \text{HoldPattern}\left[\epsilon_{\underline{\chi}\underline{\alpha}}^{\parallel\beta} \mathcal{T}_{1^{-}}^{\dagger_2\alpha} \mathcal{T}_{1^{+}}^{\dagger_2} +_{\underline{\beta}}^{\chi}\right] \Rightarrow \text{Module}[\{\}, \\ & \quad \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger], \\ & \text{HoldPattern}\left[\epsilon_{\underline{\alpha}}^{\parallel\frac{\beta\chi}{\underline{\alpha}}} \mathcal{T}_{1^{-}}^{\dagger_2\alpha} \mathcal{T}_{1^{+}}^{\dagger_2} +_{\underline{\beta}}^{\beta\chi}\right] \Rightarrow \text{Module}[\{\}, \\ & \quad \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger], \\ & \text{HoldPattern}\left[\epsilon_{\underline{\beta}\underline{\chi}}^{\parallel\frac{\alpha}{\underline{\beta}}} \mathcal{T}_{1^{-}}^{\dagger_2\alpha} \mathcal{T}_{1^{+}}^{\dagger_2} +_{\underline{\beta}}^{\beta\chi}\right] \Rightarrow \text{Module}[\{\}, \\ & \quad \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger], \\ & \text{HoldPattern}\left[\epsilon_{\underline{\beta}}^{\parallel\frac{\chi\alpha}{\underline{\beta}}} \mathcal{T}_{1^{-}}^{\dagger_2\alpha} \mathcal{T}_{1^{+}}^{\dagger_2} +_{\underline{\chi}}^{\beta}\right] \Rightarrow \text{Module}[\{\}, \\ & \quad \text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger], \end{aligned}$$

$\text{HoldPattern}\left[\epsilon^{\parallel \underline{\beta} \underline{\alpha}}_{\underline{\chi}} \mathcal{J}^{\sharp_2}_{1^-} \underline{\alpha} \mathcal{J}^{\sharp_2}_{1^+} \dagger_{\underline{\beta}}^{\underline{\chi}}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \underline{\beta \chi \alpha}} \mathcal{J}^{\sharp_2}_{1^-} \underline{\alpha} \mathcal{J}^{\sharp_2}_{1^+} \dagger_{\underline{\beta \chi}}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_2}_{1^+} \dagger_{\underline{\chi}}^{\underline{\beta}} \epsilon^{\parallel \underline{\beta \chi \alpha}} \mathcal{J}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_2}_{1^+} \dagger_{\underline{\chi}}^{\underline{\beta}} \epsilon^{\parallel \underline{\chi}}_{\underline{\beta}} \mathcal{J}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_2}_{1^+} \dagger_{\underline{\beta}}^{\underline{\chi}} \epsilon^{\parallel \underline{\beta}}_{\underline{\chi \alpha}} \mathcal{J}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_2}_{1^+} \dagger_{\underline{\chi \beta}} \epsilon^{\parallel \underline{\beta \chi}}_{\underline{\alpha}} \mathcal{J}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_2}_{1^+} \dagger_{\underline{\chi}}^{\underline{\beta}} \epsilon^{\parallel \underline{\chi \alpha}}_{\underline{\beta}} \mathcal{J}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_2}_{1^+} \dagger_{\underline{\beta}}^{\underline{\chi}} \epsilon^{\parallel \underline{\beta}}_{\underline{\chi}} \mathcal{J}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\mathcal{J}^{\sharp_2}_{1^+} \dagger_{\underline{\chi \beta}} \epsilon^{\parallel \underline{\beta \chi \alpha}} \mathcal{J}^{\sharp_2}_{1^-} \underline{\alpha}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \underline{\chi \alpha \beta}} \mathcal{J}^{\sharp_2}_{1^-} \underline{\alpha} \mathcal{J}^{\sharp_2}_{1^+} \dagger_{\underline{\chi}}^{\underline{\beta}}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$
 $\text{HoldPattern}\left[\epsilon^{\parallel \underline{\chi}}_{\underline{\alpha \beta}} \mathcal{J}^{\sharp_2}_{1^-} \underline{\alpha} \mathcal{J}^{\sharp_2}_{1^+} \dagger_{\underline{\chi}}^{\underline{\beta}}\right] \Rightarrow \text{Module}[\{\},$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol } \dagger],$

[illegible]

[illegible]

[illegible]


```

HoldPattern[ $\mathcal{K}_{1^+}^{\sharp_2 \underline{\chi}} \underline{\beta} \epsilon^{\underline{\underline{\beta \alpha}}} \underline{\underline{\chi}} \mathcal{K}_{1^-}^{\sharp_1} \dagger \underline{\alpha}$ ] := Module[{},
  Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1^+}^{\sharp_2} \underline{\underline{\chi \beta}} \epsilon^{\underline{\underline{\beta \alpha \chi}}} \mathcal{K}_{1^-}^{\sharp_1} \dagger \underline{\alpha}$ ] := Module[{},
  Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\epsilon^{\underline{\underline{\alpha \chi \beta}}} \mathcal{K}_{1^-}^{\sharp_1} \dagger \underline{\alpha} \mathcal{K}_{1^+}^{\sharp_2 \underline{\beta \chi}}$ ] := Module[{},
  Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\epsilon^{\underline{\underline{\chi}}} \underline{\underline{\alpha}} \underline{\underline{\beta}} \mathcal{K}_{1^-}^{\sharp_1} \dagger \underline{\alpha} \mathcal{K}_{1^+}^{\sharp_2 \underline{\beta}} \underline{\underline{\chi}}$ ] := Module[{},
  Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\epsilon^{\underline{\underline{\alpha}}} \underline{\underline{\chi \beta}} \mathcal{K}_{1^-}^{\sharp_1} \dagger \underline{\alpha} \mathcal{K}_{1^+}^{\sharp_2} \underline{\underline{\beta \chi}}$ ] := Module[{},
  Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\epsilon^{\underline{\underline{\alpha \chi}}} \underline{\underline{\beta}} \mathcal{K}_{1^-}^{\sharp_1} \dagger \underline{\alpha} \mathcal{K}_{1^+}^{\sharp_2 \underline{\beta}} \underline{\underline{\chi}}$ ] := Module[{},
  Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\epsilon^{\underline{\underline{\alpha}}} \underline{\underline{\chi \beta}} \mathcal{K}_{1^-}^{\sharp_1} \dagger \underline{\alpha} \mathcal{K}_{1^+}^{\sharp_2} \underline{\underline{\beta \chi}}$ ] := Module[{},
  Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\epsilon^{\underline{\underline{\alpha \chi}}} \underline{\underline{\beta}} \mathcal{K}_{1^-}^{\sharp_1} \dagger \underline{\alpha} \mathcal{K}_{1^+}^{\sharp_2 \underline{\beta}} \underline{\underline{\chi}}$ ] := Module[{},
  Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\epsilon^{\underline{\underline{\alpha}}} \underline{\underline{\chi \beta}} \mathcal{K}_{1^-}^{\sharp_1} \dagger \underline{\alpha} \mathcal{K}_{1^+}^{\sharp_2} \underline{\underline{\beta \chi}}$ ] := Module[{},
  Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1^+}^{\sharp_2 \underline{\chi \beta}} \epsilon^{\underline{\underline{\alpha \chi \beta}}} \mathcal{K}_{1^-}^{\sharp_1} \dagger \underline{\alpha}$ ] := Module[{},
  Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1^+}^{\sharp_2} \underline{\underline{\beta}} \epsilon^{\underline{\underline{\chi}}} \underline{\underline{\alpha}} \underline{\underline{\beta}} \mathcal{K}_{1^-}^{\sharp_1} \dagger \underline{\alpha}$ ] := Module[{},
  Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],

```

[illegible]


```

HoldPattern[ $\mathcal{K}_{1^+}^{\sharp_2 \underline{\chi}} \underset{\underline{\beta}}{\epsilon} \underset{\underline{\chi}}{\epsilon}^{\underline{\alpha \beta}} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}}$ ] := Module[{} ,
  -Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1^+}^{\sharp_2} \underline{\chi \beta} \underset{\underline{\epsilon}}{\epsilon}^{\underline{\chi \alpha \beta}} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}}$ ] := Module[{} ,
  -Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\underset{\underline{\epsilon}}{\epsilon}^{\underline{\chi \beta \alpha}} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\sharp_2 \underline{\beta \chi}}$ ] := Module[{} ,
  -Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\underset{\underline{\epsilon}}{\epsilon}^{\underline{\chi}} \underset{\underline{\beta \alpha}}{\epsilon} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\sharp_2 \underline{\beta}} \underset{\underline{\chi}}{\epsilon}$ ] := Module[{} ,
  -Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\underset{\underline{\epsilon}}{\epsilon}^{\underline{\beta}} \underset{\underline{\chi}}{\epsilon} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\sharp_2} \underset{\underline{\beta}}{\epsilon} \underset{\underline{\chi}}{\epsilon}$ ] := Module[{} ,
  -Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\underset{\underline{\epsilon}}{\epsilon}^{\underline{\chi \beta}} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\sharp_2} \underline{\beta \chi}$ ] := Module[{} ,
  -Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\underset{\underline{\epsilon}}{\epsilon}^{\underline{\chi \beta \alpha}} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\sharp_2 \underline{\beta \chi}}$ ] := Module[{} ,
  -Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\underset{\underline{\epsilon}}{\epsilon}^{\underline{\chi}} \underset{\underline{\beta}}{\epsilon} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\sharp_2 \underline{\beta}} \underset{\underline{\chi}}{\epsilon}$ ] := Module[{} ,
  -Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\underset{\underline{\epsilon}}{\epsilon}^{\underline{\beta \alpha}} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\sharp_2} \underset{\underline{\beta}}{\epsilon} \underset{\underline{\chi}}{\epsilon}$ ] := Module[{} ,
  -Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\underset{\underline{\epsilon}}{\epsilon}^{\underline{\chi \beta \alpha}} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\sharp_2} \underline{\beta \chi}$ ] := Module[{} ,
  -Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1^+}^{\sharp_2 \underline{\chi \beta}} \underset{\underline{\epsilon}}{\epsilon}^{\underline{\chi \beta \alpha}} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}}$ ] := Module[{} ,
  Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{K}_{1^+}^{\sharp_2} \underset{\underline{\chi}}{\epsilon} \underset{\underline{\beta \alpha}}{\epsilon} \mathcal{K}_{1^-}^{\sharp_1} \dagger_{\underline{\alpha}}$ ] := Module[{} ,
  Rank3AntisymmetricParamConstantSymbol † Rank3AntisymmetricPerp1pConstantSymbol],

```


[illegible]

[illegible]


```

HoldPattern $\left[\mathcal{K}_{1^+}^{\dagger_2 \underline{\chi}} \underline{\beta} \epsilon^{\underline{\beta \alpha}} \underline{\chi} \mathcal{K}_{1^-}^{\dagger_2} \dagger_{\underline{\alpha}}\right] \Rightarrow \text{Module}[\{\},$ 
    Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern $\left[\mathcal{K}_{1^+}^{\dagger_2} \underline{\chi \beta} \epsilon^{\underline{\beta \alpha \chi}} \mathcal{K}_{1^-}^{\dagger_2} \dagger_{\underline{\alpha}}\right] \Rightarrow \text{Module}[\{\},$ 
    Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern $\left[\epsilon^{\underline{\alpha \chi \beta}} \mathcal{K}_{1^-}^{\dagger_2} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\dagger_2 \underline{\beta \chi}}\right] \Rightarrow \text{Module}[\{\},$ 
    Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern $\left[\epsilon^{\underline{\chi}} \underline{\alpha} \underline{\beta} \mathcal{K}_{1^-}^{\dagger_2} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\dagger_2 \underline{\beta}} \underline{\chi}\right] \Rightarrow \text{Module}[\{\},$ 
    Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern $\left[\epsilon^{\underline{\alpha \chi}} \underline{\beta} \mathcal{K}_{1^-}^{\dagger_2} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\dagger_2} \underline{\chi \beta}\right] \Rightarrow \text{Module}[\{\},$ 
    Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern $\left[\epsilon^{\underline{\alpha \chi \beta}} \mathcal{K}_{1^-}^{\dagger_2} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\dagger_2} \underline{\beta \chi}\right] \Rightarrow \text{Module}[\{\},$ 
    Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern $\left[\epsilon^{\underline{\alpha}} \underline{\chi \beta} \mathcal{K}_{1^-}^{\dagger_2} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\dagger_2 \underline{\beta \chi}}\right] \Rightarrow \text{Module}[\{\},$ 
    Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern $\left[\epsilon^{\underline{\alpha \chi}} \underline{\beta} \mathcal{K}_{1^-}^{\dagger_2} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\dagger_2 \underline{\beta}} \underline{\chi}\right] \Rightarrow \text{Module}[\{\},$ 
    Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern $\left[\epsilon^{\underline{\alpha}} \underline{\chi} \underline{\beta} \mathcal{K}_{1^-}^{\dagger_2} \dagger_{\underline{\alpha}} \mathcal{K}_{1^+}^{\dagger_2} \underline{\chi \beta}\right] \Rightarrow \text{Module}[\{\},$ 
    Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern $\left[\mathcal{K}_{1^+}^{\dagger_2 \underline{\chi \beta}} \epsilon^{\underline{\alpha \chi \beta}} \mathcal{K}_{1^-}^{\dagger_2} \dagger_{\underline{\alpha}}\right] \Rightarrow \text{Module}[\{\},$ 
    Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern $\left[\mathcal{K}_{1^+}^{\dagger_2} \underline{\beta} \epsilon^{\underline{\chi}} \underline{\alpha} \underline{\beta} \mathcal{K}_{1^-}^{\dagger_2} \dagger_{\underline{\alpha}}\right] \Rightarrow \text{Module}[\{\},$ 
    Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],

```

[illegible]

[illegible]


```

HoldPattern[ $\epsilon^{\underline{\alpha}\underline{\beta}}_{\underline{\chi}} \mathcal{T}^{\sharp_1}_{1^-} \dagger_{\underline{\alpha}} \mathcal{T}^{\sharp_1}_{1^+} \underline{\beta}^{\underline{\chi}}$ ] := Module[{},
  Rank3AntisymmetricParamConstantSymbol  $\dagger$  Rank3AntisymmetricParamConstantSymbol],
HoldPattern[ $\epsilon^{\underline{\alpha}\underline{\beta}\underline{\chi}} \mathcal{T}^{\sharp_1}_{1^-} \dagger_{\underline{\alpha}} \mathcal{T}^{\sharp_1}_{1^+} \underline{\beta}\underline{\chi}$ ] := Module[{},
  Rank3AntisymmetricParamConstantSymbol  $\dagger$  Rank3AntisymmetricParamConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{1^+} \underline{\chi}\underline{\beta} \epsilon^{\underline{\alpha}\underline{\beta}\underline{\chi}} \mathcal{T}^{\sharp_1}_{1^-} \dagger_{\underline{\alpha}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricParamConstantSymbol  $\dagger$  Rank3AntisymmetricParamConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{1^+} \underline{\chi}^{\underline{\beta}} \epsilon^{\underline{\alpha}\underline{\beta}}_{\underline{\chi}} \mathcal{T}^{\sharp_1}_{1^-} \dagger_{\underline{\alpha}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricParamConstantSymbol  $\dagger$  Rank3AntisymmetricParamConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{1^+} \underline{\chi}^{\underline{\beta}}_{\underline{\beta}} \epsilon^{\underline{\alpha}\underline{\beta}}_{\underline{\chi}} \mathcal{T}^{\sharp_1}_{1^-} \dagger_{\underline{\alpha}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricParamConstantSymbol  $\dagger$  Rank3AntisymmetricParamConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{1^+} \underline{\chi}\underline{\beta} \epsilon^{\underline{\alpha}\underline{\beta}\underline{\chi}} \mathcal{T}^{\sharp_1}_{1^-} \dagger_{\underline{\alpha}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricParamConstantSymbol  $\dagger$  Rank3AntisymmetricParamConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{1^+} \underline{\chi}^{\underline{\beta}}_{\underline{\chi}} \epsilon^{\underline{\alpha}\underline{\beta}}_{\underline{\beta}} \mathcal{T}^{\sharp_1}_{1^-} \dagger_{\underline{\alpha}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricParamConstantSymbol  $\dagger$  Rank3AntisymmetricParamConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_1}_{1^+} \underline{\chi}\underline{\beta} \epsilon^{\underline{\alpha}\underline{\beta}\underline{\chi}} \mathcal{T}^{\sharp_1}_{1^-} \dagger_{\underline{\alpha}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricParamConstantSymbol  $\dagger$  Rank3AntisymmetricParamConstantSymbol],
HoldPattern[ $\epsilon^{\underline{\beta}\underline{\alpha}\underline{\chi}} \mathcal{T}^{\sharp_1}_{1^-} \dagger_{\underline{\alpha}} \mathcal{T}^{\sharp_1}_{1^+} \underline{\beta}\underline{\chi}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricParamConstantSymbol  $\dagger$  Rank3AntisymmetricParamConstantSymbol],
HoldPattern[ $\epsilon^{\underline{\chi}}_{\underline{\beta}\underline{\alpha}} \mathcal{T}^{\sharp_1}_{1^-} \dagger_{\underline{\alpha}} \mathcal{T}^{\sharp_1}_{1^+} \underline{\beta}_{\underline{\chi}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricParamConstantSymbol  $\dagger$  Rank3AntisymmetricParamConstantSymbol],

```


[illegible]

[illegible]


```

HoldPattern[ $\epsilon^{\underline{\alpha}\underline{\beta}}_{\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \dagger_{\underline{\alpha}} \mathcal{T}^{\sharp_2}_{1^+} \underline{\beta}^{\underline{\chi}}$ ] := Module[{},
  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\epsilon^{\underline{\alpha}\underline{\beta}\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \dagger_{\underline{\alpha}} \mathcal{T}^{\sharp_2}_{1^+} \underline{\beta}\underline{\chi}$ ] := Module[{},
  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_2}_{1^+} \underline{\chi}\underline{\beta} \epsilon^{\underline{\alpha}\underline{\beta}\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \dagger_{\underline{\alpha}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_2}_{1^+} \underline{\chi}^{\underline{\beta}} \epsilon^{\underline{\alpha}\underline{\beta}}_{\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \dagger_{\underline{\alpha}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_2}_{1^+} \underline{\beta}^{\underline{\chi}} \epsilon^{\underline{\alpha}\underline{\beta}}_{\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \dagger_{\underline{\alpha}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_2}_{1^+} \underline{\chi}\underline{\beta} \epsilon^{\underline{\alpha}\underline{\beta}\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \dagger_{\underline{\alpha}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_2}_{1^+} \underline{\chi}^{\underline{\beta}} \epsilon^{\underline{\alpha}\underline{\beta}}_{\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \dagger_{\underline{\alpha}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\mathcal{T}^{\sharp_2}_{1^+} \underline{\chi}\underline{\beta} \epsilon^{\underline{\alpha}\underline{\beta}\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \dagger_{\underline{\alpha}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\epsilon^{\underline{\beta}\underline{\alpha}\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \dagger_{\underline{\alpha}} \mathcal{T}^{\sharp_2}_{1^+} \underline{\beta}\underline{\chi}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],
HoldPattern[ $\epsilon^{\underline{\beta}\underline{\alpha}}_{\underline{\chi}} \mathcal{T}^{\sharp_2}_{1^-} \dagger_{\underline{\alpha}} \mathcal{T}^{\sharp_2}_{1^+} \underline{\beta}^{\underline{\chi}}$ ] := Module[{},
   $\neg$  Rank3AntisymmetricPerp1mConstantSymbol  $\dagger$  Rank3AntisymmetricPerp1pConstantSymbol],

```


[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]


```

Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol †}] }

** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`SymbolicLagrangian$173389 will now be printed.

 $k^2 K_{15}^{(4)}$  Rank3AntisymmetricPara0mConstantSymbol

Rank3AntisymmetricPara0mConstantSymbol † -  $k^2 K_{16}^{(4)}$ 

Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol † +
 $K_1^{(2)}$  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol † -
 $K_2^{(2)}$  Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol † -
 $k^2 K_1^{(4)}$  Rank3AntisymmetricPara0pConstantSymbol

Rank3AntisymmetricPara0pConstantSymbol † + -  $k^2 K_{15}^{(4)}$ 
3

Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol † +
 $\frac{1}{3} k^2 K_{16}^{(4)}$  Rank3AntisymmetricPara0pConstantSymbol

Rank3AntisymmetricPara0pConstantSymbol † -  $k^2 K_2^{(4)}$ 

Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol † -
 $\frac{1}{3} k^2 K_3^{(4)}$  Rank3AntisymmetricPara0pConstantSymbol

Rank3AntisymmetricPara0pConstantSymbol † -  $\frac{1}{3} k^2 K_6^{(4)}$ 

Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol † +
 $2 K_1^{(2)}$  Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †


---


3

+

 $K_2^{(2)}$  Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol †


---


3

```

$$\begin{aligned}
& \overset{(2)}{K}_3 \text{Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol} \dagger + \\
& k^2 \overset{(4)}{K}_{15} \text{Rank3AntisymmetricPara1mConstantSymbol} \\
& \text{Rank3AntisymmetricPara1mConstantSymbol} \dagger + \\
& - \frac{1}{2} k^2 \overset{(4)}{K}_{16} \text{Rank3AntisymmetricPara1mConstantSymbol} \\
& \text{Rank3AntisymmetricPara1mConstantSymbol} \dagger - k^2 \overset{(4)}{K}_2 \\
& \text{Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1mConstantSymbol} \dagger + \\
& \overset{(2)}{K}_1 \text{Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1mConstantSymbol} \dagger + \\
& \overset{(2)}{K}_2 \text{Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1mConstantSymbol} \dagger - \\
& \hline
& 2 \\
& \overset{(2)}{K}_3 \text{Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1mConstantSymbol} \dagger + \\
& 2 k^2 \overset{(4)}{K}_{15} \text{Rank3AntisymmetricPara1pConstantSymbol} \\
& \text{Rank3AntisymmetricPara1pConstantSymbol} \dagger - \\
& k^2 \overset{(4)}{K}_{16} \text{Rank3AntisymmetricPara1pConstantSymbol} \\
& \text{Rank3AntisymmetricPara1pConstantSymbol} \dagger - k^2 \overset{(4)}{K}_3 \\
& \text{Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol} \dagger + \\
& k^2 \overset{(4)}{K}_6 \text{Rank3AntisymmetricPara1pConstantSymbol} \\
& \text{Rank3AntisymmetricPara1pConstantSymbol} \dagger + 2 \overset{(2)}{K}_1 \\
& \text{Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol} \dagger - \\
& \overset{(2)}{K}_2 \text{Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol} \dagger + \\
& - \frac{16}{9} k^2 \overset{(4)}{K}_{15} \text{Rank3AntisymmetricPara2mConstantSymbol} \\
& \text{Rank3AntisymmetricPara2mConstantSymbol} \dagger + - \frac{8}{9} k^2 \overset{(4)}{K}_{16}
\end{aligned}$$

$$\begin{aligned}
& \text{Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol} \dagger + \\
& \frac{16 \overset{(2)}{K}_1 \text{Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol} \dagger}{9} \\
& + \\
& \frac{8 \overset{(2)}{K}_2 \text{Rank3AntisymmetricPara2mConstantSymbol Rank3AntisymmetricPara2mConstantSymbol} \dagger}{9} \\
& + 2 k^2 \overset{(4)}{K}_{15} \text{Rank3AntisymmetricPara2pConstantSymbol} \\
& \text{Rank3AntisymmetricPara2pConstantSymbol} \dagger + \\
& k^2 \overset{(4)}{K}_{16} \text{Rank3AntisymmetricPara2pConstantSymbol} \\
& \text{Rank3AntisymmetricPara2pConstantSymbol} \dagger - k^2 \overset{(4)}{K}_3 \\
& \text{Rank3AntisymmetricPara2pConstantSymbol Rank3AntisymmetricPara2pConstantSymbol} \dagger - \\
& k^2 \overset{(4)}{K}_6 \text{Rank3AntisymmetricPara2pConstantSymbol} \\
& \text{Rank3AntisymmetricPara2pConstantSymbol} \dagger + 2 \overset{(2)}{K}_1 \\
& \text{Rank3AntisymmetricPara2pConstantSymbol Rank3AntisymmetricPara2pConstantSymbol} \dagger + \\
& \overset{(2)}{K}_2 \text{Rank3AntisymmetricPara2pConstantSymbol Rank3AntisymmetricPara2pConstantSymbol} \dagger + \\
& k^2 \overset{(4)}{K}_2 \text{Rank3AntisymmetricPara1mConstantSymbol} \dagger \\
& \text{Rank3AntisymmetricPerp1mConstantSymbol} + \\
& \overset{(2)}{K}_3 \text{Rank3AntisymmetricPara1mConstantSymbol} \dagger \text{Rank3AntisymmetricPerp1mConstantSymbol} + \\
& k^2 \overset{(4)}{K}_2 \text{Rank3AntisymmetricPara1mConstantSymbol} \\
& \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger + k^2 \overset{(4)}{K}_7 \\
& \text{Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPerp1mConstantSymbol} \dagger + \\
& \overset{(2)}{K}_3 \text{Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPerp1mConstantSymbol} \dagger + \\
& 2 k^2 \overset{(4)}{K}_{15} \text{Rank3AntisymmetricPerp1mConstantSymbol}
\end{aligned}$$

$\text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger + k^2 K_{16}^{(4)}$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger -$
 $k^2 K_2^{(4)} \text{Rank3AntisymmetricPerp1mConstantSymbol}$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger - k^2 K_3^{(4)}$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger -$
 $k^2 K_4^{(4)} \text{Rank3AntisymmetricPerp1mConstantSymbol}$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger - k^2 K_7^{(4)}$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger +$
 $2 k^2 K_9^{(4)} \text{Rank3AntisymmetricPerp1mConstantSymbol}$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger + 2 K_1^{(2)}$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger +$
 $K_2^{(2)} \text{Rank3AntisymmetricPerp1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger -$
 $K_3^{(2)} \text{Rank3AntisymmetricPerp1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger -$
 $k^2 K_{16}^{(4)} \text{Rank3AntisymmetricPara1pConstantSymbol} \dagger$
 $\text{Rank3AntisymmetricPerp1pConstantSymbol} + k^2 K_4^{(4)}$
 $\text{Rank3AntisymmetricPara1pConstantSymbol} \dagger \text{Rank3AntisymmetricPerp1pConstantSymbol} -$
 $K_2^{(2)} \text{Rank3AntisymmetricPara1pConstantSymbol} \dagger \text{Rank3AntisymmetricPerp1pConstantSymbol} -$
 $k^2 K_{16}^{(4)} \text{Rank3AntisymmetricPara1pConstantSymbol}$
 $\text{Rank3AntisymmetricPerp1pConstantSymbol} \dagger -$
 $K_2^{(2)} \text{Rank3AntisymmetricPara1pConstantSymbol} \text{Rank3AntisymmetricPerp1pConstantSymbol} \dagger +$
 $k^2 K_{15}^{(4)} \text{Rank3AntisymmetricPerp1pConstantSymbol}$
 $\text{Rank3AntisymmetricPerp1pConstantSymbol} \dagger + k^2 K_9^{(4)}$
 $\text{Rank3AntisymmetricPerp1pConstantSymbol} \text{Rank3AntisymmetricPerp1pConstantSymbol} \dagger +$

$$\begin{aligned}
& \overset{(2)}{K_1} \text{Rank3AntisymmetricPerp1pConstantSymbol} \text{Rank3AntisymmetricPerp1pConstantSymbol} \dagger + \\
& - \frac{4}{3} k^2 \overset{(4)}{K_{15}} \mathcal{K}_{\overset{-}{0}}^{\dagger 1} \dagger_{\alpha\beta\chi} \mathcal{K}_{\overset{-}{2}}^{\dagger 1} \alpha\beta\chi - \frac{4}{3} k^2 \overset{(4)}{K_{16}} \mathcal{K}_{\overset{-}{0}}^{\dagger 1} \dagger_{\alpha\beta\chi} \mathcal{K}_{\overset{-}{2}}^{\dagger 1} \alpha\beta\chi + \\
& - \frac{4}{3} \overset{(2)}{K_1} \mathcal{K}_{\overset{-}{0}}^{\dagger 1} \dagger_{\alpha\beta\chi} \mathcal{K}_{\overset{-}{2}}^{\dagger 1} \alpha\beta\chi - \frac{4}{3} \overset{(2)}{K_2} \mathcal{K}_{\overset{-}{0}}^{\dagger 1} \dagger_{\alpha\beta\chi} \mathcal{K}_{\overset{-}{2}}^{\dagger 1} \alpha\beta\chi + \\
& - \frac{4}{3} k^2 \overset{(4)}{K_{15}} \mathcal{K}_{\overset{-}{0}}^{\dagger 1} \alpha\beta\chi \mathcal{K}_{\overset{-}{2}}^{\dagger 1} \dagger_{\alpha\beta\chi} - \frac{4}{3} k^2 \overset{(4)}{K_{16}} \mathcal{K}_{\overset{-}{0}}^{\dagger 1} \alpha\beta\chi \mathcal{K}_{\overset{-}{2}}^{\dagger 1} \dagger_{\alpha\beta\chi} + \\
& - \frac{4}{3} \overset{(2)}{K_1} \mathcal{K}_{\overset{-}{0}}^{\dagger 1} \alpha\beta\chi \mathcal{K}_{\overset{-}{2}}^{\dagger 1} \dagger_{\alpha\beta\chi} - \frac{4}{3} \overset{(2)}{K_2} \mathcal{K}_{\overset{-}{0}}^{\dagger 1} \alpha\beta\chi \mathcal{K}_{\overset{-}{2}}^{\dagger 1} \dagger_{\alpha\beta\chi}
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`SpinParityConstantSymbols\$173389 will now be printed.

```

<| xAct`PSALter`A23Field`Rank3Antisymmetric ->
  <| 0 -> <| xAct`PSALter`Private`Even -> {Rank3AntisymmetricPara0pConstantSymbol},
    xAct`PSALter`Private`Odd -> {Rank3AntisymmetricPara0mConstantSymbol} |>,
  1 -> <| xAct`PSALter`Private`Even -> {Rank3AntisymmetricPara1pConstantSymbol,
    Rank3AntisymmetricPerp1pConstantSymbol},
    xAct`PSALter`Private`Odd -> {Rank3AntisymmetricPara1mConstantSymbol,
    Rank3AntisymmetricPerp1mConstantSymbol} |>,
  2 -> <| xAct`PSALter`Private`Even -> {Rank3AntisymmetricPara2pConstantSymbol},
    xAct`PSALter`Private`Odd -> {Rank3AntisymmetricPara2mConstantSymbol} |> |> |>

```

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`SpinParityRescalingSymbols\$173389 will now be printed.

```

<| xAct`PSALter`A23Field`Rank3Antisymmetric ->
  <| 0 -> <| xAct`PSALter`Private`Even -> {Rank3AntisymmetricPara0pRescalingSymbol},
    xAct`PSALter`Private`Odd -> {Rank3AntisymmetricPara0mRescalingSymbol} |>,
  1 -> <| xAct`PSALter`Private`Even -> {Rank3AntisymmetricPara1pRescalingSymbol,
    Rank3AntisymmetricPerp1pRescalingSymbol},
    xAct`PSALter`Private`Odd -> {Rank3AntisymmetricPara1mRescalingSymbol,
    Rank3AntisymmetricPerp1mRescalingSymbol} |>,
  2 -> <| xAct`PSALter`Private`Even -> {Rank3AntisymmetricPara2pRescalingSymbol},
    xAct`PSALter`Private`Odd -> {Rank3AntisymmetricPara2mRescalingSymbol} |> |> |>

```

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`Symbols\$173389 will now be printed.

<| |>

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`Symbols\$173389 will now be printed.

<| 0 → {Rank3AntisymmetricPara0pConstantSymbol,
 Rank3AntisymmetricPara0mConstantSymbol, Rank3AntisymmetricPara0pConstantSymbol†,
 Rank3AntisymmetricPara0mConstantSymbol†},
 1 → {Rank3AntisymmetricPara1pConstantSymbol, Rank3AntisymmetricPerp1pConstantSymbol,
 Rank3AntisymmetricPara1mConstantSymbol, Rank3AntisymmetricPerp1mConstantSymbol,
 Rank3AntisymmetricPara1pConstantSymbol†,
 Rank3AntisymmetricPerp1pConstantSymbol†, Rank3AntisymmetricPara1mConstantSymbol†,
 Rank3AntisymmetricPerp1mConstantSymbol†},
 2 → {Rank3AntisymmetricPara2pConstantSymbol, Rank3AntisymmetricPara2mConstantSymbol,
 Rank3AntisymmetricPara2pConstantSymbol†,
 Rank3AntisymmetricPara2mConstantSymbol†}|>

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`DummySectors\$173389 will now be printed.

<| 0 → <| xAct`PSALter`Private`Even →
 Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol†,
 xAct`PSALter`Private`Odd → Rank3AntisymmetricPara0mConstantSymbol
 Rank3AntisymmetricPara0mConstantSymbol†|> ,
 1 → <| xAct`PSALter`Private`Even →
 Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol†+
 Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol†,
 xAct`PSALter`Private`Odd → Rank3AntisymmetricPara1mConstantSymbol
 Rank3AntisymmetricPara1mConstantSymbol†+
 Rank3AntisymmetricPerp1mConstantSymbol
 Rank3AntisymmetricPerp1mConstantSymbol†|> ,
 2 → <| xAct`PSALter`Private`Even → Rank3AntisymmetricPara2pConstantSymbol
 Rank3AntisymmetricPara2pConstantSymbol†,
 xAct`PSALter`Private`Odd → Rank3AntisymmetricPara2mConstantSymbol
 Rank3AntisymmetricPara2mConstantSymbol†|> |>

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`Rescalings will now be printed.

$$\begin{aligned}
\langle | \rangle_0 &\rightarrow \left\{ \frac{1}{\text{Rank3AntisymmetricPara0pRescalingSymbol}}, \right. \\
&\quad \left. \frac{1}{\text{Rank3AntisymmetricPara0mRescalingSymbol}} \right\}, \\
1 &\rightarrow \left\{ \frac{1}{\text{Rank3AntisymmetricPara1pRescalingSymbol}}, \right. \\
&\quad \frac{1}{\text{Rank3AntisymmetricPerp1pRescalingSymbol}}, \frac{1}{\text{Rank3AntisymmetricPara1mRescalingSymbol}}, \\
&\quad \left. \frac{1}{\text{Rank3AntisymmetricPerp1mRescalingSymbol}} \right\}, \\
2 &\rightarrow \left\{ \frac{1}{\text{Rank3AntisymmetricPara2pRescalingSymbol}}, \right. \\
&\quad \left. \frac{1}{\text{Rank3AntisymmetricPara2mRescalingSymbol}} \right\} | \rangle
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`RaisedIndexSources will now be printed.

$$\begin{aligned}
\langle | \rangle_0 &\rightarrow \{ \mathcal{T}_{0^+}^{\sharp_1}, \mathcal{T}_{0^-}^{\sharp_1} \alpha \beta \chi \}, \\
1 &\rightarrow \{ \mathcal{T}_{1^+}^{\sharp_1} \alpha \beta, \mathcal{T}_{1^+}^{\sharp_2} \alpha \beta, \mathcal{T}_{1^-}^{\sharp_1} \alpha, \mathcal{T}_{1^-}^{\sharp_2} \alpha \}, 2 \rightarrow \{ \mathcal{T}_{2^+}^{\sharp_1} \alpha \beta, \mathcal{T}_{2^-}^{\sharp_1} \alpha \beta \chi \} | \rangle
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`LoweredIndexSources will now be printed.

$$\begin{aligned}
\langle | \rangle_0 &\rightarrow \{ \mathcal{T}_{0^+}^{\sharp_1}, \mathcal{T}_{0^-}^{\sharp_1} \alpha \beta \chi \}, \\
1 &\rightarrow \{ \mathcal{T}_{1^+}^{\sharp_1} \alpha \beta, \mathcal{T}_{1^+}^{\sharp_2} \alpha \beta, \mathcal{T}_{1^-}^{\sharp_1} \alpha, \mathcal{T}_{1^-}^{\sharp_2} \alpha \}, 2 \rightarrow \{ \mathcal{T}_{2^+}^{\sharp_1} \alpha \beta, \mathcal{T}_{2^-}^{\sharp_1} \alpha \beta \chi \} | \rangle
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`RaisedIndexFields\$173389 will now be printed.

$$\begin{aligned}
\langle | \rangle_0 &\rightarrow \{ \mathcal{K}_{0^+}^{\sharp_1}, \mathcal{K}_{0^-}^{\sharp_1} \alpha \beta \chi \}, \\
1 &\rightarrow \{ \mathcal{K}_{1^+}^{\sharp_1} \alpha \beta, \mathcal{K}_{1^+}^{\sharp_2} \alpha \beta, \mathcal{K}_{1^-}^{\sharp_1} \alpha, \mathcal{K}_{1^-}^{\sharp_2} \alpha \}, 2 \rightarrow \{ \mathcal{K}_{2^+}^{\sharp_1} \alpha \beta, \mathcal{K}_{2^-}^{\sharp_1} \alpha \beta \chi \} | \rangle
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`LoweredIndexFields\$173389 will now be printed.

$$\langle | \rightarrow \{ \mathcal{K}_{\theta^+}^{\dagger\dagger_1}, \mathcal{K}_{\theta^-}^{\dagger\dagger_1} \alpha \beta \chi \}, \\ 1 \rightarrow \{ \mathcal{K}_{1^+}^{\dagger\dagger_1} \alpha \beta, \mathcal{K}_{1^+}^{\dagger\dagger_2} \alpha \beta, \mathcal{K}_{1^-}^{\dagger\dagger_1} \alpha, \mathcal{K}_{1^-}^{\dagger\dagger_2} \alpha \}, 2 \rightarrow \{ \mathcal{K}_{2^+}^{\dagger\dagger_1} \alpha \beta, \mathcal{K}_{2^-}^{\dagger\dagger_1} \alpha \beta \chi \} | \rangle$$

****** xAct`PSALter`Private`Diagnostic: the value of
the variable xAct`PSALter`Private`\$Neglect will now be printed.

$$\langle | \rightarrow \langle | \text{xAct`PSALter`Private`Even} \rightarrow \text{False}, \text{xAct`PSALter`Private`Odd} \rightarrow \text{False} | \rangle, \\ 1 \rightarrow \langle | \text{xAct`PSALter`Private`Even} \rightarrow \text{False}, \text{xAct`PSALter`Private`Odd} \rightarrow \text{False} | \rangle, \\ 2 \rightarrow \langle | \text{xAct`PSALter`Private`Even} \rightarrow \text{False}, \text{xAct`PSALter`Private`Odd} \rightarrow \text{False} | \rangle | \rangle$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable
xAct`PSALter`Private`SymbolicLagrangian\$173389 will now be printed.

$$k^2 K_{15}^{(4)} \text{Rank3AntisymmetricPara0mConstantSymbol}$$

$$\text{Rank3AntisymmetricPara0mConstantSymbol} \dagger - k^2 K_{16}^{(4)}$$

$$\text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol} \dagger +$$

$$K_1^{(2)} \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol} \dagger -$$

$$K_2^{(2)} \text{Rank3AntisymmetricPara0mConstantSymbol Rank3AntisymmetricPara0mConstantSymbol} \dagger -$$

$$k^2 K_1^{(4)} \text{Rank3AntisymmetricPara0pConstantSymbol}$$

$$\text{Rank3AntisymmetricPara0pConstantSymbol} \dagger + - \frac{2}{3} k^2 K_{15}^{(4)}$$

$$\text{Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol} \dagger +$$

$$- \frac{1}{3} k^2 K_{16}^{(4)} \text{Rank3AntisymmetricPara0pConstantSymbol}$$

$$\text{Rank3AntisymmetricPara0pConstantSymbol} \dagger - k^2 K_2^{(4)}$$

$$\text{Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol} \dagger -$$

$$- \frac{1}{3} k^2 K_3^{(4)} \text{Rank3AntisymmetricPara0pConstantSymbol}$$

$$\text{Rank3AntisymmetricPara0pConstantSymbol} \dagger - \frac{1}{3} k^2 K_6^{(4)}$$

$$\text{Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol} \dagger +$$

$$\begin{aligned}
& \frac{2 \overset{(2)}{K}_1 \text{Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol} \dagger}{3} \\
& + \\
& \frac{\overset{(2)}{K}_2 \text{Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol} \dagger}{3} \\
& \overset{(2)}{K}_3 \text{Rank3AntisymmetricPara0pConstantSymbol Rank3AntisymmetricPara0pConstantSymbol} \dagger + \\
& k^2 \overset{(4)}{K}_{15} \text{Rank3AntisymmetricPara1mConstantSymbol} \\
& \text{Rank3AntisymmetricPara1mConstantSymbol} \dagger + \\
& - \frac{1}{2} k^2 \overset{(4)}{K}_{16} \text{Rank3AntisymmetricPara1mConstantSymbol} \\
& \text{Rank3AntisymmetricPara1mConstantSymbol} \dagger - k^2 \overset{(4)}{K}_2 \\
& \text{Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1mConstantSymbol} \dagger + \\
& \overset{(2)}{K}_1 \text{Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1mConstantSymbol} \dagger + \\
& \overset{(2)}{K}_2 \text{Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1mConstantSymbol} \dagger \\
& \frac{}{2} \\
& \overset{(2)}{K}_3 \text{Rank3AntisymmetricPara1mConstantSymbol Rank3AntisymmetricPara1mConstantSymbol} \dagger + \\
& 2 k^2 \overset{(4)}{K}_{15} \text{Rank3AntisymmetricPara1pConstantSymbol} \\
& \text{Rank3AntisymmetricPara1pConstantSymbol} \dagger - \\
& k^2 \overset{(4)}{K}_{16} \text{Rank3AntisymmetricPara1pConstantSymbol} \\
& \text{Rank3AntisymmetricPara1pConstantSymbol} \dagger - k^2 \overset{(4)}{K}_3 \\
& \text{Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol} \dagger + \\
& k^2 \overset{(4)}{K}_6 \text{Rank3AntisymmetricPara1pConstantSymbol} \\
& \text{Rank3AntisymmetricPara1pConstantSymbol} \dagger + 2 \overset{(2)}{K}_1 \\
& \text{Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPara1pConstantSymbol} \dagger -
\end{aligned}$$

$$\begin{aligned}
& K_2^{(2)} \text{Rank3AntisymmetricPara1pConstantSymbol} \text{Rank3AntisymmetricPara1pConstantSymbol} \dagger + \\
& - \frac{16}{9} k^2 K_{15}^{(4)} \text{Rank3AntisymmetricPara2mConstantSymbol} \\
& \text{Rank3AntisymmetricPara2mConstantSymbol} \dagger + - \frac{8}{9} k^2 K_{16}^{(4)} \\
& \text{Rank3AntisymmetricPara2mConstantSymbol} \text{Rank3AntisymmetricPara2mConstantSymbol} \dagger + \\
& \frac{16}{9} K_1^{(2)} \text{Rank3AntisymmetricPara2mConstantSymbol} \text{Rank3AntisymmetricPara2mConstantSymbol} \dagger \\
& + \\
& \frac{8}{9} K_2^{(2)} \text{Rank3AntisymmetricPara2mConstantSymbol} \text{Rank3AntisymmetricPara2mConstantSymbol} \dagger \\
& + 2 k^2 K_{15}^{(4)} \text{Rank3AntisymmetricPara2pConstantSymbol} \\
& \text{Rank3AntisymmetricPara2pConstantSymbol} \dagger + \\
& k^2 K_{16}^{(4)} \text{Rank3AntisymmetricPara2pConstantSymbol} \\
& \text{Rank3AntisymmetricPara2pConstantSymbol} \dagger - k^2 K_3^{(4)} \\
& \text{Rank3AntisymmetricPara2pConstantSymbol} \text{Rank3AntisymmetricPara2pConstantSymbol} \dagger - \\
& k^2 K_6^{(4)} \text{Rank3AntisymmetricPara2pConstantSymbol} \\
& \text{Rank3AntisymmetricPara2pConstantSymbol} \dagger + 2 K_1^{(2)} \\
& \text{Rank3AntisymmetricPara2pConstantSymbol} \text{Rank3AntisymmetricPara2pConstantSymbol} \dagger + \\
& K_2^{(2)} \text{Rank3AntisymmetricPara2pConstantSymbol} \text{Rank3AntisymmetricPara2pConstantSymbol} \dagger + \\
& k^2 K_2^{(4)} \text{Rank3AntisymmetricPara1mConstantSymbol} \dagger \\
& \text{Rank3AntisymmetricPerp1mConstantSymbol} + \\
& K_3^{(2)} \text{Rank3AntisymmetricPara1mConstantSymbol} \dagger \text{Rank3AntisymmetricPerp1mConstantSymbol} + \\
& k^2 K_2^{(4)} \text{Rank3AntisymmetricPara1mConstantSymbol}
\end{aligned}$$

$\text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger + k^2 \overset{(4)}{K}_7$
 $\text{Rank3AntisymmetricPara1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger +$
 $\overset{(2)}{K}_3 \text{Rank3AntisymmetricPara1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger +$
 $2 k^2 \overset{(4)}{K}_{15} \text{Rank3AntisymmetricPerp1mConstantSymbol}$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger + k^2 \overset{(4)}{K}_{16}$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger -$
 $k^2 \overset{(4)}{K}_2 \text{Rank3AntisymmetricPerp1mConstantSymbol}$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger - k^2 \overset{(4)}{K}_3$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger -$
 $k^2 \overset{(4)}{K}_4 \text{Rank3AntisymmetricPerp1mConstantSymbol}$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger - k^2 \overset{(4)}{K}_7$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger +$
 $2 k^2 \overset{(4)}{K}_9 \text{Rank3AntisymmetricPerp1mConstantSymbol}$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger + 2 \overset{(2)}{K}_1$
 $\text{Rank3AntisymmetricPerp1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger +$
 $\overset{(2)}{K}_2 \text{Rank3AntisymmetricPerp1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger -$
 $\overset{(2)}{K}_3 \text{Rank3AntisymmetricPerp1mConstantSymbol} \text{Rank3AntisymmetricPerp1mConstantSymbol} \dagger -$
 $k^2 \overset{(4)}{K}_{16} \text{Rank3AntisymmetricPara1pConstantSymbol} \dagger$
 $\text{Rank3AntisymmetricPerp1pConstantSymbol} + k^2 \overset{(4)}{K}_4$
 $\text{Rank3AntisymmetricPara1pConstantSymbol} \dagger \text{Rank3AntisymmetricPerp1pConstantSymbol} -$
 $\overset{(2)}{K}_2 \text{Rank3AntisymmetricPara1pConstantSymbol} \dagger \text{Rank3AntisymmetricPerp1pConstantSymbol} -$
 $k^2 \overset{(4)}{K}_{16} \text{Rank3AntisymmetricPara1pConstantSymbol}$
 $\text{Rank3AntisymmetricPerp1pConstantSymbol} \dagger -$

$$K_2^{(2)} \text{Rank3AntisymmetricPara1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol} \dagger +$$

$$k^2 K_{15}^{(4)} \text{Rank3AntisymmetricPerp1pConstantSymbol}$$

$$\text{Rank3AntisymmetricPerp1pConstantSymbol} \dagger + k^2 K_9^{(4)}$$

$$\text{Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol} \dagger +$$

$$K_1^{(2)} \text{Rank3AntisymmetricPerp1pConstantSymbol Rank3AntisymmetricPerp1pConstantSymbol} \dagger +$$

$$-\frac{4}{3} k^2 K_{15}^{(4)} \mathcal{K}_{0^-}^{\dagger\dagger} \dagger_{\alpha\beta\chi} \mathcal{K}_{2^-}^{\dagger\dagger} \alpha\beta\chi - \frac{4}{3} k^2 K_{16}^{(4)} \mathcal{K}_{0^-}^{\dagger\dagger} \dagger_{\alpha\beta\chi} \mathcal{K}_{2^-}^{\dagger\dagger} \alpha\beta\chi +$$

$$-\frac{4}{3} K_1^{(2)} \mathcal{K}_{0^-}^{\dagger\dagger} \dagger_{\alpha\beta\chi} \mathcal{K}_{2^-}^{\dagger\dagger} \alpha\beta\chi - \frac{4}{3} K_2^{(2)} \mathcal{K}_{0^-}^{\dagger\dagger} \dagger_{\alpha\beta\chi} \mathcal{K}_{2^-}^{\dagger\dagger} \alpha\beta\chi +$$

$$-\frac{4}{3} k^2 K_{15}^{(4)} \mathcal{K}_{0^-}^{\dagger\dagger} \alpha\beta\chi \mathcal{K}_{2^-}^{\dagger\dagger} \dagger_{\alpha\beta\chi} - \frac{4}{3} k^2 K_{16}^{(4)} \mathcal{K}_{0^-}^{\dagger\dagger} \alpha\beta\chi \mathcal{K}_{2^-}^{\dagger\dagger} \dagger_{\alpha\beta\chi} +$$

$$-\frac{4}{3} K_1^{(2)} \mathcal{K}_{0^-}^{\dagger\dagger} \alpha\beta\chi \mathcal{K}_{2^-}^{\dagger\dagger} \dagger_{\alpha\beta\chi} - \frac{4}{3} K_2^{(2)} \mathcal{K}_{0^-}^{\dagger\dagger} \alpha\beta\chi \mathcal{K}_{2^-}^{\dagger\dagger} \dagger_{\alpha\beta\chi}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`MatrixLagrangian will now be printed.

$$\left\langle \begin{array}{l} 0 \end{array} \rightarrow \text{SparseArray} \left[\begin{array}{c} \text{Specified elements: 2} \\ \text{Dimensions: \{2, 2\}} \end{array} \right], \right.$$

$$1 \rightarrow \text{SparseArray} \left[\begin{array}{c} \text{Specified elements: 8} \\ \text{Dimensions: \{4, 4\}} \end{array} \right], 2 \rightarrow \text{SparseArray} \left[\begin{array}{c} \text{Specified elements: 2} \\ \text{Dimensions: \{2, 2\}} \end{array} \right] \right\rangle$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`MatrixLagrangian will now be printed.

$$\left\langle \begin{array}{l} 0 \end{array} \rightarrow \text{SparseArray} \left[\begin{array}{c} \text{Specified elements: 2} \\ \text{Dimensions: \{2, 2\}} \end{array} \right], \right.$$

$$1 \rightarrow \text{SparseArray} \left[\begin{array}{c} \text{Specified elements: 8} \\ \text{Dimensions: \{4, 4\}} \end{array} \right], 2 \rightarrow \text{SparseArray} \left[\begin{array}{c} \text{Specified elements: 2} \\ \text{Dimensions: \{2, 2\}} \end{array} \right] \right\rangle$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable

xAct`PSALter`Private`Class\$173389[xAct`PSALter`Private`InverseRescalingMatrix] will now be printed.

$$\left\langle \begin{array}{l} 0 \end{array} \rightarrow \left\{ \left\{ \frac{1}{\text{Rank3AntisymmetricPara0pRescalingSymbol}^2}, \right. \right.$$

$$\begin{aligned}
& \left\{ \frac{i \sqrt{6}}{\text{Rank3AntisymmetricPara0mRescalingSymbol Rank3AntisymmetricPara0pRescalingSymbol}} \right\}, \\
& \left\{ \frac{i \sqrt{6}}{\text{Rank3AntisymmetricPara0mRescalingSymbol Rank3AntisymmetricPara0pRescalingSymbol}} \right. \\
& \left. , \frac{1}{\text{Rank3AntisymmetricPara0mRescalingSymbol}^2} \right\}, \\
& 1 \rightarrow \left\{ \frac{1}{\text{Rank3AntisymmetricPara1pRescalingSymbol}^2}, \right. \\
& \left. \frac{1}{\text{Rank3AntisymmetricPara1pRescalingSymbol Rank3AntisymmetricPerp1pRescalingSymbol}} \right\}, \\
& \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara1mRescalingSymbol Rank3AntisymmetricPara1pRescalingSymbol}}, \\
& \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara1pRescalingSymbol Rank3AntisymmetricPerp1mRescalingSymbol}}, \\
& \left\{ \frac{1}{\text{Rank3AntisymmetricPara1pRescalingSymbol Rank3AntisymmetricPerp1pRescalingSymbol}} \right. \\
& \left. \frac{1}{\text{Rank3AntisymmetricPerp1pRescalingSymbol}^2}, \right. \\
& \left. \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara1mRescalingSymbol Rank3AntisymmetricPerp1pRescalingSymbol}} \right\},
\end{aligned}$$

$$\begin{aligned}
& \left\{ \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPerp1mRescalingSymbol Rank3AntisymmetricPerp1pRescalingSymbol}} \right\}, \\
& \left\{ \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara1mRescalingSymbol Rank3AntisymmetricPara1pRescalingSymbol}} \right\}, \\
& \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara1mRescalingSymbol Rank3AntisymmetricPerp1pRescalingSymbol}} \\
& \frac{1}{\text{Rank3AntisymmetricPara1mRescalingSymbol}^2}, \\
& \left. \frac{1}{\text{Rank3AntisymmetricPara1mRescalingSymbol Rank3AntisymmetricPerp1mRescalingSymbol}} \right\}, \\
& \left\{ \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara1pRescalingSymbol Rank3AntisymmetricPerp1mRescalingSymbol}} \right\}, \\
& \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPerp1mRescalingSymbol Rank3AntisymmetricPerp1pRescalingSymbol}}, \\
& \frac{1}{\text{Rank3AntisymmetricPara1mRescalingSymbol Rank3AntisymmetricPerp1mRescalingSymbol}}, \\
& \frac{1}{\text{Rank3AntisymmetricPerp1mRescalingSymbol}^2} \left. \right\} \left. \right\}, \\
& 2 \rightarrow \left\{ \frac{1}{\text{Rank3AntisymmetricPara2pRescalingSymbol}^2}, \right. \\
& \left. \frac{i \sqrt{2}}{\text{Rank3AntisymmetricPara2mRescalingSymbol Rank3AntisymmetricPara2pRescalingSymbol}} \right\}
\end{aligned}$$

$$\left. \left\{ -\frac{i\sqrt{2}}{\text{Rank3AntisymmetricPara2mRescalingSymbol} \text{Rank3AntisymmetricPara2pRescalingSymbol}}, \frac{1}{\text{Rank3AntisymmetricPara2mRescalingSymbol}^2} \right\} \right| \rangle$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`MatrixLagrangian will now be printed.

$$\langle \left| \begin{aligned} & \left\{ \left\{ \frac{-2k^2 \binom{4}{K_1} + \frac{4k^2 \binom{4}{K_{15}}}{3} + \frac{2k^2 \binom{4}{K_{16}}}{3} - 2k^2 \binom{4}{K_2} - \frac{2k^2 \binom{4}{K_3}}{3} - \frac{2k^2 \binom{4}{K_6}}{3} + \frac{4 \binom{2}{K_1}}{3} + \frac{2 \binom{2}{K_2}}{3} - 2 \binom{2}{K_3}}{2 \text{Rank3AntisymmetricPara0pRescalingSymbol}^2}, 0 \right\}, \right. \\ & \left. \left\{ 0, \frac{2k^2 \binom{4}{K_{15}} - 2k^2 \binom{4}{K_{16}} + 2 \binom{2}{K_1} - 2 \binom{2}{K_2}}{2 \text{Rank3AntisymmetricPara0mRescalingSymbol}^2} \right\}, \right. \\ & 1 \rightarrow \left\{ \left\{ \frac{4k^2 \binom{4}{K_{15}} - 2k^2 \binom{4}{K_{16}} - 2k^2 \binom{4}{K_3} + 2k^2 \binom{4}{K_6} + 4 \binom{2}{K_1} - 2 \binom{2}{K_2}}{2 \text{Rank3AntisymmetricPara1pRescalingSymbol}^2}, \right. \right. \\ & \quad \left. \left. \frac{-2k^2 \binom{4}{K_{16}} + k^2 \binom{4}{K_4} - 2 \binom{2}{K_2}}{2 \text{Rank3AntisymmetricPara1pRescalingSymbol} \text{Rank3AntisymmetricPerp1pRescalingSymbol}}, 0, 0 \right\}, \right. \\ & \left. \left\{ \frac{-2k^2 \binom{4}{K_{16}} + k^2 \binom{4}{K_4} - 2 \binom{2}{K_2}}{2 \text{Rank3AntisymmetricPara1pRescalingSymbol} \text{Rank3AntisymmetricPerp1pRescalingSymbol}}, \right. \\ & \quad \left. \frac{2k^2 \binom{4}{K_{15}} + 2k^2 \binom{4}{K_9} + 2 \binom{2}{K_1}}{2 \text{Rank3AntisymmetricPerp1pRescalingSymbol}^2}, 0, 0 \right\}, \end{aligned} \right\rangle$$

$$\begin{aligned}
& \left\{ 0, 0, \frac{{}_2 k^2 \binom{4}{K_{15}} + k^2 \binom{4}{K_{16}} - {}_2 k^2 \binom{4}{K_2} + {}_2 \binom{2}{K_1} + \binom{2}{K_2} - {}_2 \binom{2}{K_3}}{2 \text{Rank3AntisymmetricPara1mRescalingSymbol}^2}, \right. \\
& \quad \left. \frac{{}_2 k^2 \binom{4}{K_2} + k^2 \binom{4}{K_7} + {}_2 \binom{2}{K_3}}{2 \text{Rank3AntisymmetricPara1mRescalingSymbol} \text{Rank3AntisymmetricPerp1mRescalingSymbol}} \right\}, \left\{ 0, 0, \right. \\
& \quad \left. \frac{{}_2 k^2 \binom{4}{K_2} + k^2 \binom{4}{K_7} + {}_2 \binom{2}{K_3}}{2 \text{Rank3AntisymmetricPara1mRescalingSymbol} \text{Rank3AntisymmetricPerp1mRescalingSymbol}} \right. \\
& \quad \left. , \left({}_4 k^2 \binom{4}{K_{15}} + {}_2 k^2 \binom{4}{K_{16}} - {}_2 k^2 \binom{4}{K_2} - {}_2 k^2 \binom{4}{K_3} - {}_2 k^2 \binom{4}{K_4} - {}_2 k^2 \binom{4}{K_7} + {}_4 k^2 \binom{4}{K_9} + \right. \right. \\
& \quad \left. \left. {}_4 \binom{2}{K_1} + {}_2 \binom{2}{K_2} - {}_2 \binom{2}{K_3} \right) / \left(2 \text{Rank3AntisymmetricPerp1mRescalingSymbol}^2 \right) \right\} \Bigg\}, \\
& {}_2 \rightarrow \left\{ \left\{ \frac{{}_4 k^2 \binom{4}{K_{15}} + {}_2 k^2 \binom{4}{K_{16}} - {}_2 k^2 \binom{4}{K_3} - {}_2 k^2 \binom{4}{K_6} + {}_4 \binom{2}{K_1} + {}_2 \binom{2}{K_2}}{2 \text{Rank3AntisymmetricPara2pRescalingSymbol}^2}, 0 \right\}, \right. \\
& \quad \left. \left\{ 0, \frac{\frac{32 k^2 \binom{4}{K_{15}}}{9} + \frac{16 k^2 \binom{4}{K_{16}}}{9} + \frac{32 \binom{2}{K_1}}{9} + \frac{16 \binom{2}{K_2}}{9}}{2 \text{Rank3AntisymmetricPara2mRescalingSymbol}^2} \right\} \right\} \Big| \rangle
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`MatrixLagrangian will now be printed.

$$\langle \Big|_0 \rightarrow$$

$$\left\{ \left\{ \frac{-{}_2 k^2 \binom{4}{K_1} + \frac{{}_4 k^2 \binom{4}{K_{15}}}{3} + \frac{{}_2 k^2 \binom{4}{K_{16}}}{3} - {}_2 k^2 \binom{4}{K_2} - \frac{{}_2 k^2 \binom{4}{K_3}}{3} - \frac{{}_2 k^2 \binom{4}{K_6}}{3} + \frac{{}_4 \binom{2}{K_1}}{3} + \frac{{}_2 \binom{2}{K_2}}{3} - {}_2 \binom{2}{K_3}}{2 \sqrt{\frac{2}{3}}}, 0 \right\}, \right\}$$

$$\begin{aligned}
& \left\{ 0, \frac{{}_2 k^2 \binom{4}{K_{15}} - {}_2 k^2 \binom{4}{K_{16}} + {}_2 \binom{2}{K_1} - {}_2 \binom{2}{K_2}}{{}_2 \times 1^2} \right\}, \\
& \xrightarrow{1} \left\{ \left\{ \frac{{}_4 k^2 \binom{4}{K_{15}} - {}_2 k^2 \binom{4}{K_{16}} - {}_2 k^2 \binom{4}{K_3} + {}_2 k^2 \binom{4}{K_6} + {}_4 \binom{2}{K_1} - {}_2 \binom{2}{K_2}}{{}_2 \sqrt{2}^2}, \right. \right. \\
& \quad \left. \left. \frac{- {}_2 k^2 \binom{4}{K_{16}} + {}_2 k^2 \binom{4}{K_4} - {}_2 \binom{2}{K_2}}{{}_2 \sqrt{2}}, 0, 0 \right\}, \right. \\
& \quad \left\{ \frac{- {}_2 k^2 \binom{4}{K_{16}} + {}_2 k^2 \binom{4}{K_4} - {}_2 \binom{2}{K_2}}{{}_2 \sqrt{2}}, \frac{{}_2 k^2 \binom{4}{K_{15}} + {}_2 k^2 \binom{4}{K_9} + {}_2 \binom{2}{K_1}}{{}_2 \times 1^2}, 0, 0 \right\}, \\
& \quad \left\{ 0, 0, \frac{{}_2 k^2 \binom{4}{K_{15}} + {}_2 k^2 \binom{4}{K_{16}} - {}_2 k^2 \binom{4}{K_2} + {}_2 \binom{2}{K_1} + \binom{2}{K_2} - {}_2 \binom{2}{K_3}}{{}_2 \times 1^2}, \frac{{}_2 k^2 \binom{4}{K_2} + {}_2 k^2 \binom{4}{K_7} + {}_2 \binom{2}{K_3}}{{}_2 \sqrt{2}} \right\}, \\
& \quad \left\{ 0, 0, \frac{{}_2 k^2 \binom{4}{K_2} + {}_2 k^2 \binom{4}{K_7} + {}_2 \binom{2}{K_3}}{{}_2 \sqrt{2}}, \frac{1}{{}_2 \sqrt{2}^2} \left({}_4 k^2 \binom{4}{K_{15}} + {}_2 k^2 \binom{4}{K_{16}} - {}_2 k^2 \binom{4}{K_2} - \right. \right. \\
& \quad \left. \left. {}_2 k^2 \binom{4}{K_3} - {}_2 k^2 \binom{4}{K_4} - {}_2 k^2 \binom{4}{K_7} + {}_4 k^2 \binom{4}{K_9} + {}_4 \binom{2}{K_1} + {}_2 \binom{2}{K_2} - {}_2 \binom{2}{K_3} \right) \right\}, \\
& \xrightarrow{2} \left\{ \left\{ \frac{{}_4 k^2 \binom{4}{K_{15}} + {}_2 k^2 \binom{4}{K_{16}} - {}_2 k^2 \binom{4}{K_3} - {}_2 k^2 \binom{4}{K_6} + {}_4 \binom{2}{K_1} + {}_2 \binom{2}{K_2}}{{}_2 \sqrt{2}^2}, 0 \right\}, \right. \\
& \quad \left. \left\{ 0, \frac{\frac{32 k^2 \binom{4}{K_{15}}}{9} + \frac{16 k^2 \binom{4}{K_{16}}}{9} + \frac{32 \binom{2}{K_1}}{9} + \frac{16 \binom{2}{K_2}}{9}}{2 \left(\frac{4}{3} \right)^2} \right\} \right\} \Big| \rangle
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the
variable xAct`PSALter`Private`MatrixLagrangian will now be printed.

$$\begin{aligned}
& \langle \left| \begin{aligned} & 0 \rightarrow \left\{ \left\{ \frac{1}{2} \left(-k^2 \left({}^3 K_1^{(4)} - {}^2 K_{15}^{(4)} - K_{16}^{(4)} + {}^3 K_2^{(4)} + K_3^{(4)} + K_6^{(4)} \right) + {}^2 K_1^{(2)} + K_2^{(2)} - {}^3 K_3^{(2)} \right), 0 \right\}, \right. \\ & \left. \left\{ 0, k^2 \left(K_{15}^{(4)} - K_{16}^{(4)} \right) + K_1^{(2)} - K_2^{(2)} \right\} \right\}, \\ & 1 \rightarrow \left\{ \left\{ \frac{1}{2} k^2 \left({}^2 K_{15}^{(4)} - K_{16}^{(4)} - K_3^{(4)} + K_6^{(4)} \right) + K_1^{(2)} - \frac{K_2^{(2)}}{2}, \frac{k^2 \left(-{}^2 K_{16}^{(4)} + K_4^{(4)} \right) - {}^2 K_2^{(2)}}{{}^2 \sqrt{2}}, 0, 0 \right\}, \right. \\ & \left. \left\{ \frac{k^2 \left(-{}^2 K_{16}^{(4)} + K_4^{(4)} \right) - {}^2 K_2^{(2)}}{{}^2 \sqrt{2}}, k^2 \left(K_{15}^{(4)} + K_9^{(4)} \right) + K_1^{(2)}, 0, 0 \right\}, \right. \\ & \left. \left\{ 0, 0, \frac{1}{2} \left(k^2 \left({}^2 K_{15}^{(4)} + K_{16}^{(4)} - {}^2 K_2^{(4)} \right) + {}^2 K_1^{(2)} + K_2^{(2)} - {}^2 K_3^{(2)} \right), \frac{k^2 \left({}^2 K_2^{(4)} + K_7^{(4)} \right) + {}^2 K_3^{(2)}}{{}^2 \sqrt{2}} \right\}, \right. \\ & \left. \left\{ 0, 0, \frac{k^2 \left({}^2 K_2^{(4)} + K_7^{(4)} \right) + {}^2 K_3^{(2)}}{{}^2 \sqrt{2}}, \right. \right. \\ & \left. \left. \frac{1}{2} \left(k^2 \left({}^2 K_{15}^{(4)} + K_{16}^{(4)} - K_2^{(4)} - K_3^{(4)} - K_4^{(4)} - K_7^{(4)} + {}^2 K_9^{(4)} \right) + {}^2 K_1^{(2)} + K_2^{(2)} - K_3^{(2)} \right) \right\} \right\}, \\ & 2 \rightarrow \left\{ \left\{ \frac{1}{2} \left(k^2 \left({}^2 K_{15}^{(4)} + K_{16}^{(4)} - K_3^{(4)} - K_6^{(4)} \right) + {}^2 K_1^{(2)} + K_2^{(2)} \right), 0 \right\}, \right. \\ & \left. \left\{ 0, \frac{1}{2} \left(k^2 \left({}^2 K_{15}^{(4)} + K_{16}^{(4)} \right) + {}^2 K_1^{(2)} + K_2^{(2)} \right) \right\} \right\} \Big| \rangle
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the
variable xAct`PSALter`Private`AntiMaskMatrixValue will now be printed.

```

< | 0 → < | xAct`PSALter`Private`Even → {{1, 0}, {0, 0}},
      xAct`PSALter`Private`Odd → {{0, 0}, {0, 1}} | > ,
1 → < | xAct`PSALter`Private`Even →
      {{1, 1, 0, 0}, {1, 1, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}}, xAct`PSALter`Private`Odd →
      {{0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 1, 1}, {0, 0, 1, 1}} | > ,
2 → < | xAct`PSALter`Private`Even → {{1, 0}, {0, 0}},
      xAct`PSALter`Private`Odd → {{0, 0}, {0, 1}} | > | >

** xAct`PSALter`Private`Diagnostic: the value of the
variable xAct`PSALter`Private`$BMatricesValues will now be printed.

```


$$\begin{aligned}
& \langle \left| \right\rangle_0 \rightarrow \\
& \left\{ \left\{ \left\{ \frac{1}{2} \left(-k^2 \left({}_3 \binom{(4)}{K_1} - {}_2 \binom{(4)}{K_{15}} - \binom{(4)}{K_{16}} + {}_3 \binom{(4)}{K_2} + \binom{(4)}{K_3} + \binom{(4)}{K_6} \right) + {}_2 \binom{(2)}{K_1} + \binom{(2)}{K_2} - {}_3 \binom{(2)}{K_3} \right), 0 \right\}, \right. \right. \\
& \quad \left. \left. \left\{ 0, 0 \right\} \right\}, \left\{ \left\{ 0, 0 \right\}, \left\{ 0, k^2 \left(\binom{(4)}{K_{15}} - \binom{(4)}{K_{16}} \right) + \binom{(2)}{K_1} - \binom{(2)}{K_2} \right\} \right\} \right\}, 1 \rightarrow \\
& \left\{ \left\{ \left\{ \frac{1}{2} k^2 \left({}_2 \binom{(4)}{K_{15}} - \binom{(4)}{K_{16}} - \binom{(4)}{K_3} + \binom{(4)}{K_6} \right) + \binom{(2)}{K_1} - \frac{\binom{(2)}{K_2}}{2}, \frac{k^2 \left(-{}_2 \binom{(4)}{K_{16}} + \binom{(4)}{K_4} \right) - {}_2 \binom{(2)}{K_2}}{{}_2 \sqrt{2}}, 0, 0 \right\}, \right. \right. \\
& \quad \left. \left. \left\{ \frac{k^2 \left(-{}_2 \binom{(4)}{K_{16}} + \binom{(4)}{K_4} \right) - {}_2 \binom{(2)}{K_2}}{{}_2 \sqrt{2}}, k^2 \left(\binom{(4)}{K_{15}} + \binom{(4)}{K_9} \right) + \binom{(2)}{K_1}, 0, 0 \right\}, \right. \right. \\
& \quad \left. \left. \left\{ 0, 0, 0, 0 \right\}, \left\{ 0, 0, 0, 0 \right\} \right\}, \left\{ \left\{ 0, 0, 0, 0 \right\}, \left\{ 0, 0, 0, 0 \right\}, \right. \right. \\
& \quad \left. \left. \left\{ 0, 0, \frac{1}{2} \left(k^2 \left({}_2 \binom{(4)}{K_{15}} + \binom{(4)}{K_{16}} - {}_2 \binom{(4)}{K_2} \right) + {}_2 \binom{(2)}{K_1} + \binom{(2)}{K_2} - {}_2 \binom{(2)}{K_3} \right), \frac{k^2 \left({}_2 \binom{(4)}{K_2} + \binom{(4)}{K_7} \right) + {}_2 \binom{(2)}{K_3}}{{}_2 \sqrt{2}} \right\}, \right. \right. \\
& \quad \left. \left. \left\{ 0, 0, \frac{k^2 \left({}_2 \binom{(4)}{K_2} + \binom{(4)}{K_7} \right) + {}_2 \binom{(2)}{K_3}}{{}_2 \sqrt{2}}, \right. \right. \\
& \quad \left. \left. \frac{1}{2} \left(k^2 \left({}_2 \binom{(4)}{K_{15}} + \binom{(4)}{K_{16}} - \binom{(4)}{K_2} - \binom{(4)}{K_3} - \binom{(4)}{K_4} - \binom{(4)}{K_7} + {}_2 \binom{(4)}{K_9} \right) + {}_2 \binom{(2)}{K_1} + \binom{(2)}{K_2} - \binom{(2)}{K_3} \right) \right\} \right\} \right\}, \\
& {}_2 \rightarrow \left\{ \left\{ \left\{ \frac{1}{2} \left(k^2 \left({}_2 \binom{(4)}{K_{15}} + \binom{(4)}{K_{16}} - \binom{(4)}{K_3} - \binom{(4)}{K_6} \right) + {}_2 \binom{(2)}{K_1} + \binom{(2)}{K_2} \right), 0 \right\}, \left\{ 0, 0 \right\} \right\}, \right. \\
& \quad \left. \left\{ \left\{ 0, 0 \right\}, \left\{ 0, \frac{1}{2} \left(k^2 \left({}_2 \binom{(4)}{K_{15}} + \binom{(4)}{K_{16}} \right) + {}_2 \binom{(2)}{K_1} + \binom{(2)}{K_2} \right) \right\} \right\} \right\} \left| \right\rangle
\end{aligned}$$

****** xAct`PSALter`Private`Diagnostic: the value of the variable xAct`PSALter`Private`\$ValuesAllMatrices will now be printed.

$$\begin{aligned}
& \left\{ \left\{ \left\{ \frac{1}{2} \left(-k^2 \left({}_3 K_1^{(4)} - {}_2 K_{15}^{(4)} - K_{16}^{(4)} + {}_3 K_2^{(4)} + K_3^{(4)} + K_6^{(4)} \right) + {}_2 K_1^{(2)} + K_2^{(2)} - {}_3 K_3^{(2)} \right), 0 \right\}, \right. \\
& \quad \left. \left\{ 0, k^2 \left(K_{15}^{(4)} - K_{16}^{(4)} \right) + K_1^{(2)} - K_2^{(2)} \right\} \right\}, \\
& \left\{ \left\{ \frac{1}{2} k^2 \left({}_2 K_{15}^{(4)} - K_{16}^{(4)} - K_3^{(4)} + K_6^{(4)} \right) + K_1^{(2)} - \frac{K_2^{(2)}}{2}, \frac{k^2 \left(-{}_2 K_{16}^{(4)} + K_4^{(4)} \right) - {}_2 K_2^{(2)}}{{}_2 \sqrt{2}}, 0, 0 \right\}, \right. \\
& \quad \left. \left\{ \frac{k^2 \left(-{}_2 K_{16}^{(4)} + K_4^{(4)} \right) - {}_2 K_2^{(2)}}{{}_2 \sqrt{2}}, k^2 \left(K_{15}^{(4)} + K_9^{(4)} \right) + K_1^{(2)}, 0, 0 \right\}, \right. \\
& \quad \left. \left\{ 0, 0, \frac{1}{2} \left(k^2 \left({}_2 K_{15}^{(4)} + K_{16}^{(4)} - {}_2 K_2^{(4)} \right) + {}_2 K_1^{(2)} + K_2^{(2)} - {}_2 K_3^{(2)} \right), \frac{k^2 \left({}_2 K_2^{(4)} + K_7^{(4)} \right) + {}_2 K_3^{(2)}}{{}_2 \sqrt{2}} \right\}, \right. \\
& \quad \left. \left\{ 0, 0, \frac{k^2 \left({}_2 K_2^{(4)} + K_7^{(4)} \right) + {}_2 K_3^{(2)}}{{}_2 \sqrt{2}}, \right. \right. \\
& \quad \left. \left. \frac{1}{2} \left(k^2 \left({}_2 K_{15}^{(4)} + K_{16}^{(4)} - K_2^{(4)} - K_3^{(4)} - K_4^{(4)} - K_7^{(4)} + {}_2 K_9^{(4)} \right) + {}_2 K_1^{(2)} + K_2^{(2)} - K_3^{(2)} \right) \right\} \right\}, \\
& \left\{ \left\{ \frac{1}{2} \left(k^2 \left({}_2 K_{15}^{(4)} + K_{16}^{(4)} - K_3^{(4)} - K_6^{(4)} \right) + {}_2 K_1^{(2)} + K_2^{(2)} \right), 0 \right\}, \right. \\
& \quad \left. \left\{ 0, \frac{1}{2} \left(k^2 \left({}_2 K_{15}^{(4)} + K_{16}^{(4)} \right) + {}_2 K_1^{(2)} + K_2^{(2)} \right) \right\} \right\}
\end{aligned}$$

****** xAct`PSALTER`Private`Diagnostic: the value of the variable
xAct`PSALTER`Private`EmptyDeterminants\$173389 will now be printed.

$\{\{\}, \{\}, \{\}\}$

Key observation: This is the end of the calibration.